

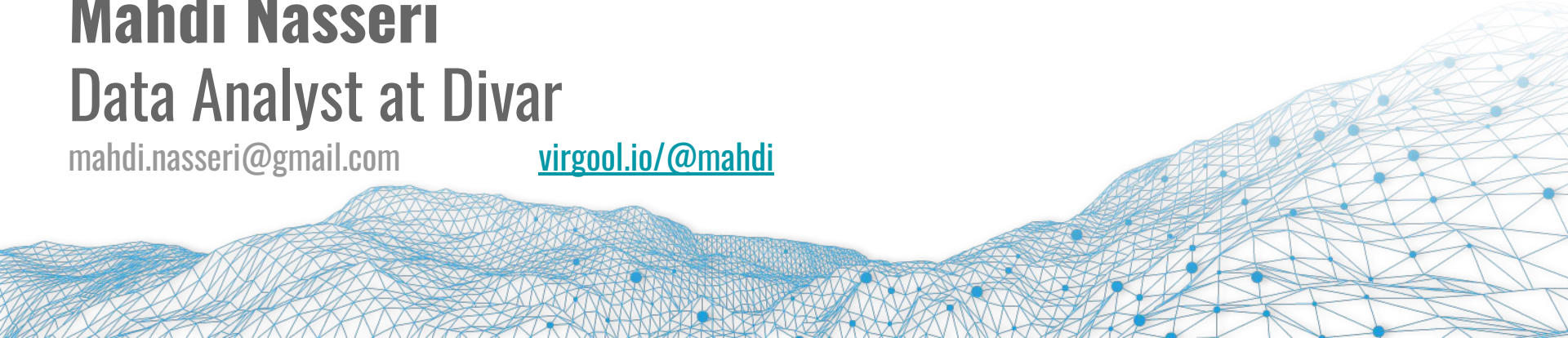
Storytelling With Data

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Part 0.

HOW TO USE YOUR OWN STORIES
TO COMMUNICATE
WITH POWER AND IMPACT

Whoever Tells
the Best Story
WINS

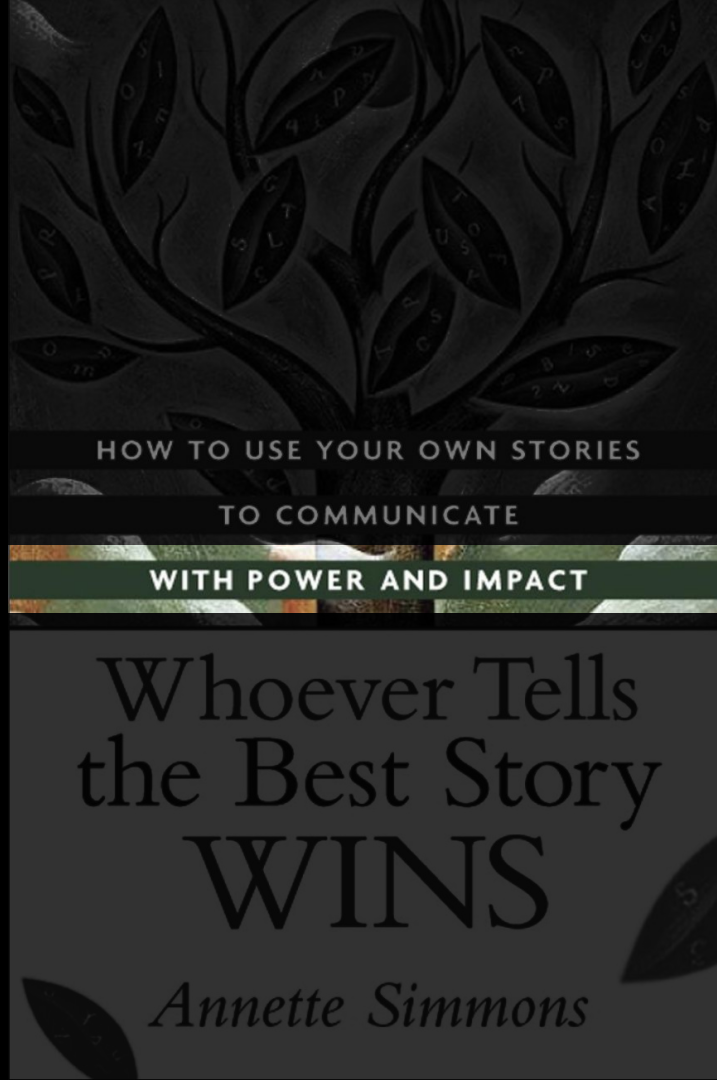
Annette Simmons

WITH POWER AND IMPACT

Annette Simmons

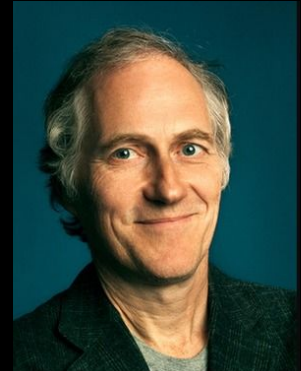
Why Storytelling with **Data**?

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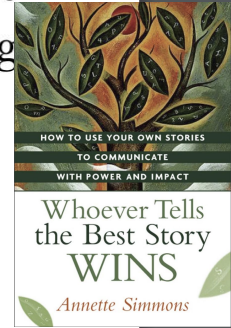


Who has the data has the power.

— *Tim O'Reilly* —



To be a good storyteller you must be a good story listener. I've never met a good storyteller who wasn't equally good at listening to stories. Where do you think they find their stories?

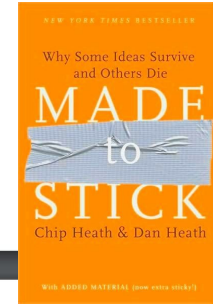


Experience Reconstituted

But what is a story? There are a thousand definitions, mostly academic, quite accurate but not very instructional. For our purposes our working definition of story is:

Story is a reimagined experience narrated with enough detail and feeling to cause your listeners' imaginations to experience it as real.

PRINCIPLE 3: CONCRETENESS



How do we make our ideas clear? We must explain our ideas in terms of human actions, in terms of sensory information. This is where so much business communication goes awry. Mission statements, synergies, strategies, visions—they are often ambiguous to the point of being meaningless. Naturally sticky ideas are full of concrete images—ice-filled bathtubs, apples with razors—because our brains are wired to remember concrete data. In proverbs, abstract truths are often encoded in concrete language: “A bird in hand is worth two in the bush.” Speaking concretely is the only way to ensure that our idea will mean the same thing to everyone in our audience.



Show, don't tell

🌐 11 languages ▾

Article [Talk](#)

[Read](#) [Edit](#) [View history](#) [Tools](#) ▾

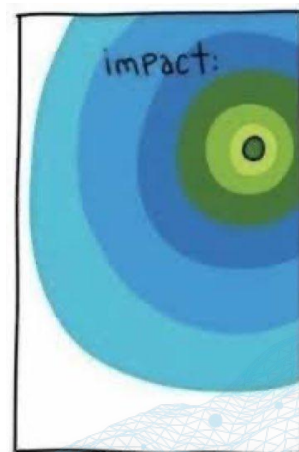
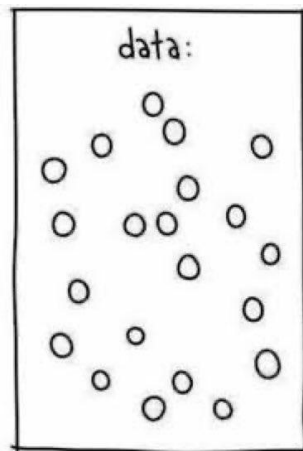
From Wikipedia, the free encyclopedia

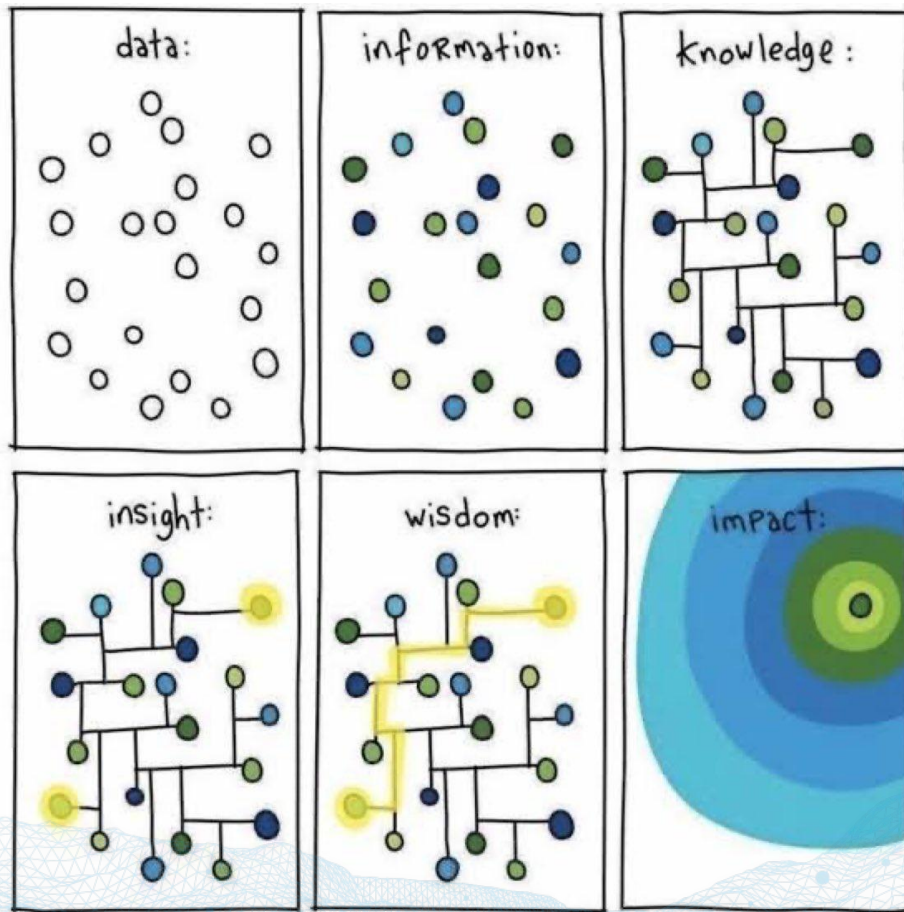
For the song, see [Show Don't Tell](#).

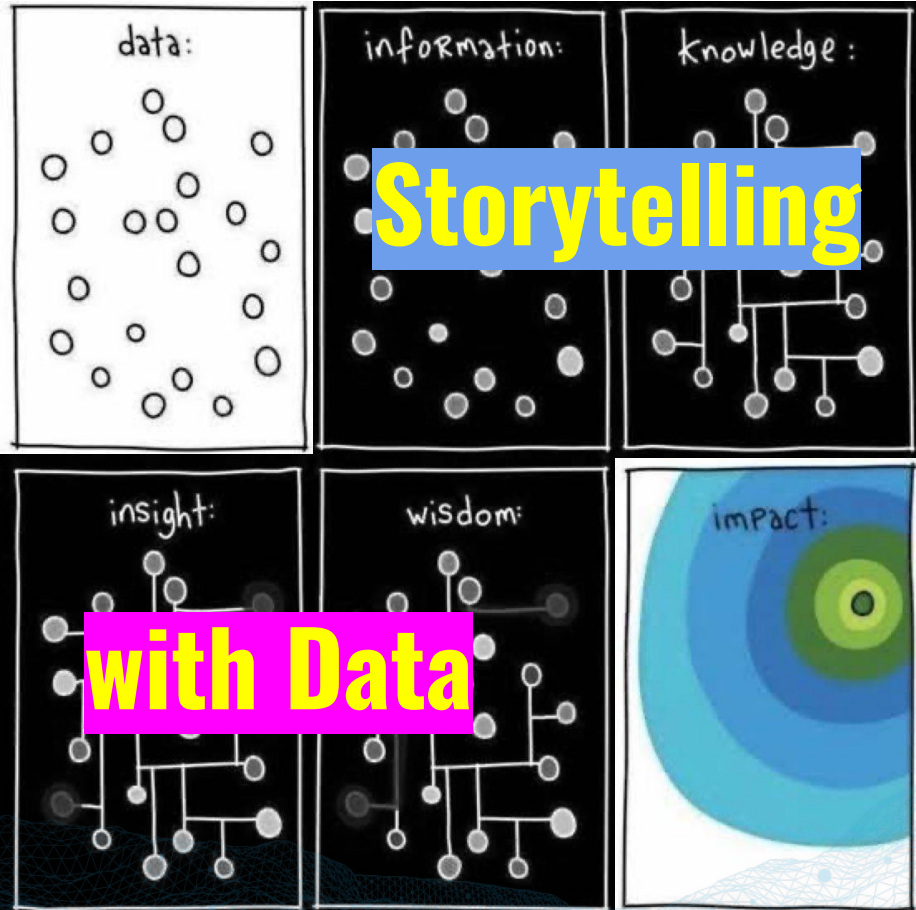
Show, don't tell is a technique used in various kinds of texts to allow the reader to experience the story through [actions](#), words, [subtext](#), thoughts, senses, and feelings rather than through the author's [exposition](#), summarization, and [description](#).^[1] It avoids adjectives describing the author's analysis, but instead describes the scene in such a way that readers can draw their own conclusions. The technique applies equally to [nonfiction](#) and all forms of [fiction](#), literature including [haiku](#)^[2] and [Imagism](#) poetry in particular, speech, movie making, and playwriting.^{[3][4][5][6]}

The concept is often attributed to Russian playwright [Anton Chekhov](#), reputed to have said "[Don't tell me the moon is shining; show me the glint of light on broken glass](#)." What Chekhov actually said, in a letter to his brother, was "In descriptions of Nature one must seize on small details, grouping them so that when the reader closes his eyes he gets a picture. For instance, you'll have a moonlit night if you write that on the mill dam a piece of glass from a broken bottle glittered like a bright little star, and that the black shadow of a dog or a wolf rolled past like a ball."^[7]

Its having become, by the mid-twentieth century, an important element in Anglo-Saxon [narratological theory](#), according to dramatist and author [Arthur E. Krows](#), the American dramatist [Mark Swan](#) told Krows about the playwriting motto "Show – not tell" on an occasion during the 1910s.^[8] In 1921, the same distinction, but in the form picture-versus-drama, was utilized in a chapter of [Percy Lubbock](#)'s analysis of fiction, *The Craft of Fiction*. In 1927, Swan published a playwriting manual that made prominent use of the showing-versus-telling distinction throughout.^[9]

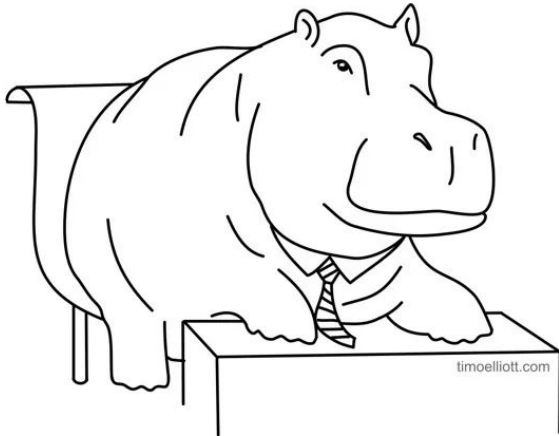




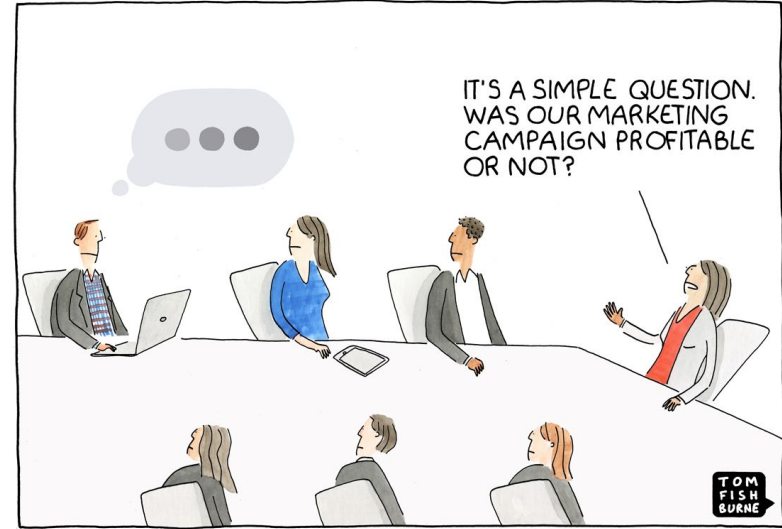


Intuitive Decision Making Rational

No Analytics? Welcome to the HIPPO*

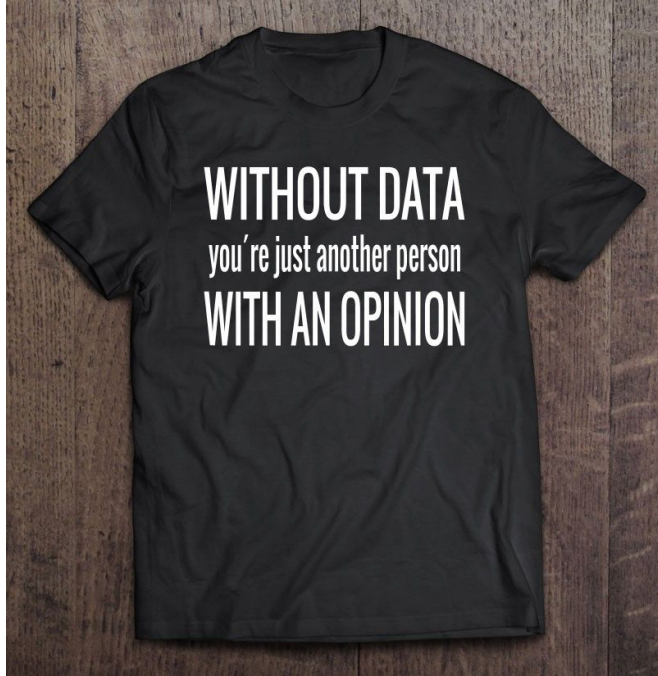


*Highest Paid Person's Opinion



© marketoonist.com

Intuitive Decision Making Rational



"When you two have finished arguing your opinions, I actually have data!"

Notice!



People

'Data holds power. But the most important thing is the story you tell with it.'

Tell a Data Story vs. Show the Statistics

Memorability

63% could remember stories, but only 5% could remember a single statistic.

Persuasiveness

In another study, researchers tested two variations of a brochure for the [Save the Children](#) charity organization. The story-based version [outperformed the infographic version](#) by \$2.38 to \$1.14 in terms of per participant donations.

Engagement

Researchers also discovered people enter into a trance-like state, where they [drop their intellectual guard and are less critical and skeptical](#). Rather than nitpicking over the details, the audience wants to see where the story leads them.

Tell a Data Story vs. Show the Statistics

Memorability

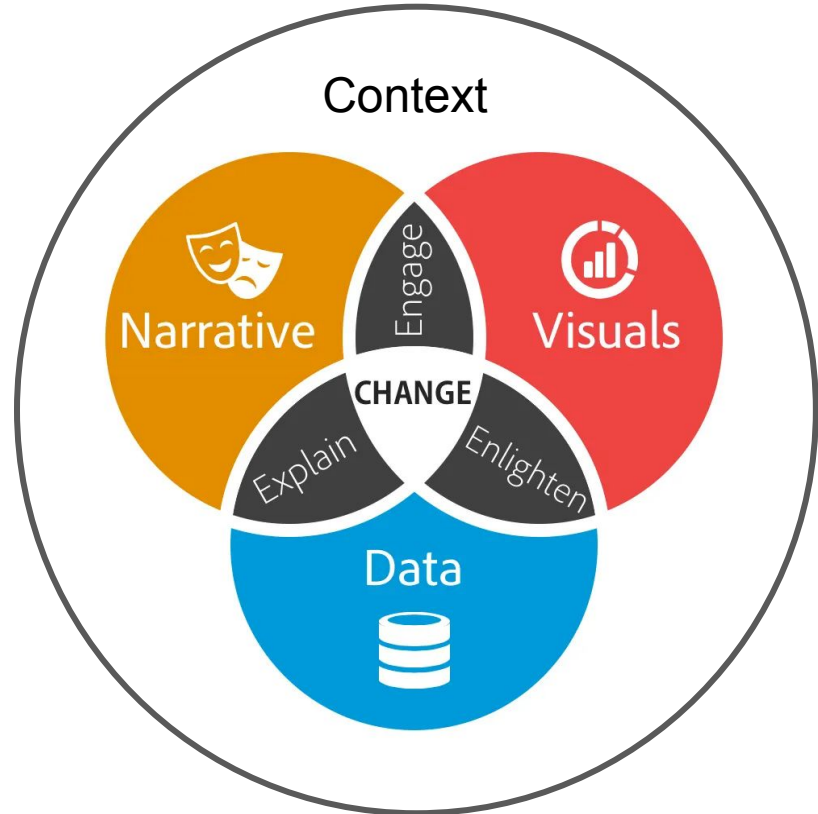
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Data Story

Id	price	created_at	user_id	status
...				
292970	1001000.0	2019-05-27T17:11:38.390Z	350663	PAID
292971	870000.0	2019-05-27T17:31:32.627Z	286190	PAID
292973	99000.0	2019-05-27T17:43:08.090Z	383168	UNPAID
292974	756000.0	2019-05-27T17:43:56.027Z	251070	UNPAID
292975	379000.0	2019-05-27T18:20:45.837Z	270144	UNPAID
292977	425000.0	2019-05-27T19:46:46.250Z	300993	UNPAID
292978	104000.0	2019-05-27T19:56:43.250Z	383173	UNPAID
292979	118000.0	2019-05-27T20:16:13.043Z	355577	PAID
292980	249000.0	2019-05-27T21:03:48.973Z	263163	UNPAID
292982	379000.0	2019-05-27T23:04:57.990Z	383170	PAID
292983	152000.0	2019-05-28T02:32:46.843Z	383175	UNPAID
292985	425000.0	2019-05-28T03:33:33.923Z	325013	UNPAID
292986	361000.0	2019-05-28T06:19:35.213Z	329752	UNPAID
292987	416000.0	2019-05-28T09:10:07.847Z	318198	PAID
292988	1043000.0	2019-05-28T08:36:44.873Z	372225	PAID
...				

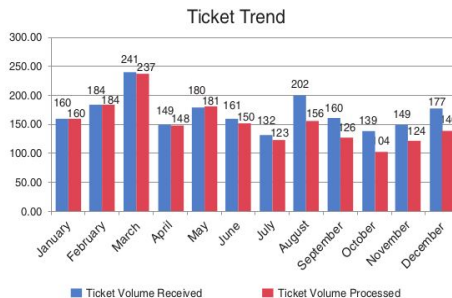
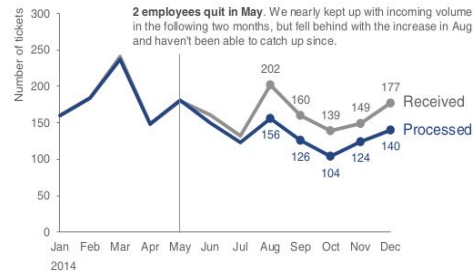


FIGURE 0.2 Example 1 (before): showing data

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

Ticket volume over time



Data source: XYZ Dashboard, as of 12/31/2014 | A detailed analysis on tickets processed per person and time to resolve issues was undertaken to inform this request and can be provided if needed.

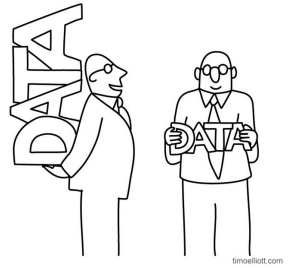
FIGURE 0.3 Example 1 (after): storytelling with data

DATA

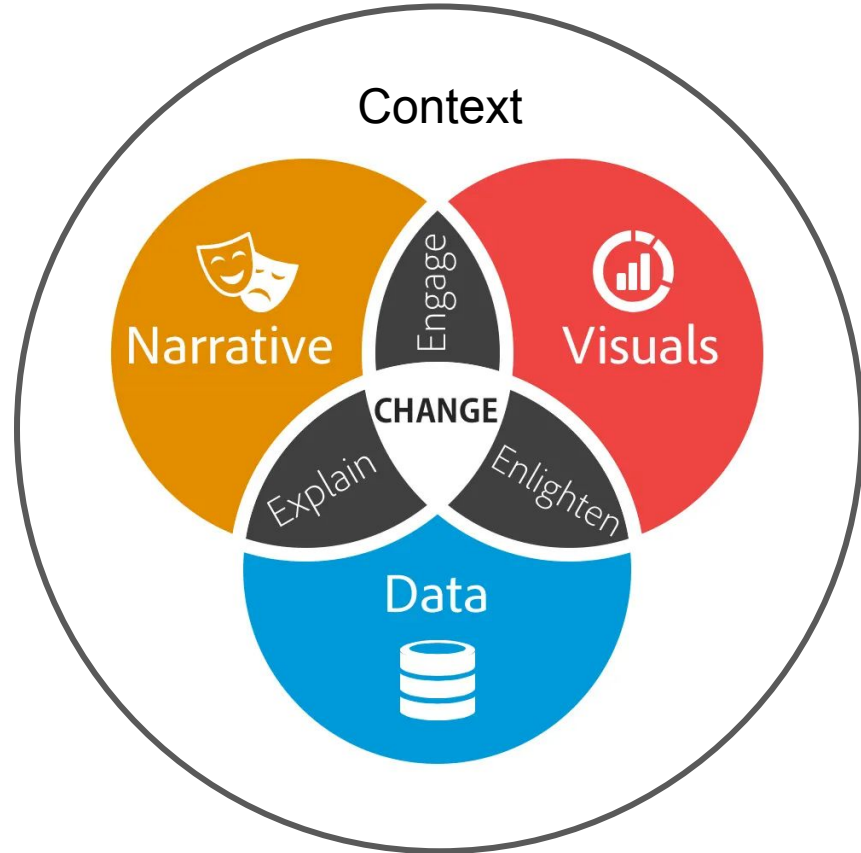
Visualize

Narrative

Part 1.



Data Story Context



Context

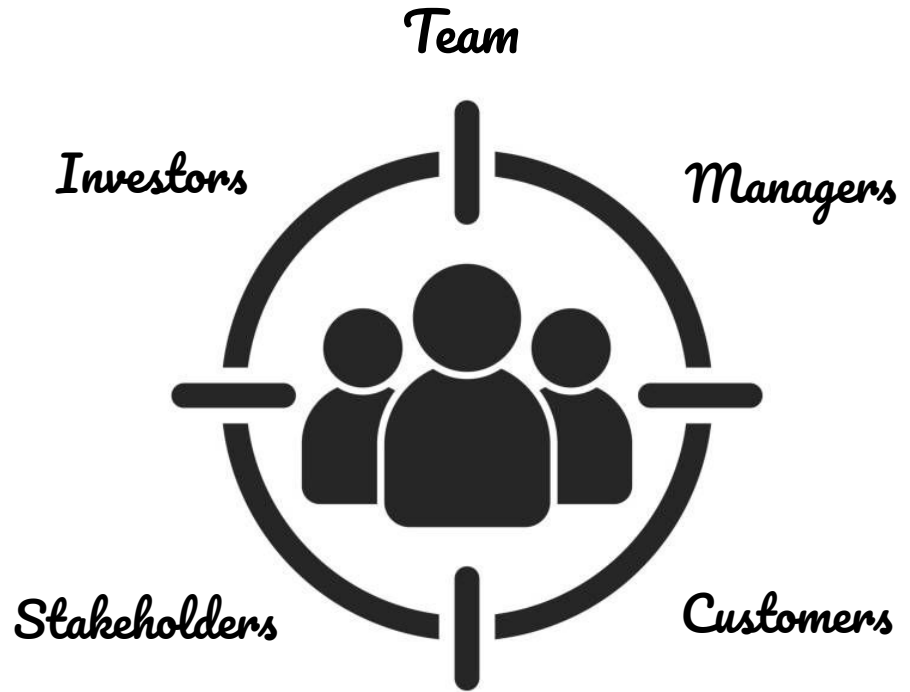
1. Who is Your Audience?

What is their **relationship** to you?

WHat **motivates** them?

What **keeps** them up to night?

Be Specific!

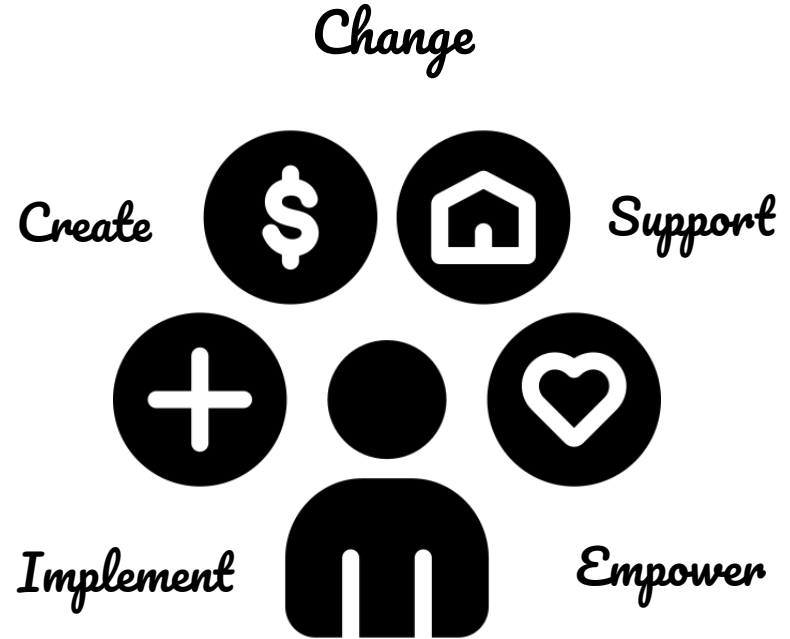


Context

1. What do You Need them to Do?

Don't assume they will connect dots!

Be Explicit!

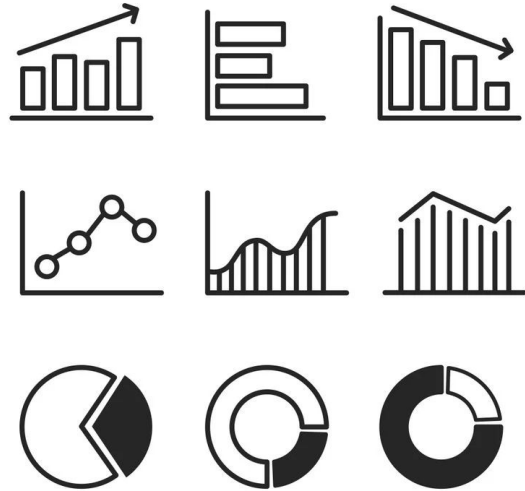


Context

1. How will Data Help Make Your Point?

What data will act as **evidence** for the case?

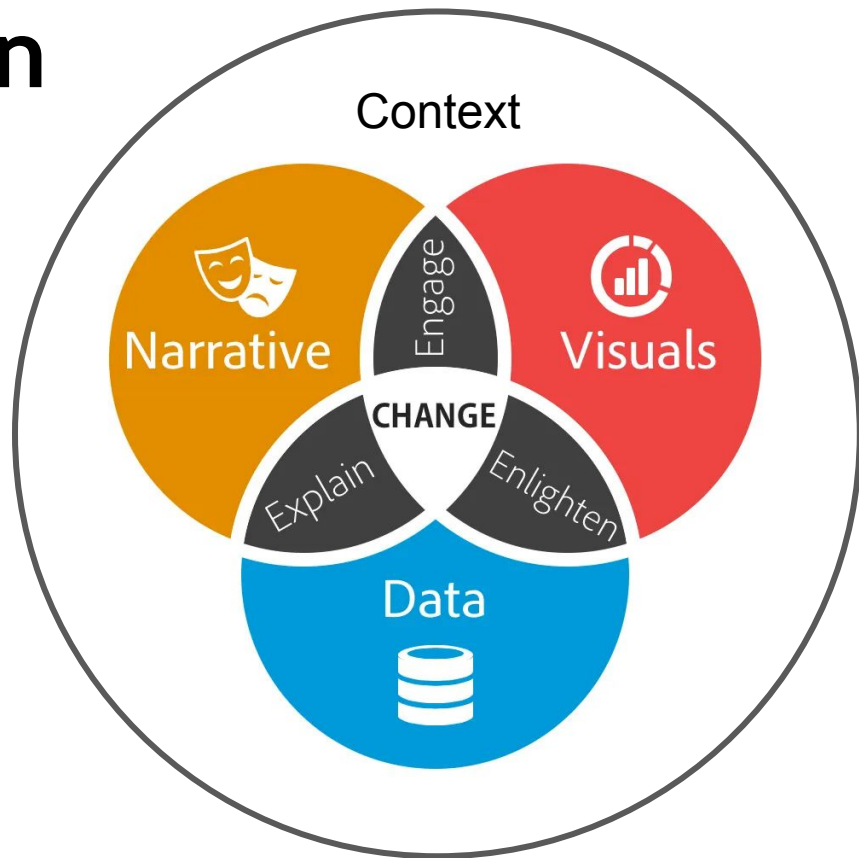
Be Discerning!



Part 2. Visual Data

Data Story

Data Visualization



Bad graphs are everywhere.

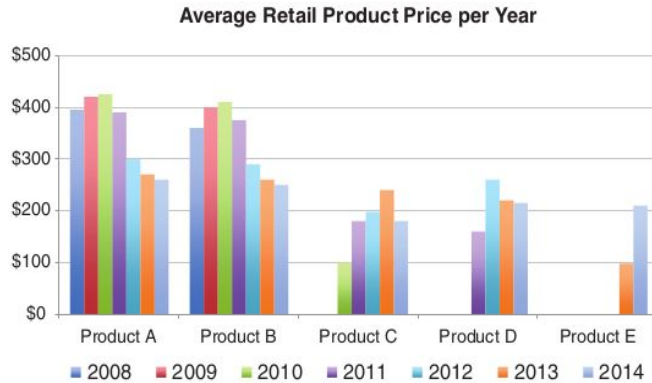


FIGURE 0.6 Example 3 (before): showing data

To be competitive, we recommend introducing our product *below* the \$223 average price point in the \$150–\$200 range

Retail price over time by product

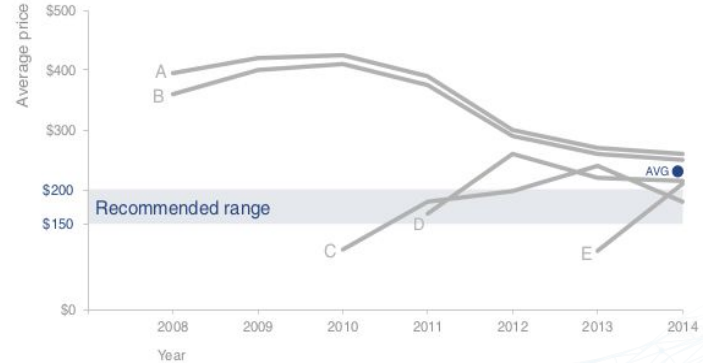
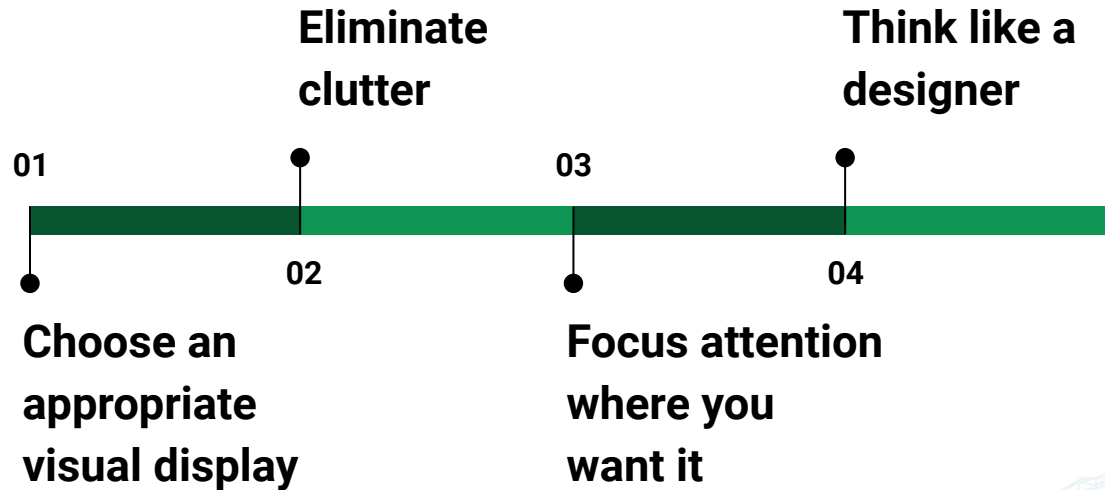


FIGURE 0.7 Example 3 (after): storytelling with data

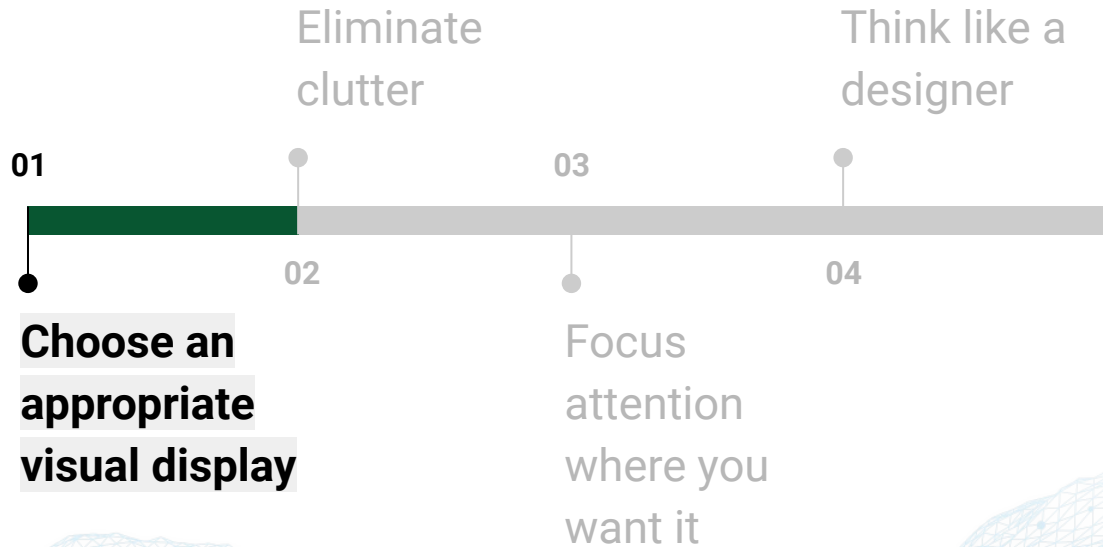
✖ Suffering with data

✔ Storytelling with data

How you'll learn to tell stories with data



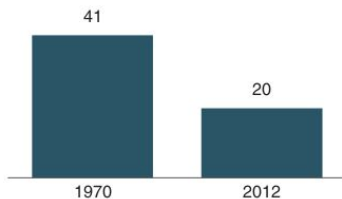
How you'll learn to tell stories with data



Simple text

Children with a "Traditional" Stay-at- Home Mother

*% of children with a married
stay-at-home mother with a
working husband*



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

 **Suffering with data**

 **Storytelling with data**

Tables

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

FIGURE 2.4 Table borders

 **Suffering with data**

 **Storytelling with data**

Heatmap

Table

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap

LOW-HIGH

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
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FIGURE 2.5 Two views of the same data

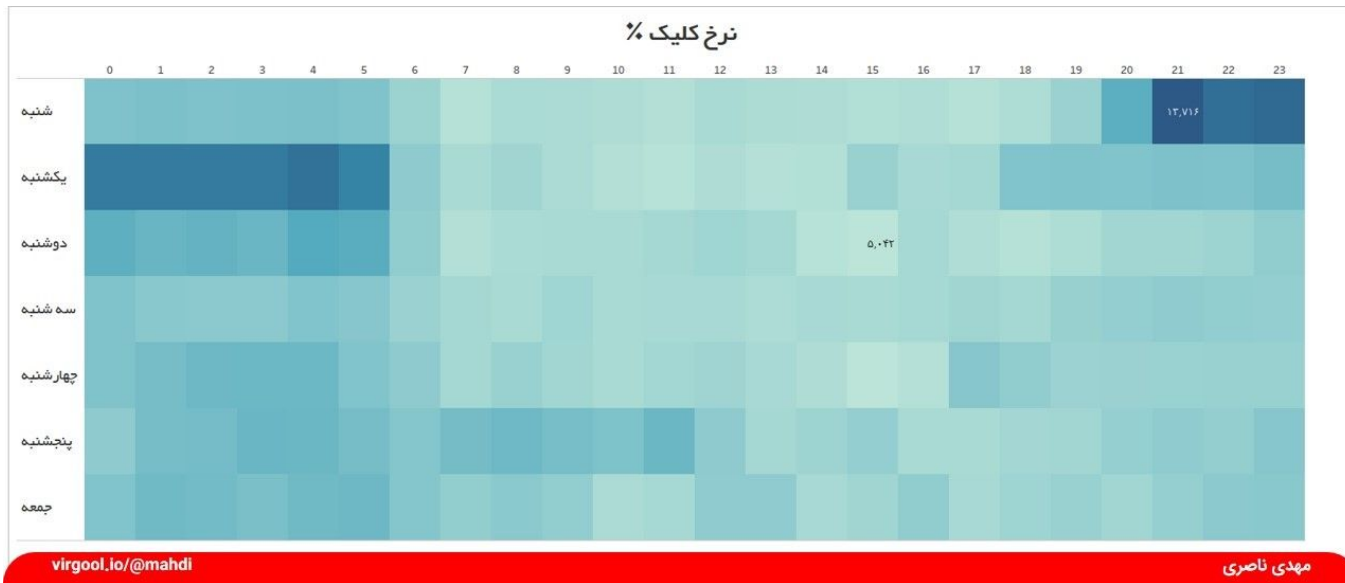


Suffering with data



Storytelling with data

Heatmap



Source: <https://vrgl.ir/fmkjn>



Storytelling with data

Points: Scatterplot

Cost per mile by miles driven

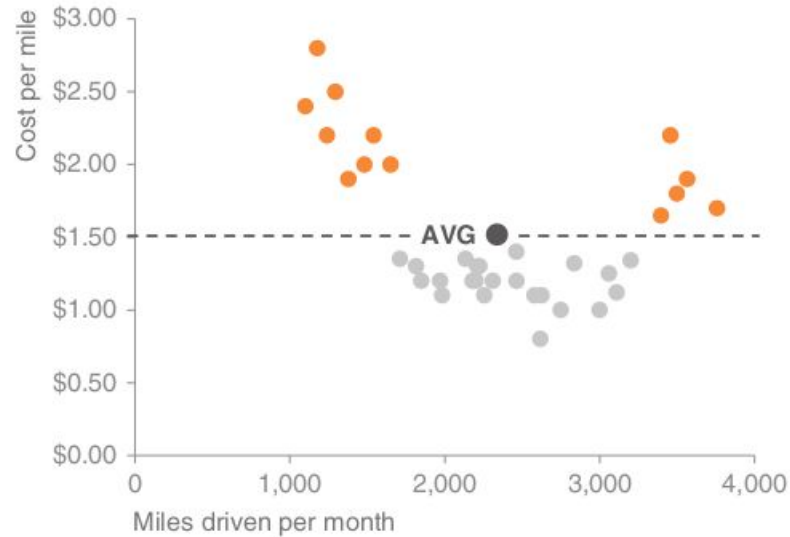
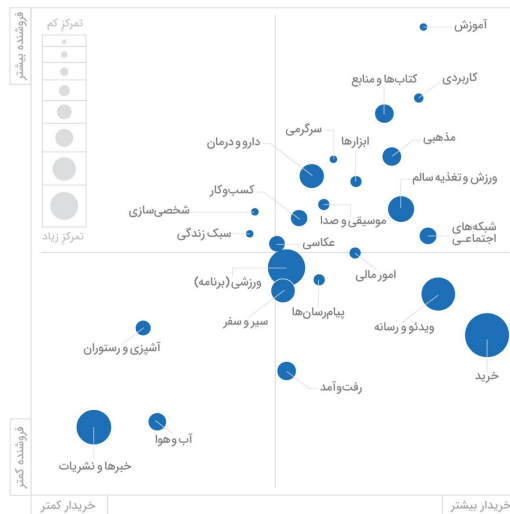


FIGURE 2.7 Modified scatterplot

Points: Bubble Chart



نقشه‌ی رقابت برنامه‌ها
در صنعت محصولات دیجیتال



منبع: گزارش سال ۹۸ بازار

میزان عرضه، میزان تقاضا و تمرکز درآمد در دسته‌های مختلف برنامه‌های بازار

Line graph

Single series



Two series



Multiple series

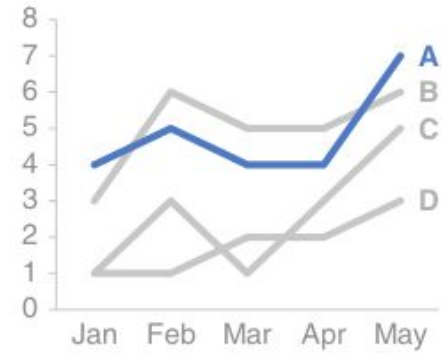
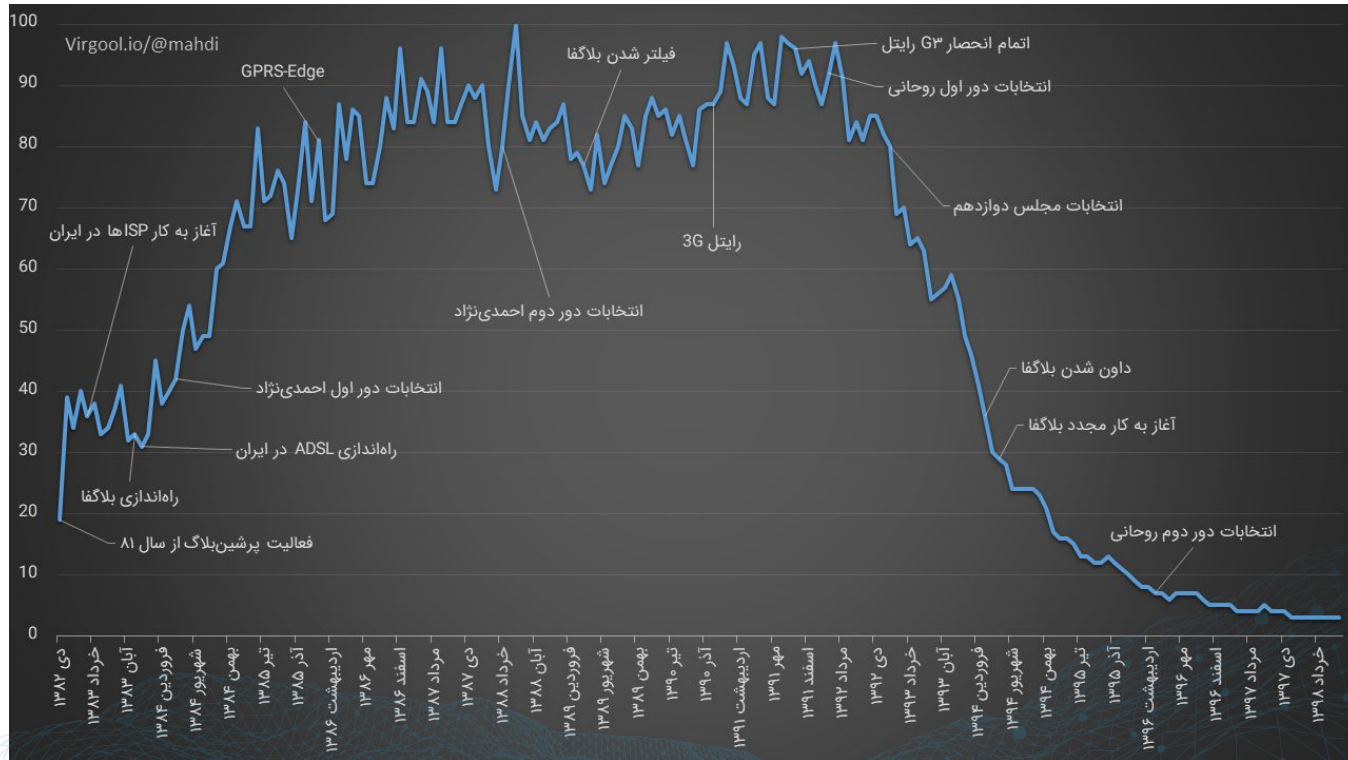


FIGURE 2.8 Line graphs

Line graph: Best for TRENDS



Source: <https://vrql.ir/HlnGx>

Slopegraph



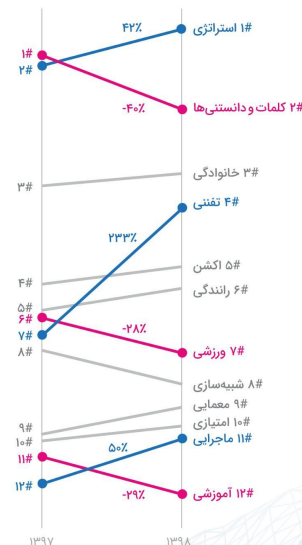
بیشترین و کمترین رشد
در تعداد تراکنش دسته‌های مختلف بازی

Employee feedback over time



FIGURE 2.11 Modified slopegraph

رنجه و رشد تراکنش بازی‌ها



منبع: گزارش سال ۹۸ بازار

بازی‌های دسته‌ی تفننی، ماجراجویی و استراتژی بیشترین رشد تراکنش موفق را در مقایسه با سال ۹۷ داشتند.

Bars

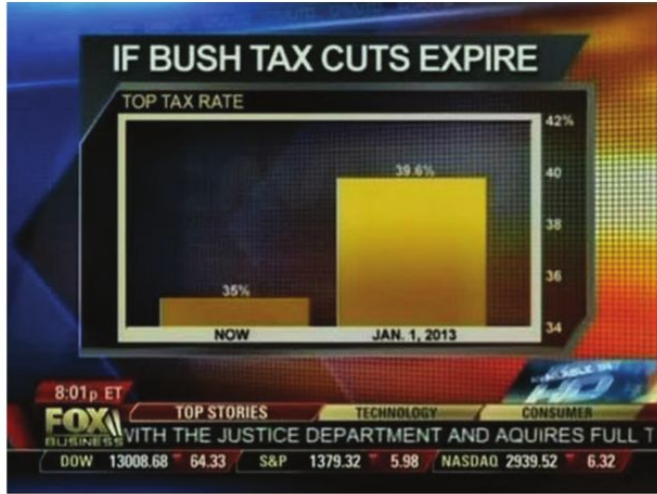


FIGURE 2.12 Fox News bar chart

Zero baseline: as it should be graphed

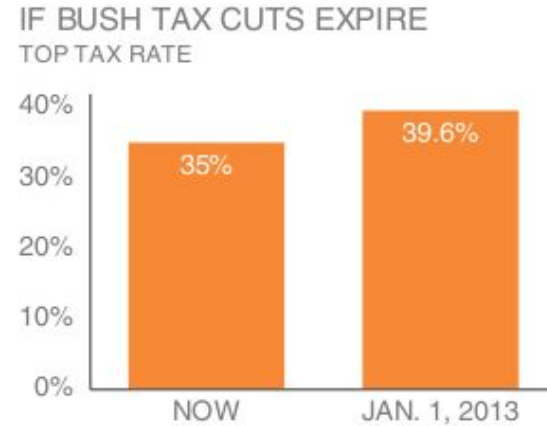


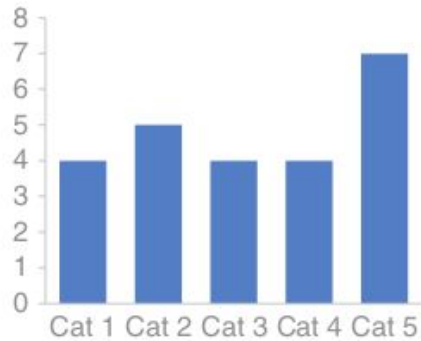
FIGURE 2.13 Bar charts must have a zero baseline

✗ **Suffering with data**

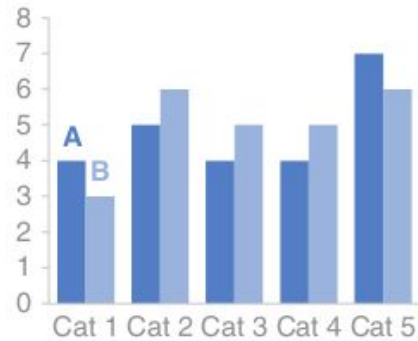
✓ **Storytelling with data**

Vertical bar chart

Single series



Two series



Multiple series

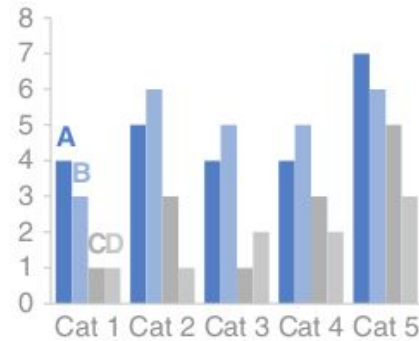


FIGURE 2.15 Bar charts

Stacked vertical bar chart

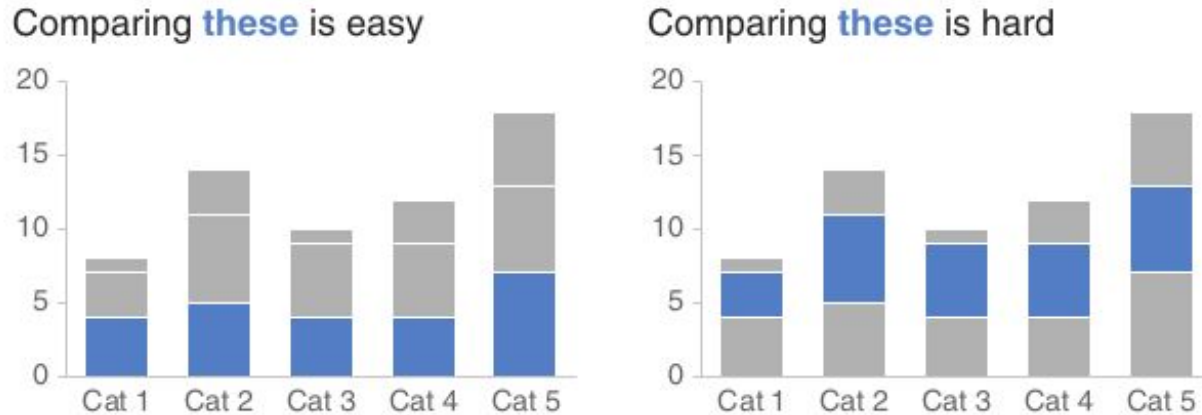


FIGURE 2.16 Comparing series with stacked bar charts

Waterfall chart

2014 Headcount math

Though more employees transferred out of the team than transferred in, aggressive hiring means overall headcount (HC) increased 16% over the course of the year.

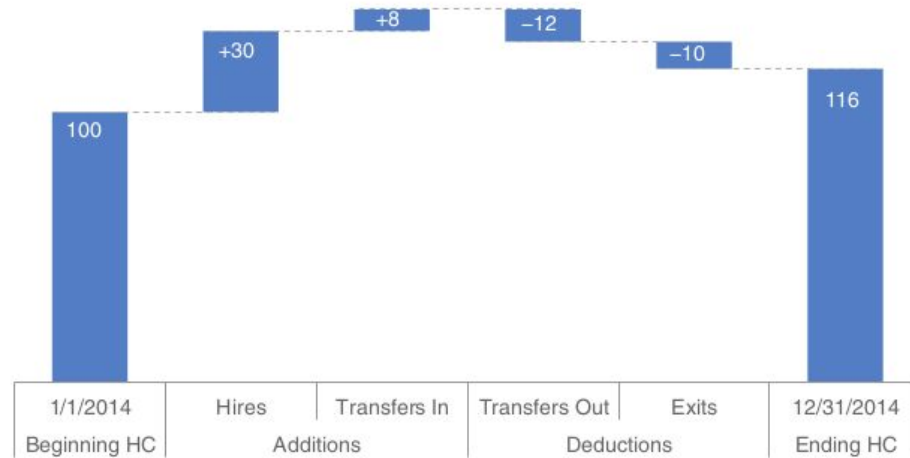
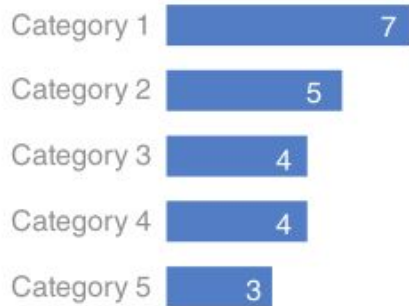


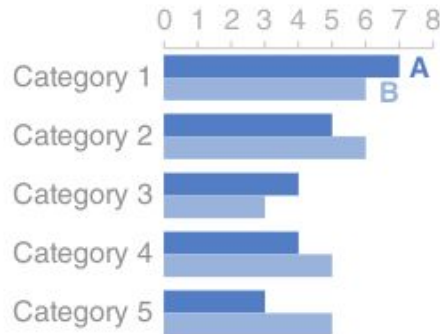
FIGURE 2.17 Waterfall chart

Horizontal bar chart

Single series



Two series



Multiple series

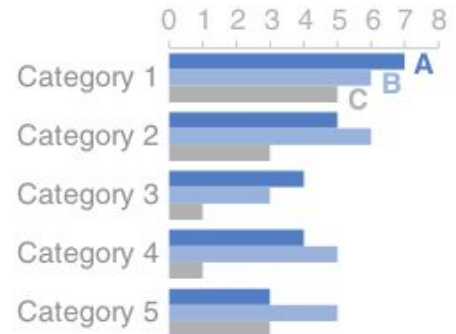


FIGURE 2.18 Horizontal bar charts

Stacked horizontal bar chart

Survey results

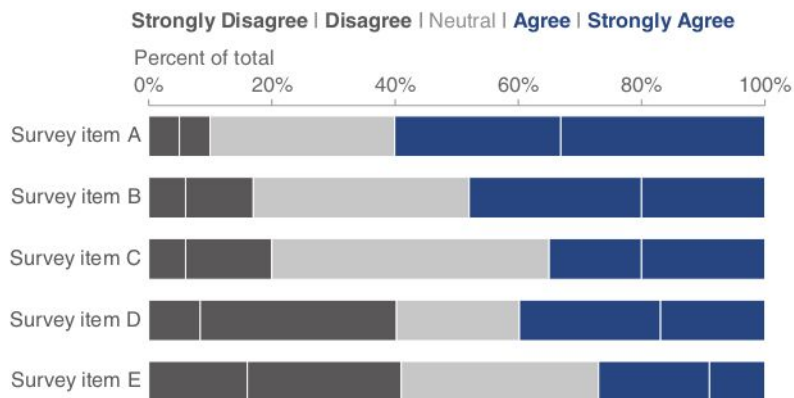


FIGURE 2.19 100% stacked horizontal bar chart



سهم بازی‌ها:
یک پنجم تراکنش، یک سوم درآمد



منبع: گزارش سال ۹۸ بازار

در مقایسه با سایر برنامه‌های فروشنده، بازی‌های فروشنده میانگین نصب بیشتر و درآمد بالاتر به ازای هر تراکنش دارند.

Racing horizontal bar chart



Area



۱/۲ میلیارد بار دانلود و به روز رسانی برنامه‌ها

Interview breakdown

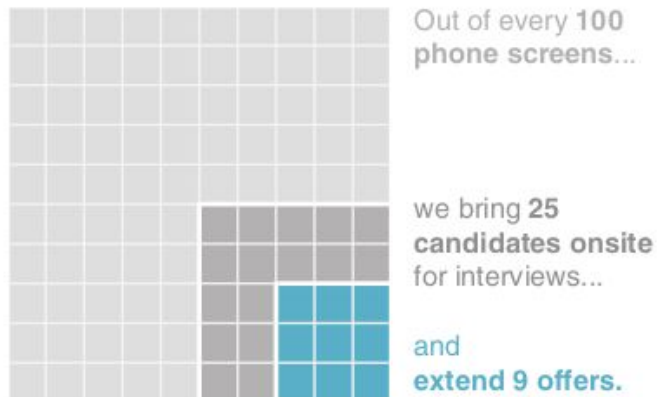
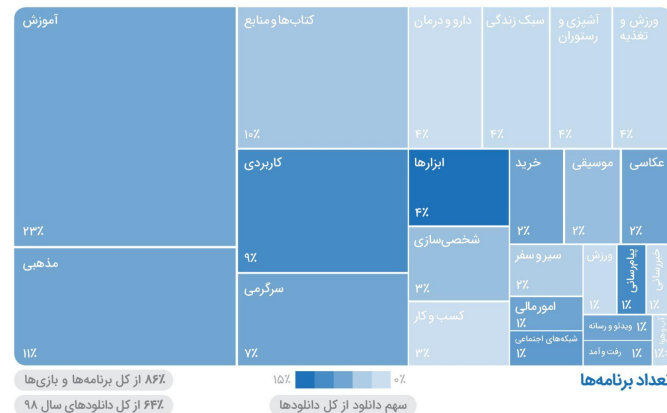


FIGURE 2.20 Square area graph



منبع: گزارش سال ۹۸ بازار

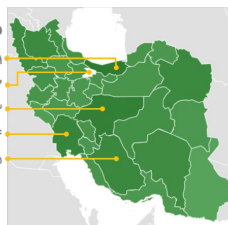
دسته‌ی آموزش بیشترین تعداد برنامه و دسته‌ی ابزارها بیشترین تعداد دانلود و به روز رسانی را دارد.

Maps

محبوبیت ولتاین

- ۱- مازندران
- ۲- البرز
- ۳- اصفهان
- ۴- خوزستان
- ۵- فارس

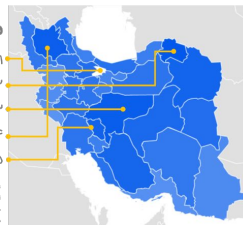
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت یلدا

- ۱- البرز
- ۲- خراسان شمالی
- ۳- اصفهان
- ۴- آذربایجان شرقی
- ۵- چهارمحال و بختیاری

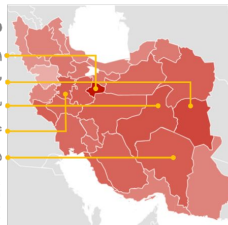
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت اربعین

- ۱- قم
- ۲- خراسان جنوبی
- ۳- یزد
- ۴- همدان
- ۵- کرمان

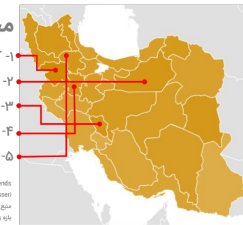
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت نوروز

- ۱- کردستان
- ۲- سمنان
- ۳- چهارمحال و بختیاری
- ۴- همدان
- ۵- زنجان

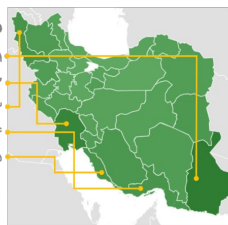
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت عید فطر

- ۱- سیستان و بلوچستان
- ۲- خوزستان
- ۳- آذربایجان غربی
- ۴- هرمزگان
- ۵- بوشهر

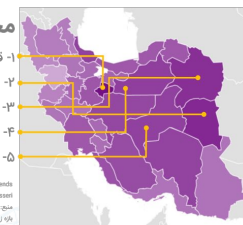
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت عید غدیر

- ۱- قم
- ۲- خراسان جنوبی
- ۳- خراسان رضوی
- ۴- سمنان
- ۵- یزد

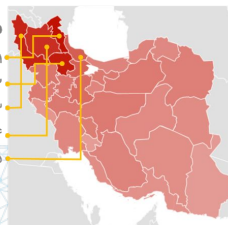
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت چهارشنبه سوری

- ۱- اردبیل
- ۲- زنجان
- ۳- آذربایجان غربی
- ۴- آذربایجان شرقی
- ۵- گیلان

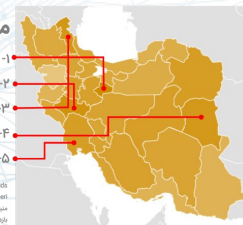
Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



محبوبیت عاشورا

- ۱- قم
- ۲- لرستان
- ۳- اردبیل
- ۴- خراسان جنوبی
- ۵- خوزستان

Virgoal.ir/trends
By: Mahdi Nazari
منبع: گوگل ترنڈز
داده زمانی از ۱۳۹۳ تا ۱۳۹۴



Other types of graphs
To be avoided



Pie charts are evil

Supplier Market Share

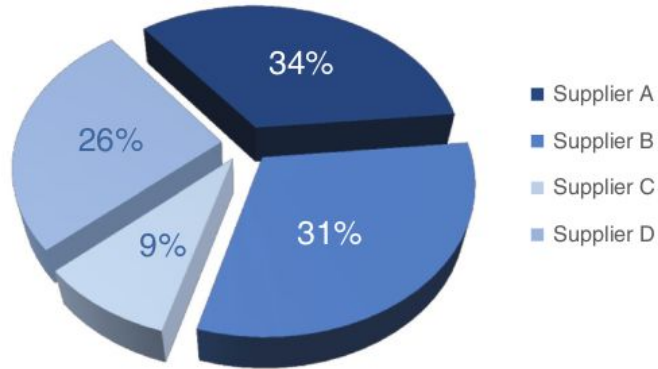


FIGURE 2.22 Pie chart with labeled segments

Supplier Market Share

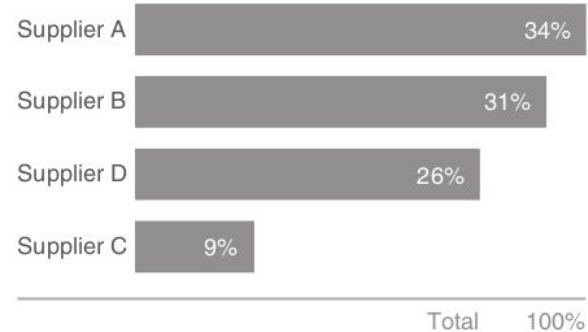


FIGURE 2.23 An alternative to the pie chart

 **Suffering with data**

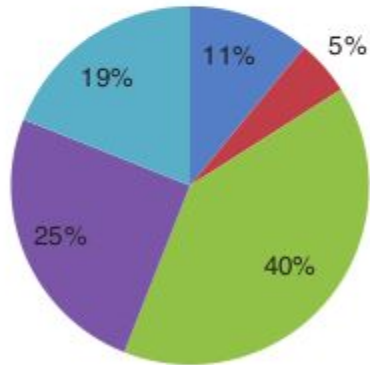
 **Storytelling with data**

Alternatives to pies

Survey results: summer learning program on science

**PRE: How do you feel
about doing science?**

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited



**POST: How do you feel
about doing science?**

■ Bored ■ Not great ■ OK ■ Kind of interested ■ Excited

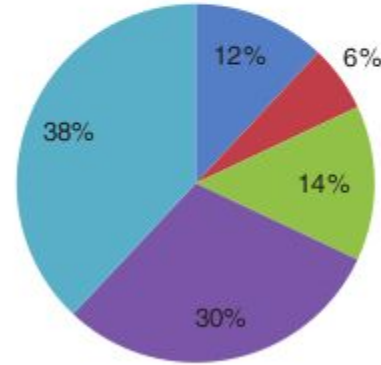


FIGURE 9.28 Original visual

Alternative #1: show the numbers directly

Pilot program was a success

After the pilot program,

68%

of kids expressed interest towards science,
compared to 44% going into the program.

Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

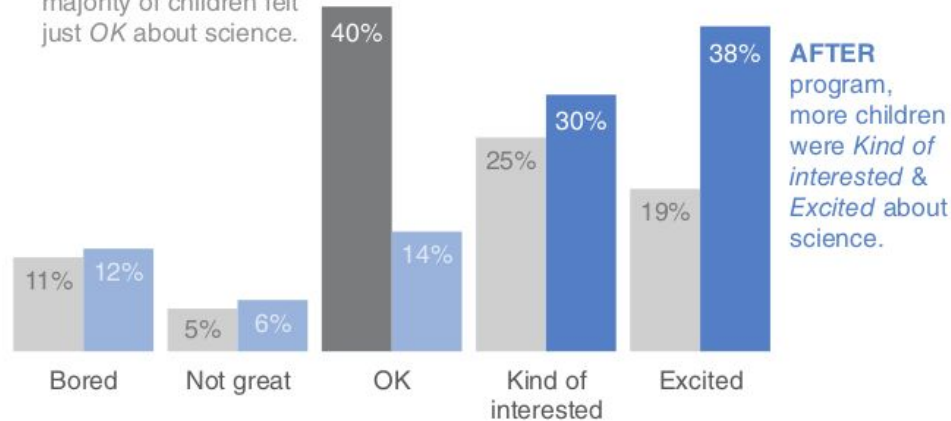
FIGURE 9.29 Show the numbers directly

Alternative #2: simple bar graph

Pilot program was a success

How do you feel about science?

BEFORE program, the majority of children felt just *OK* about science.



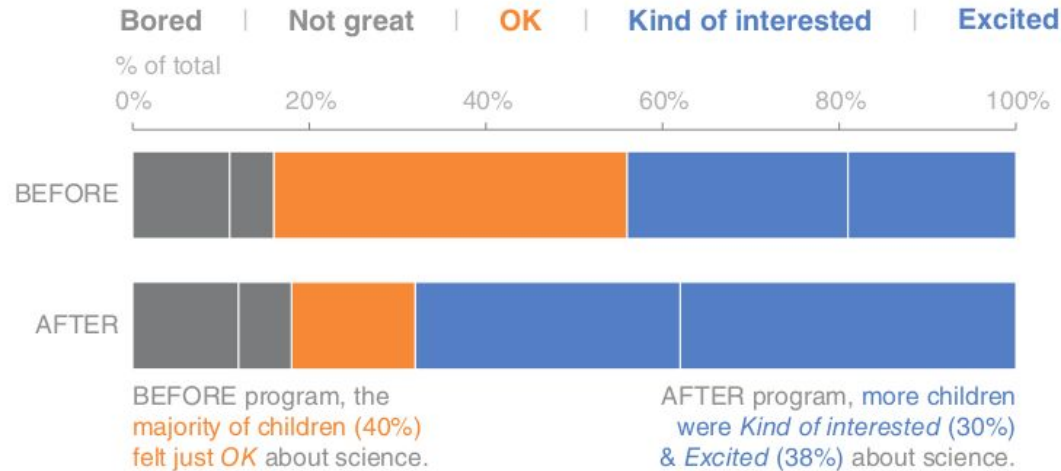
Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.30 Simple bar graph

Alternative #3: 100% stacked horizontal bar graph

Pilot program was a success

How do you feel about science?



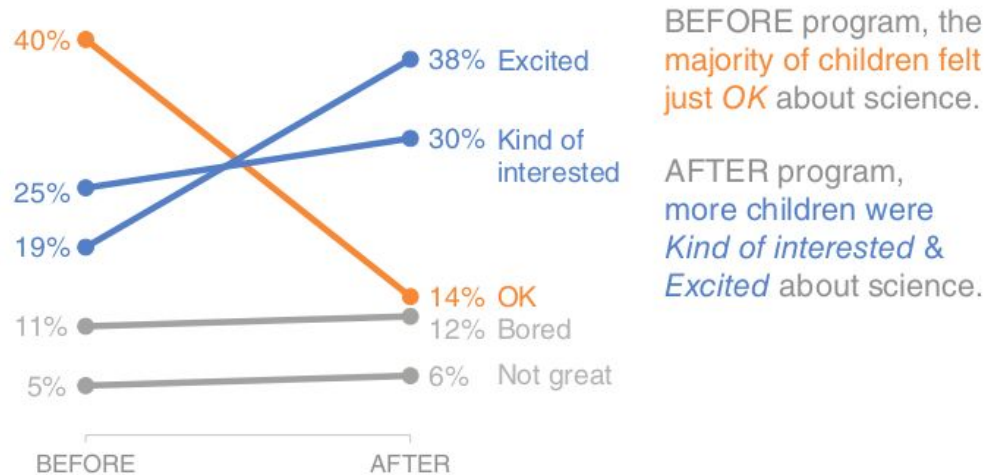
Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.31 100% stacked horizontal bar graph

Alternative #4: slopegraph

Pilot program was a success

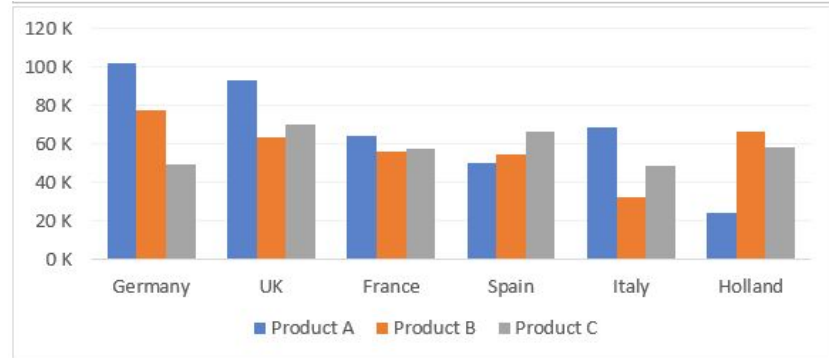
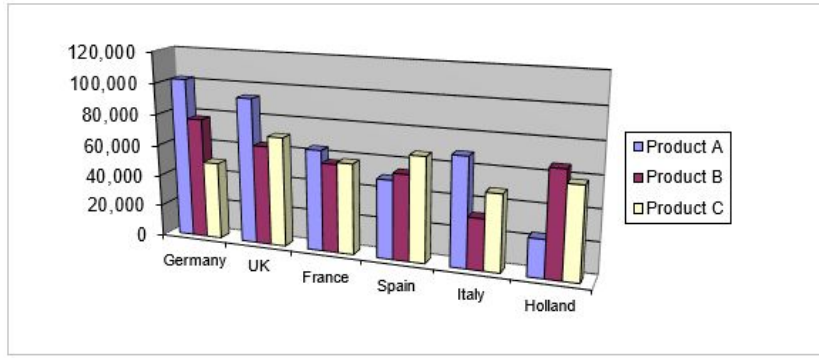
How do you feel about science?



Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.32 Slopegraph

Never use 3D



✗ **Suffering with data**

✓ **Storytelling with data**

Secondary y-axis: generally not a good idea

Alternative 1: label directly



Alternative 2: pull apart vertically

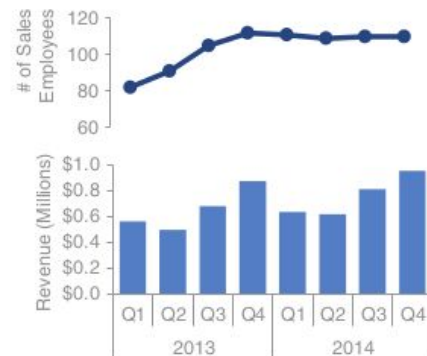
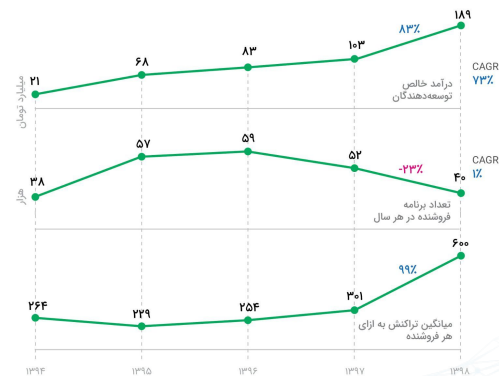


FIGURE 2.27 Strategies for avoiding a secondary y-axis



۱۸۹ میلیارد تومان
درآمد خالص توسعه‌دهنده‌ها در سال ۱۳۹۸

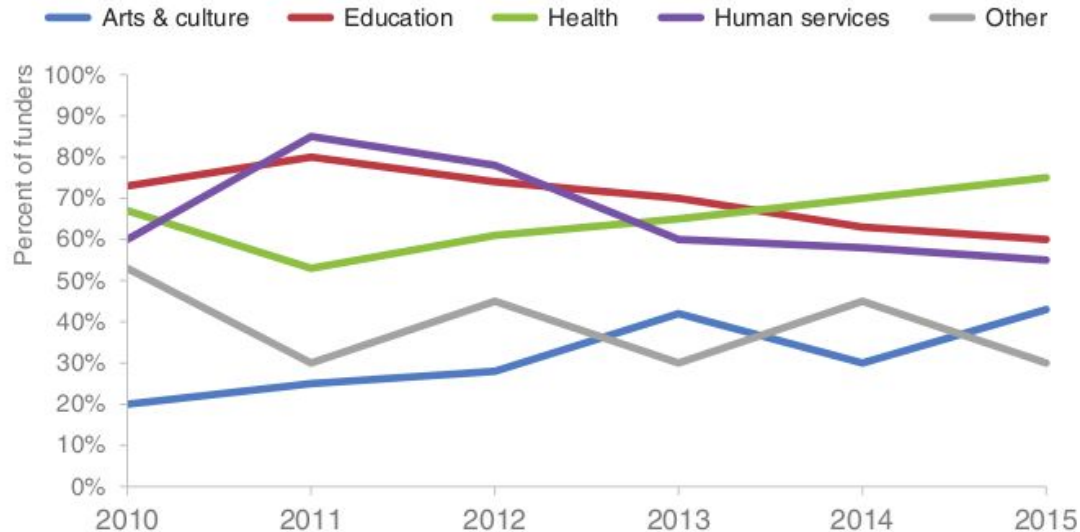


منبع: گزارش سال ۹۸ بازار

با وجود کاهش برنامه‌های فروشنده، درآمد توسعه‌دهنده‌ها و میانگین تعداد تراکنش به ازای هر فروشنده افزایش یافته است.

Strategies for avoiding the spaghetti graph

Types of non-profits supported by area funders

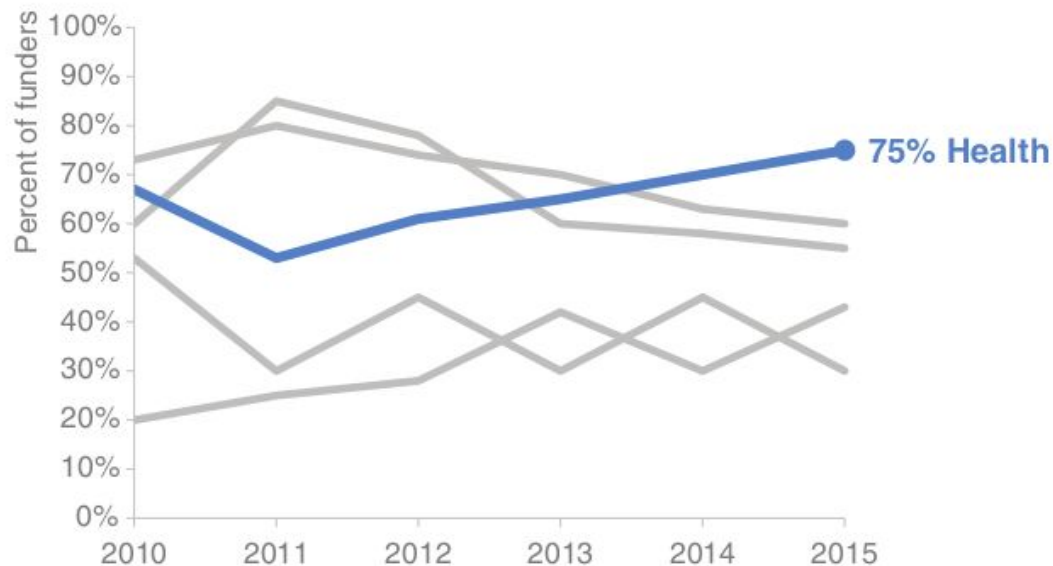


Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

FIGURE 9.21 The spaghetti graph

Emphasize one line at a time

Types of non-profits supported by area funders

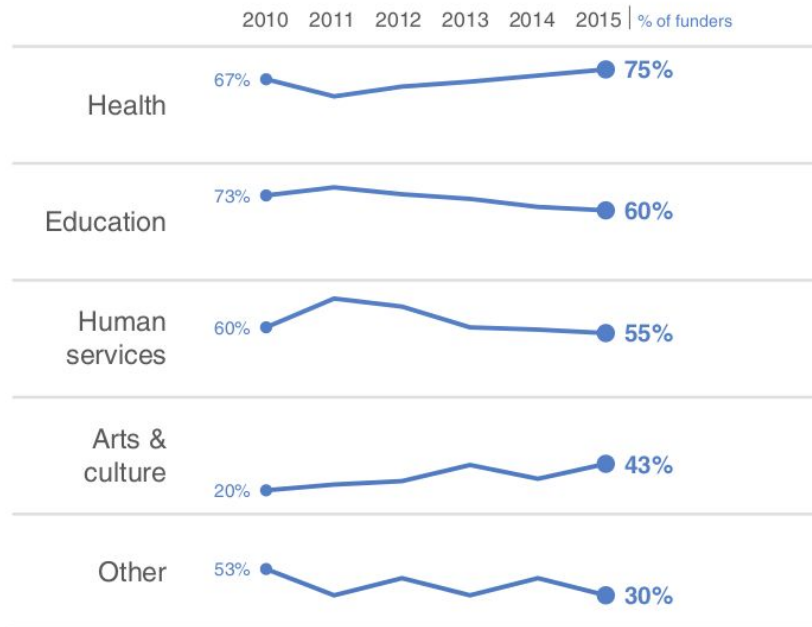


Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

FIGURE 9.22 Emphasize a single line

Separate spatially - vertical

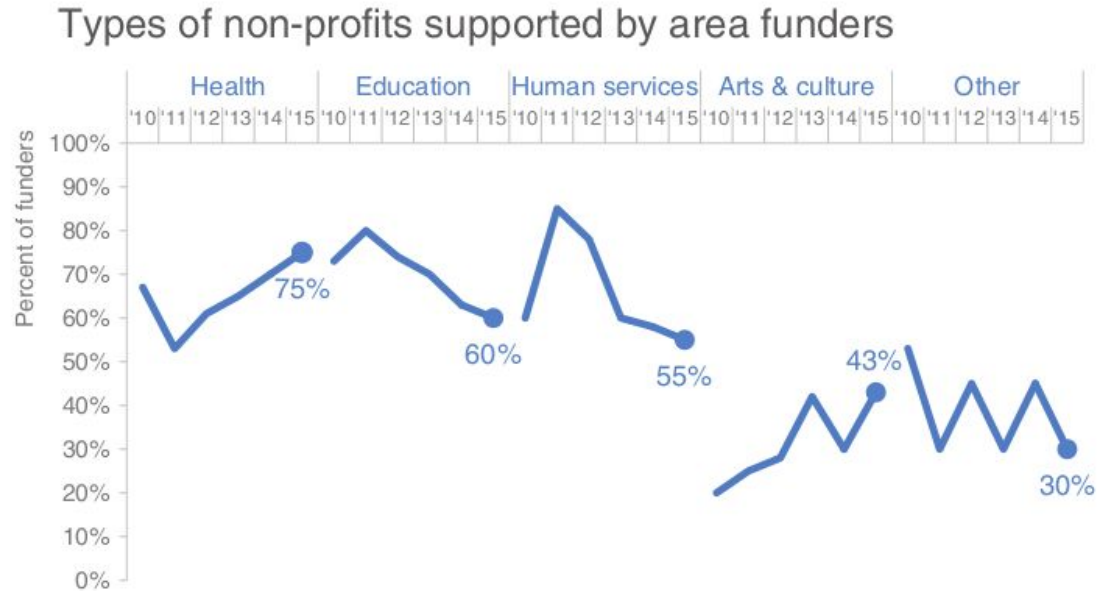
Types of non-profits supported by area funders



Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

FIGURE 9.24 Pull the lines apart vertically

Separate spatially - horizontal

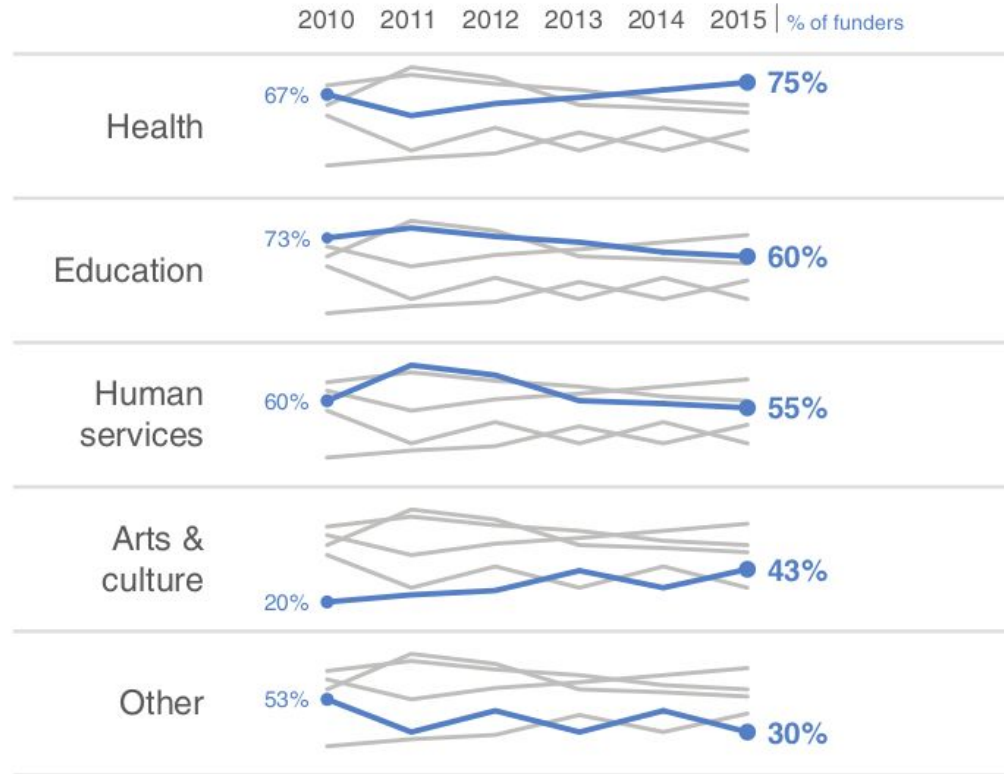


Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

FIGURE 9.25 Pull the lines apart horizontally

Combined approach

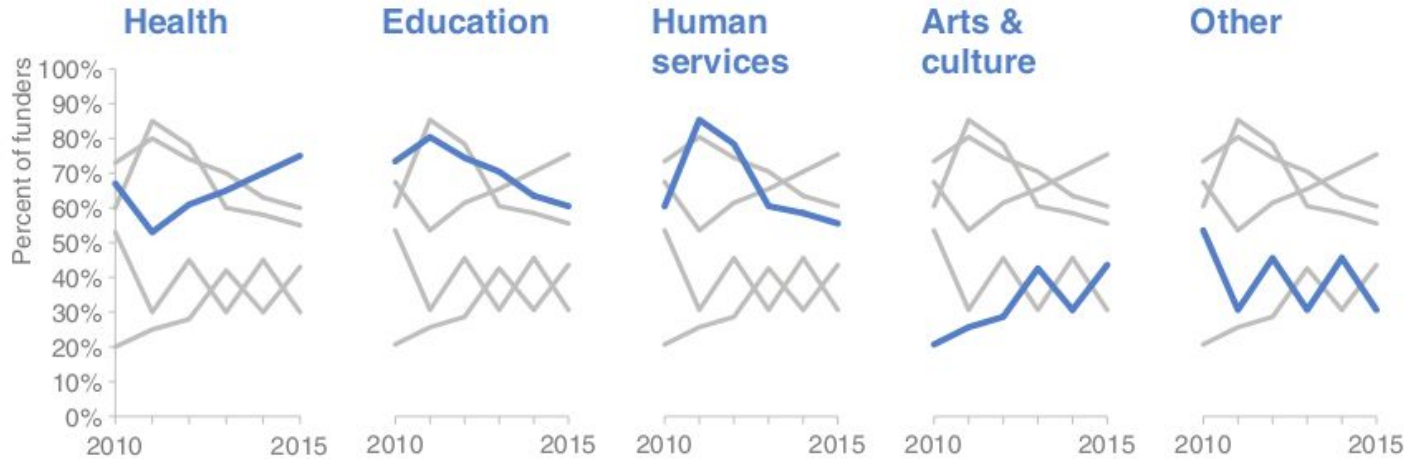
Types of non-profits supported by area funders



Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

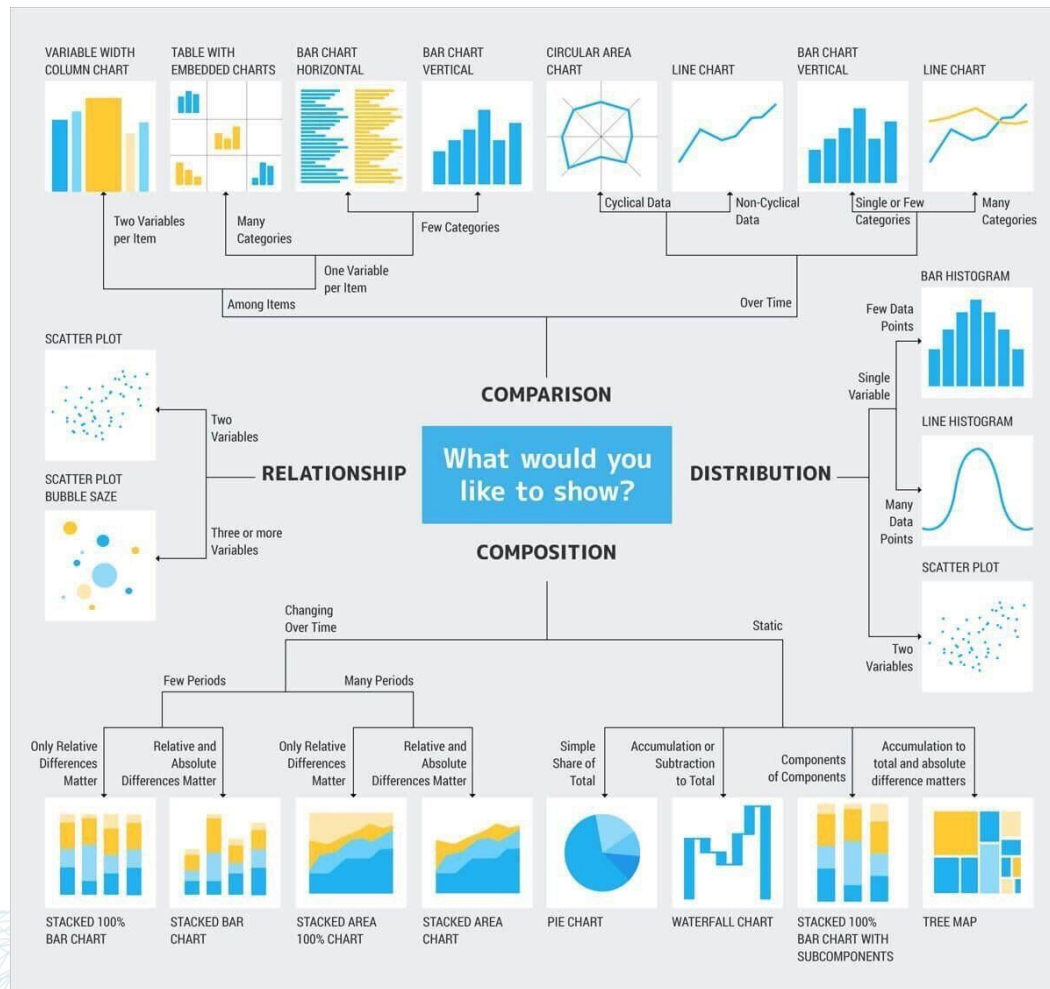
FIGURE 9.26 Combined approach, with vertical separation

Types of non-profits supported by area funders



Data is self-reported by funders; percents sum to greater than 100 because respondents can make multiple selections.

FIGURE 9.27 Combined approach, with horizontal separation



Let's practice!



Exercise: improve this table

New client tier share

Tier	# of Accounts	% Accounts	Revenue (\$M)	% Revenue
A	77	7.08%	\$4.68	25%
A+	19	1.75%	\$3.93	21%
B	338	31.07%	\$5.98	32%
C	425	39.06%	\$2.81	15%
D	24	2.21%	\$0.37	2%

FIGURE 2.1a Original table

Exercise

New client tier share

Tier	# of Accounts	% Accounts	Revenue (\$M)	% Revenue
A+	19	2%	\$3.9	21%
A	77	7%	\$4.7	25%
B	338	31%	\$6.0	32%
C	425	39%	\$2.8	15%
D	24	2%	\$0.4	2%
All other	205	19%	\$0.9	5%
TOTAL	1,088	100%	\$18.7	100%

FIGURE 2.1b Slightly improved table

Exercise

New client tier share

TIER	ACCOUNTS		REVENUE	
	#	% OF TOT	\$M	% OF TOT
A+	19	2%	\$3.9	21%
A	77	7%	\$4.7	25%
B	338	31%	\$6.0	32%
C	425	39%	\$2.8	15%
D	24	2%	\$0.4	2%
All other	205	19%	\$0.9	5%
TOTAL	1,088	100%	\$18.7	100%

FIGURE 2.1c Table with heatmapping

New client tier share

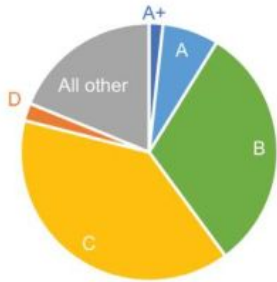
TIER	ACCOUNTS		REVENUE	
	#	% OF TOT	\$M	% OF TOT
A+	19		\$3.9	
A	77		\$4.7	
B	338		\$6.0	
C	425		\$2.8	
D	24		\$0.4	
All other	205		\$0.9	
TOTAL	1,088	100%	\$18.7	100%

FIGURE 2.1d Table with embedded bars

Exercise

New client tier share

% of Total Accounts



% of Total Revenue



FIGURE 2.1e A pair of pies

New client tier share

TIER | % OF TOTAL ACCOUNTS | % OF TOTAL REVENUE

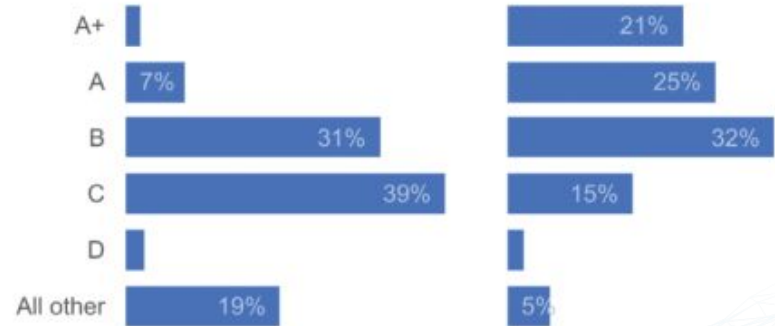


FIGURE 2.1f Two horizontal bar charts

✗ **Suffering with data**

✓ **Storytelling with data**

Exercise

New client tier share



FIGURE 2.1g Horizontal dual series bar chart

Exercise

New client tier share

% OF TOTAL ACCOUNTS vs. REVENUE

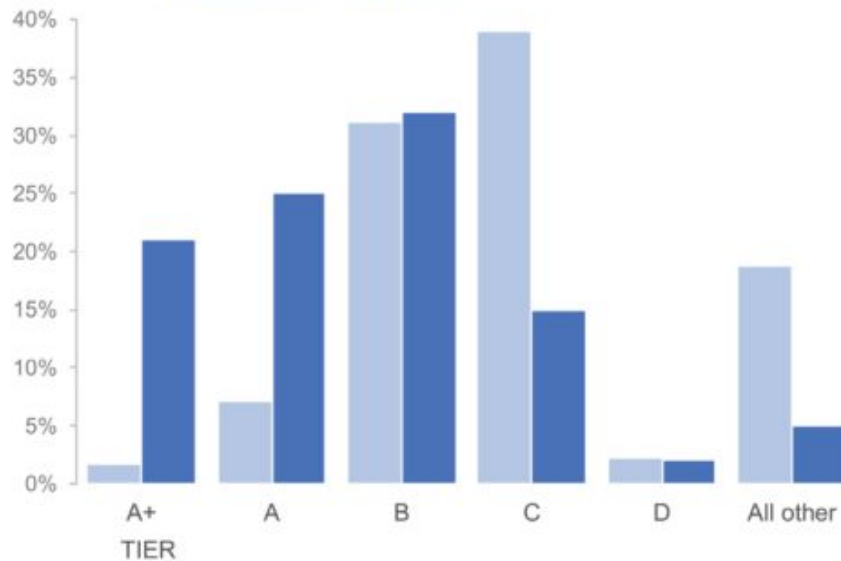


FIGURE 2.1h A vertical bar chart

Exercise

New client tier share

% OF TOTAL ACCOUNTS vs. REVENUE

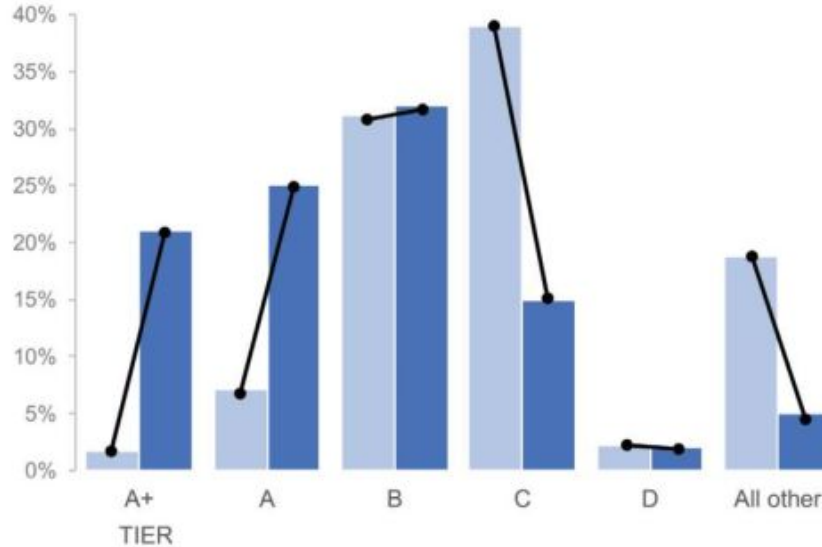


FIGURE 2.1i Let's draw some lines

Exercise

New client tier share

% OF TOTAL ACCOUNTS vs. REVENUE

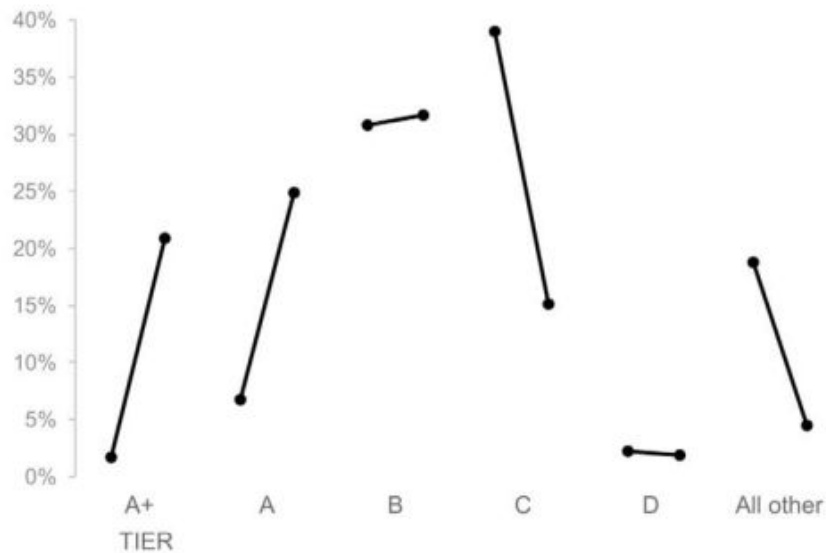


FIGURE 2.1j Take away the bars

Exercise

New client tier share

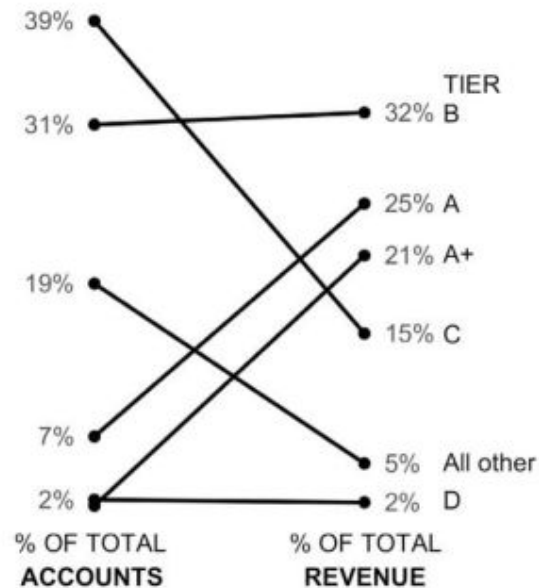
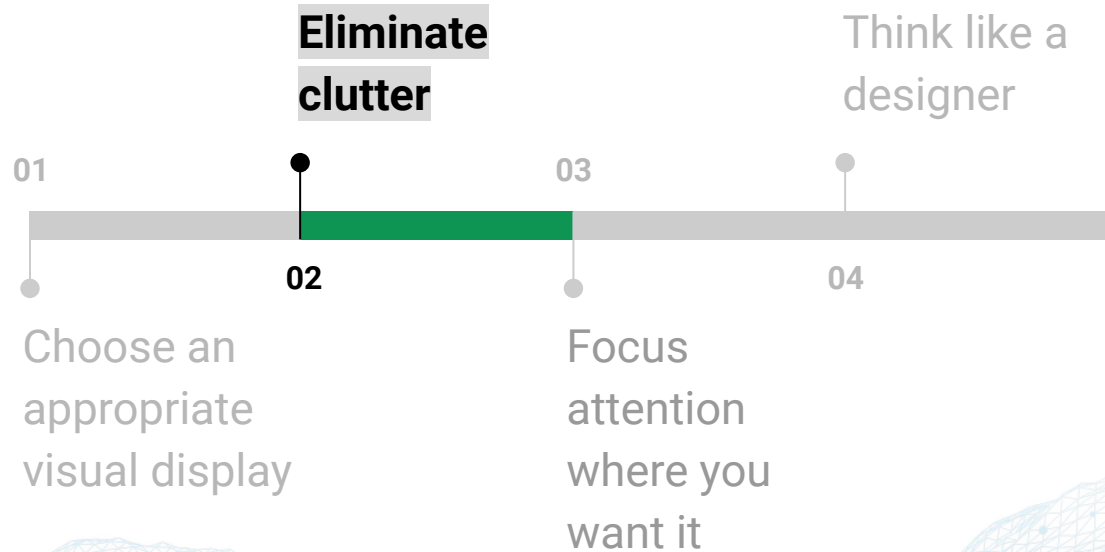


FIGURE 2.1k A slopegraph

Part 3.

How you'll learn to tell stories with data



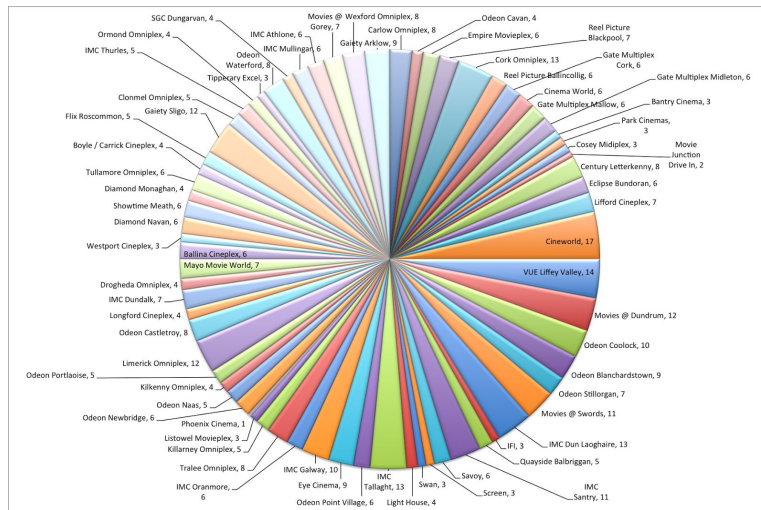
Clutter is your enemy!

Clutter

Visual elements that take up space and don't aid our understanding.

Cognitive Load

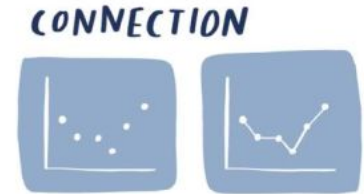
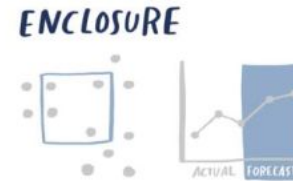
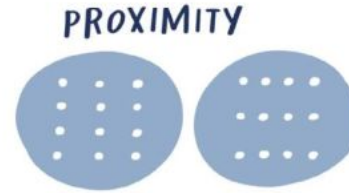
The **Mental Effort** that's required to learn new information



Remove Clutter

Gestalt Principles

Describe how we **subconsciously** order what we see in the world



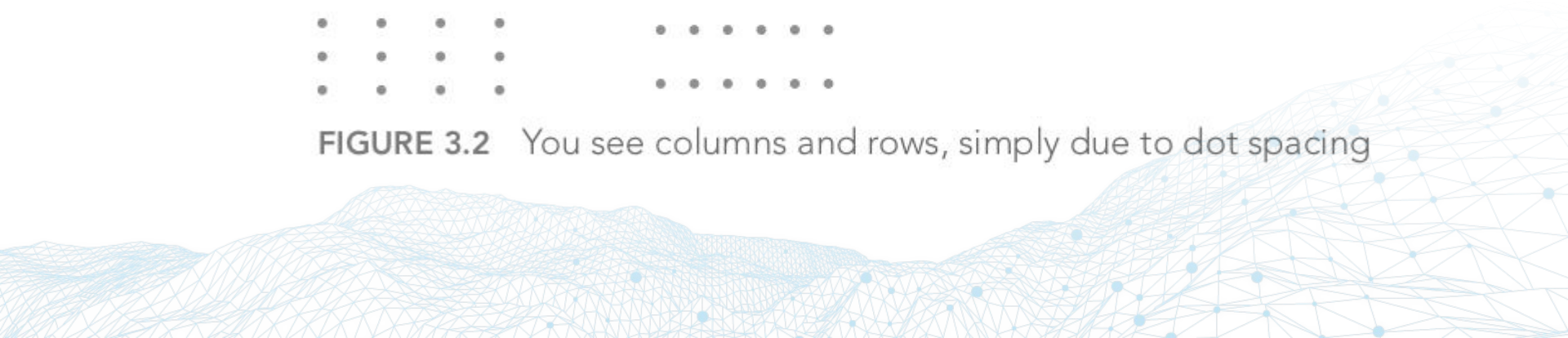
Proximity



FIGURE 3.1 Gestalt principle of proximity



FIGURE 3.2 You see columns and rows, simply due to dot spacing



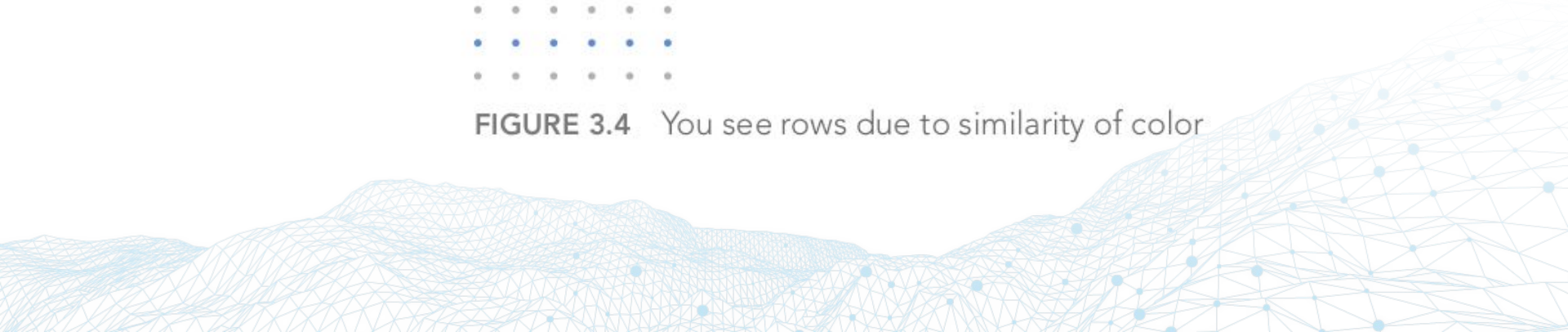
Similarity



FIGURE 3.3 Gestalt principle of similarity



FIGURE 3.4 You see rows due to similarity of color



Enclosure



FIGURE 3.5 Gestalt principle of enclosure

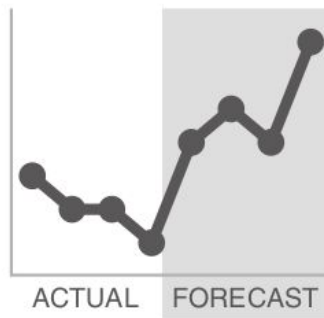
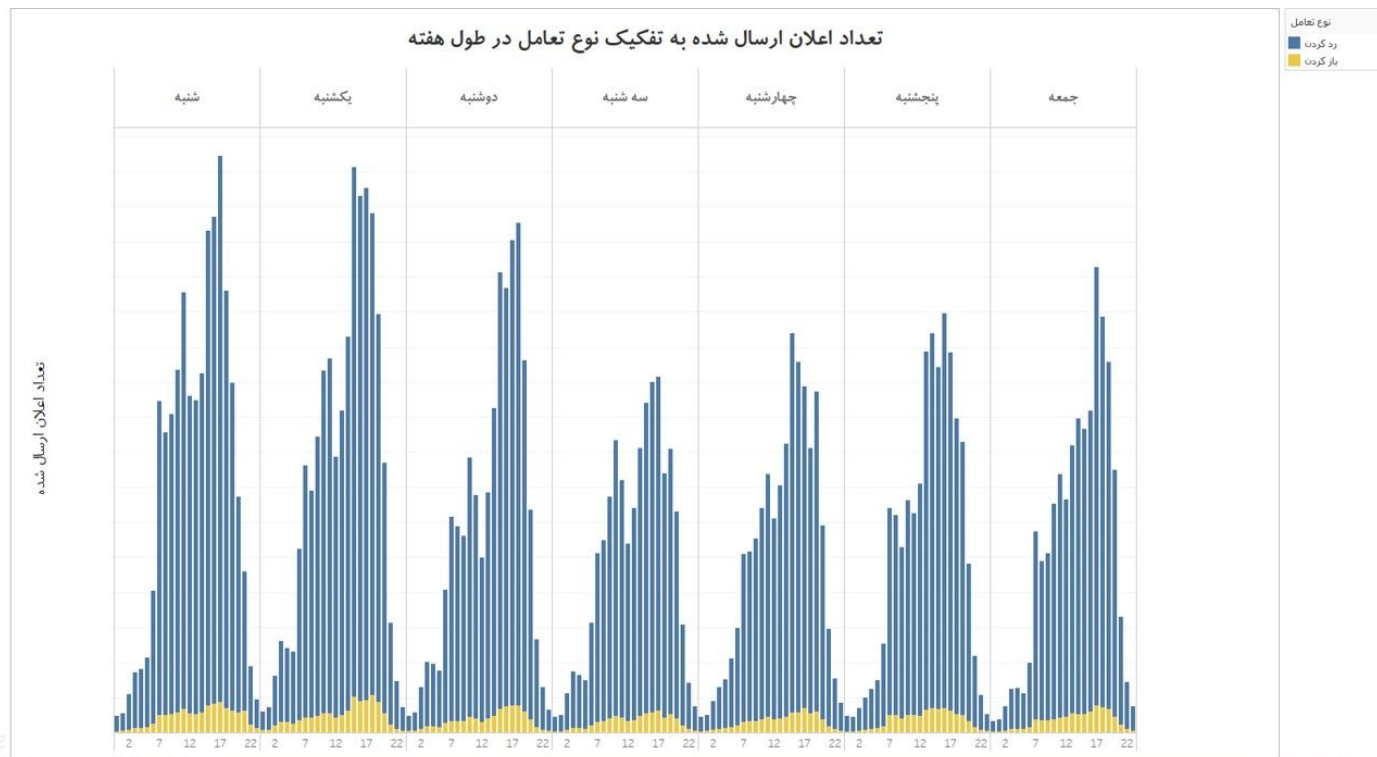


FIGURE 3.6 The shaded area separates the forecast from actual data

Enclosure



Closure



FIGURE 3.7 Gestalt principle of closure

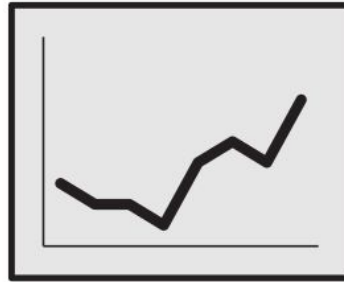


FIGURE 3.8 The graph still appears complete without the border and background shading

Continuity

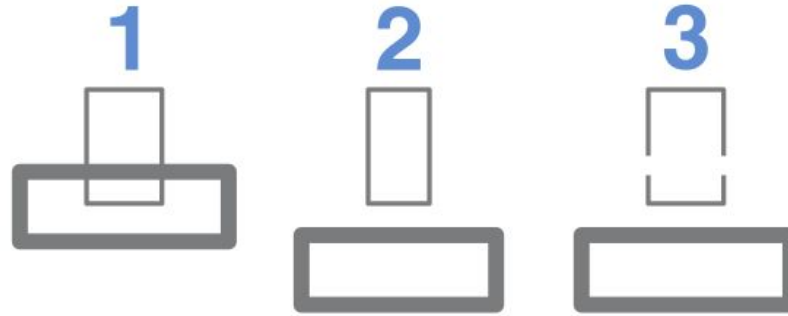


FIGURE 3.9 Gestalt principle of continuity

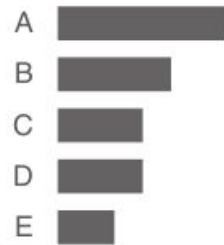


FIGURE 3.10 Graph with y-axis line removed

Connection



FIGURE 3.11 Gestalt principle of connection

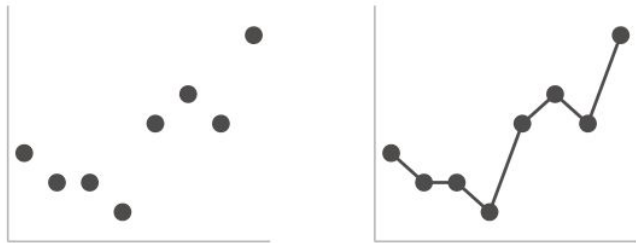


FIGURE 3.12 Lines connect the dots



Remove Clutter

Visual Order

Leverage white space and align elements

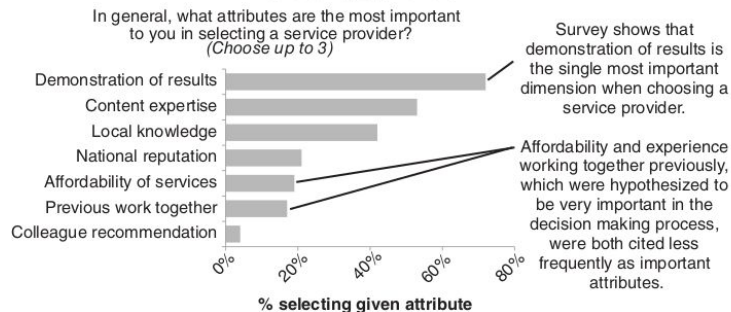
Aim for clean horizontal and vertical elements

Avoid diagonal



Remove Clutter: visual order

Demonstrating effectiveness is most important consideration when selecting a provider



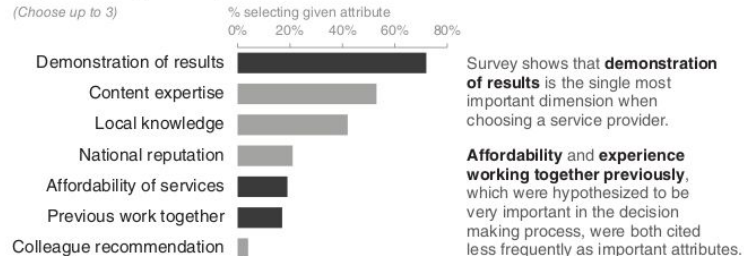
Data source: xyz; includes N number of survey respondents. Note that respondents were able to choose up to 3 options.

FIGURE 3.13 Summary of survey feedback

Demonstrating effectiveness is most important consideration when selecting a provider

In general, what attributes are the most important to you in selecting a service provider?

(Choose up to 3)



Data source: xyz; includes N number of survey respondents. Note that respondents were able to choose up to 3 options.

FIGURE 3.14 Revamped summary of survey feedback

✗ Suffering with data

✓ Storytelling with data

Remove Clutter

Strategic use of contrast

Clear contrast is a signal, indicating where to look

Don't make too many things different, or key points will get lost



Non-strategic use of contrast



FIGURE 3.15 Original graph

Performance overview



FIGURE 3.16 Revamped graph, using contrast strategically

✗ **Suffering with data**

✓ **Storytelling with data**

Let's practice!



Exercise: declutter!



FIGURE 3.4a Let's declutter!

1. Remove heavy lines.



FIGURE 3.4b Remove heavy lines

2. Remove gridlines.



FIGURE 3.4c Remove gridlines

3. Drop trailing zeros from y-axis labels.



FIGURE 3.4d Drop trailing zeros from y-axis labels

4. Eliminate diagonal text on x-axis.



FIGURE 3.4e Eliminate diagonal text on x-axis.

5. Thicken the bars!



FIGURE 3.4f Thicken the bars!

6. Pull data labels into ends of bars.



FIGURE 3.4g Pull data labels into ends of bars

7. Eliminate data labels.



FIGURE 3.4h Eliminate data labels

8. Make it a line graph.



FIGURE 3.4i Make it a line graph

9. Label the data directly.

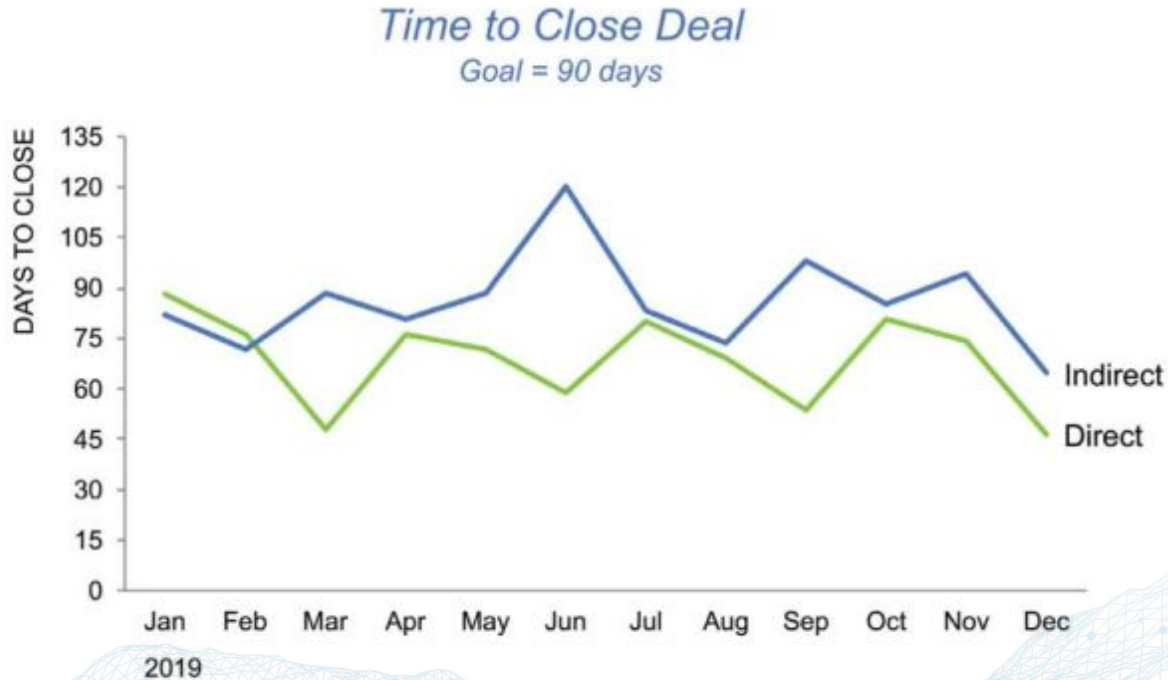


FIGURE 3.4j Label the data directly

10. Make data labels the same color as the data.

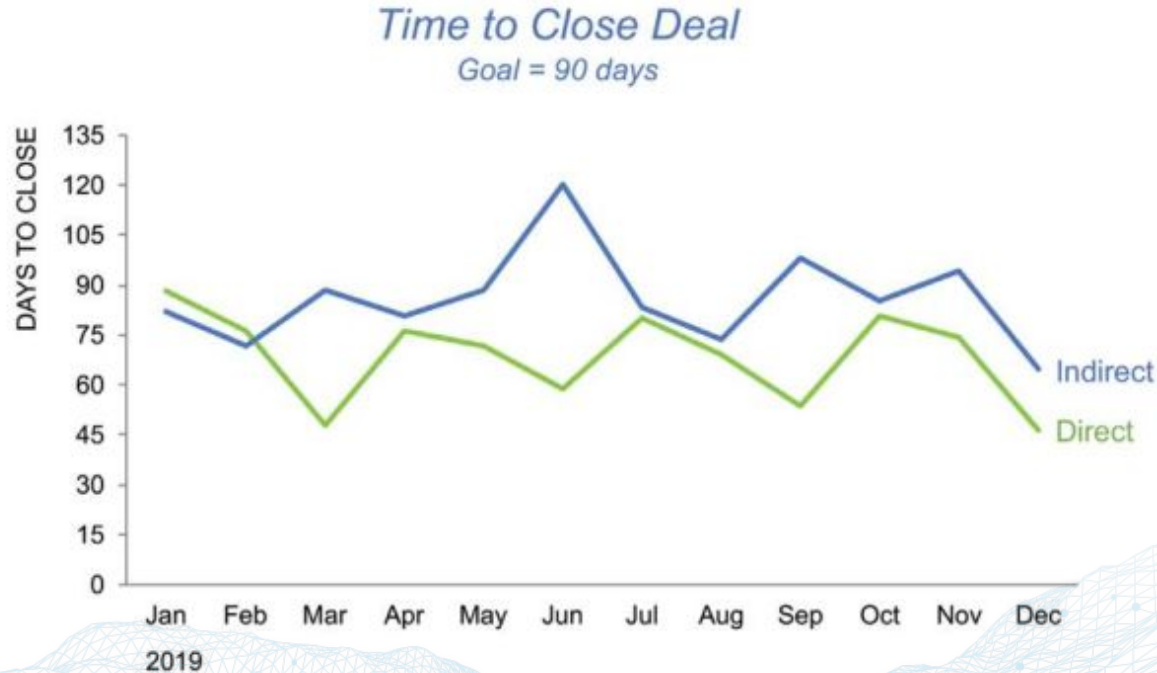


FIGURE 3.4k Make data labels the same color as the data

11. Upper-left-most orient graph title.

Time to Close Deal

Goal = 90 days

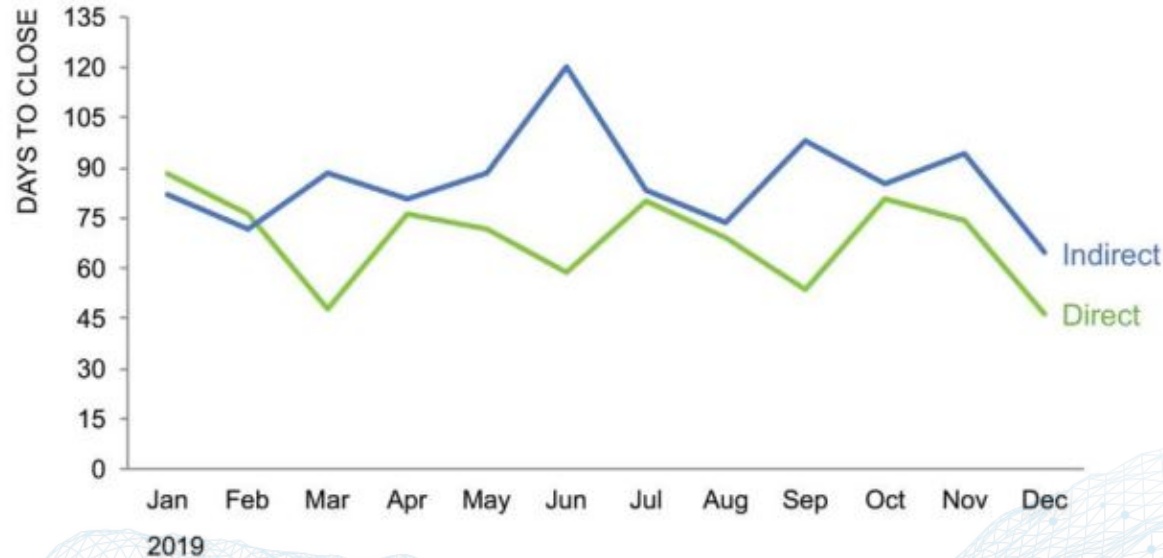


FIGURE 3.4I Upper-left-most orient graph title

12. Remove title color.

Time to close deal

Goal = 90 days

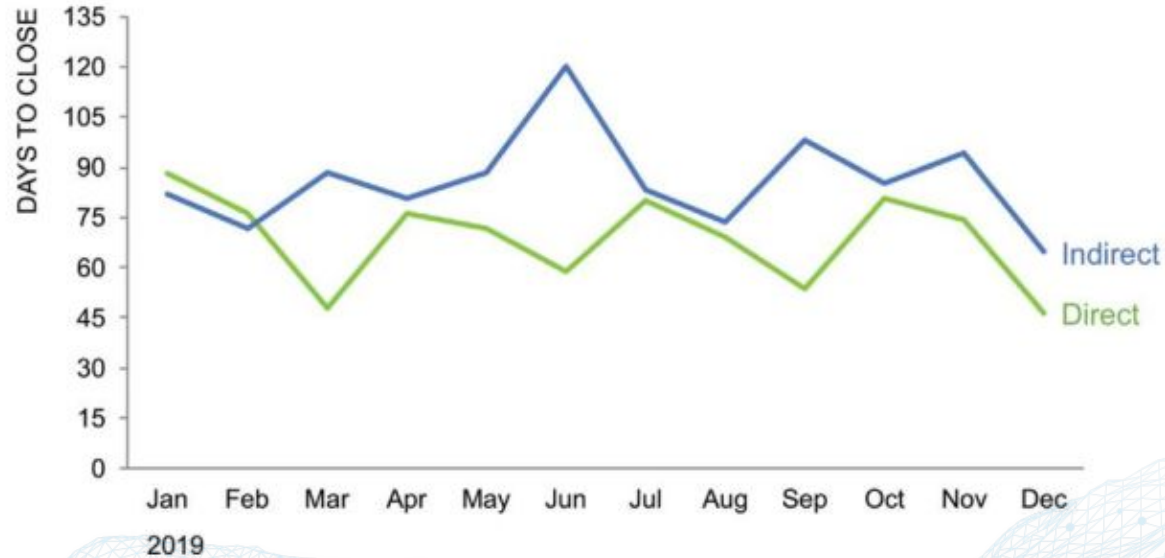


FIGURE 3.4m Remove title color

13. Put the goal in the graph.

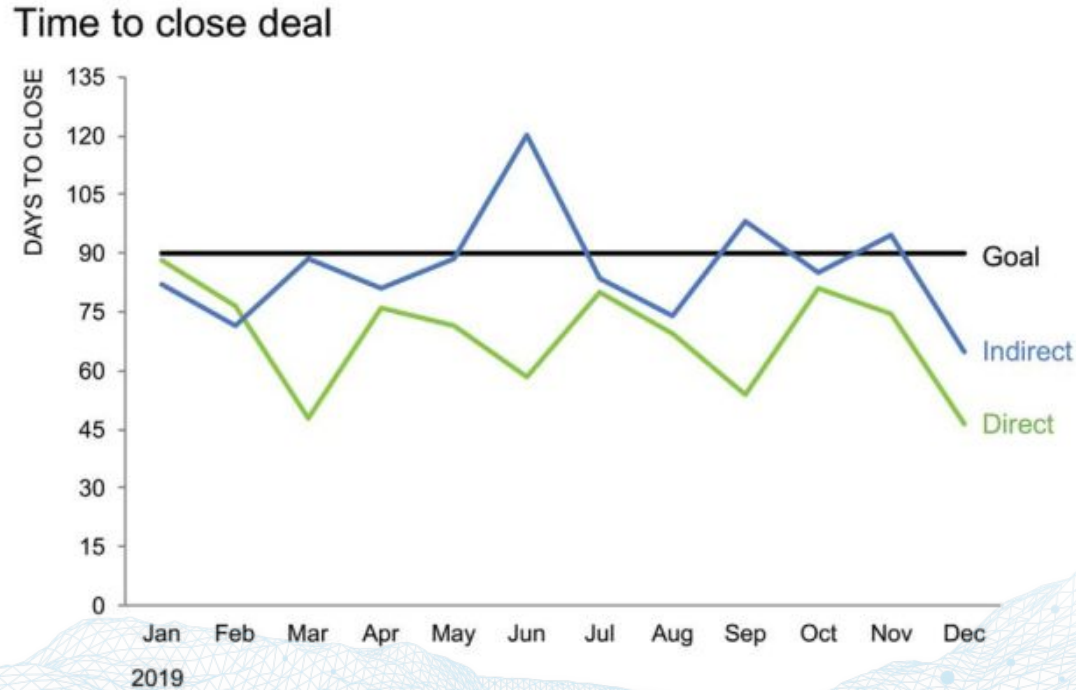


FIGURE 3.4n Put the goal in the graph

14. Iterate to best visualize Goal.

Iterating on **goal** line

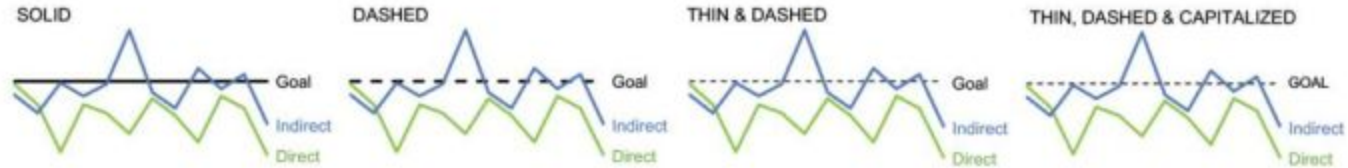


FIGURE 3.4o Iterating: different formatting for Goal line

Before, After



FIGURE 3.4a Let's declutter!

✗ **Suffering with data**

Time to close deal

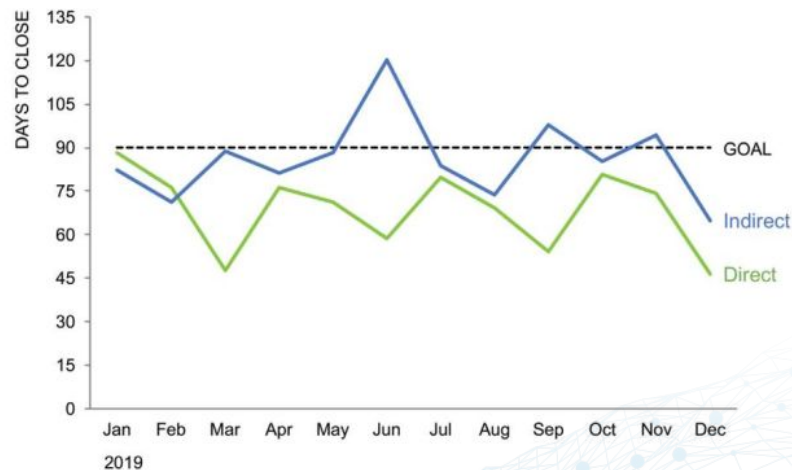
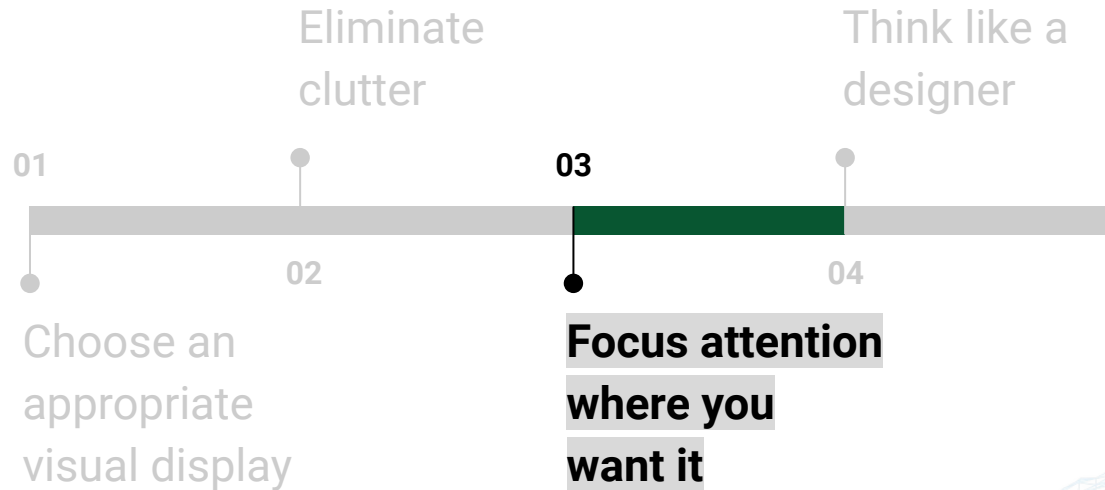


FIGURE 3.4p My favorite of the Goal line iterations

✓ **Storytelling with data**

Part 4.

How you'll learn to tell stories with data



enable your audience to see what you
want them to see before they even
know they're seeing it!



www.timoelliott.com

"I'm making a decision! Stop confusing me with facts!"

Focus Attention



Orientation



Shape



Line length



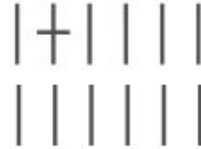
Line width



Size



Curvature



Added marks



Enclosure



Hue



Intensity



Spatial position



Motion

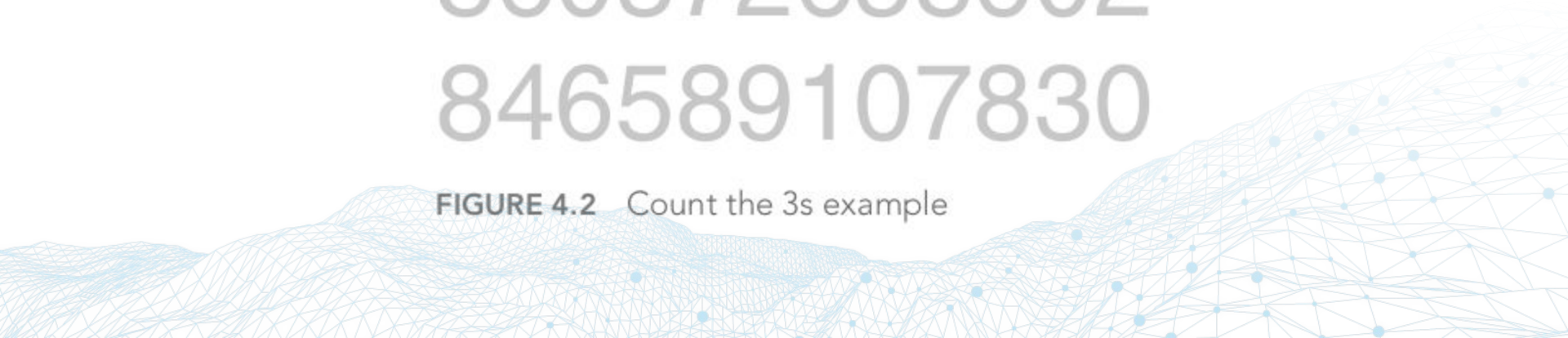
FIGURE 4.4 Preattentive attributes

Source: Adapted from Stephen Few's *Show Me the Numbers*, 2004.

Preattentive attributes signal where to look!

756395068473
658663037576
860372658602
846589107830

FIGURE 4.2 Count the 3s example

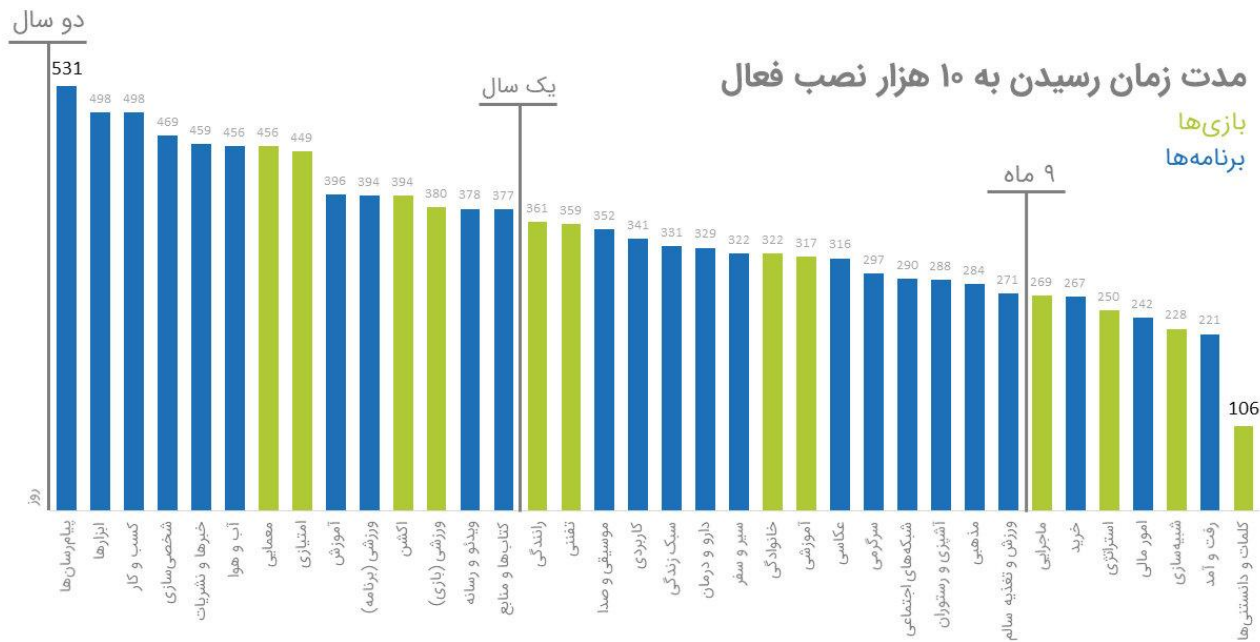


Preattentive attributes signal where to look!

756**3**95068473
65866**303**7576
860**3**72658602
8465891078**30**

FIGURE 4.3 Count the 3s example with preattentive attributes

Preattentive attributes in graphs



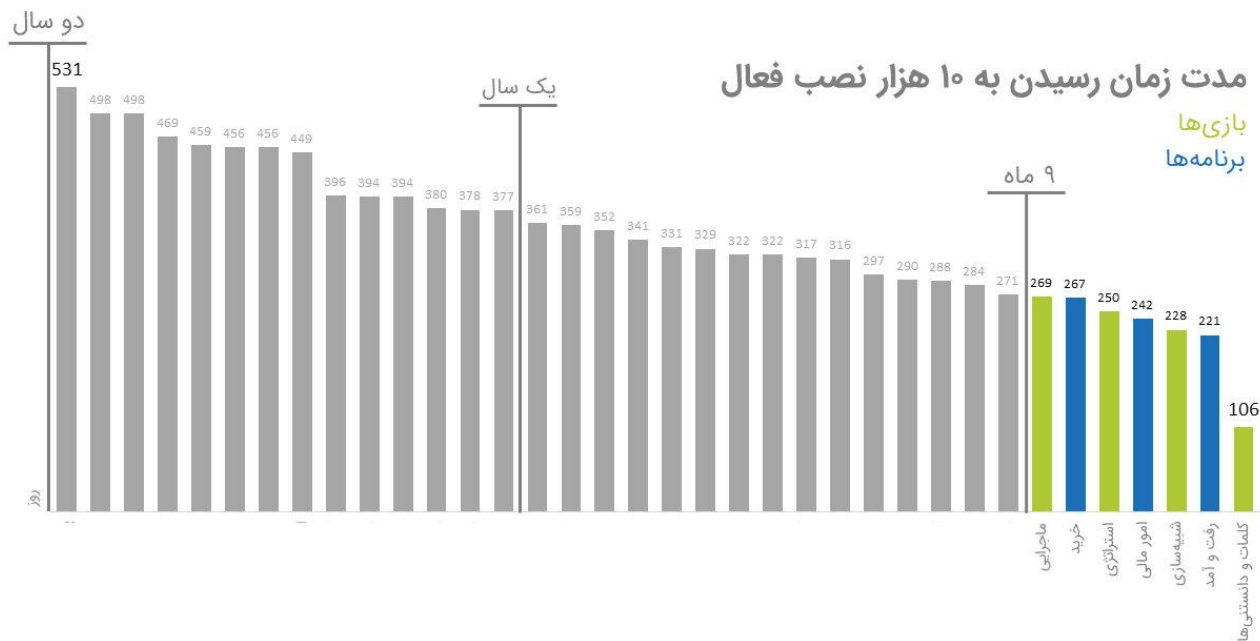
Reports

منبع داده‌ها: گزارش‌های کافه‌بار | تاریخ گزارش: پاییز ۱۳۹۸

رقابت و سودآوری صنعت برنامه‌های موبایل در ایران

Source: <https://vrgl.ir/FC0pB>

Preattentive attributes in graphs



منبع داده‌ها: گزارش‌های کافه‌بازار | تاریخ گزارش: پاییز ۱۳۹۸

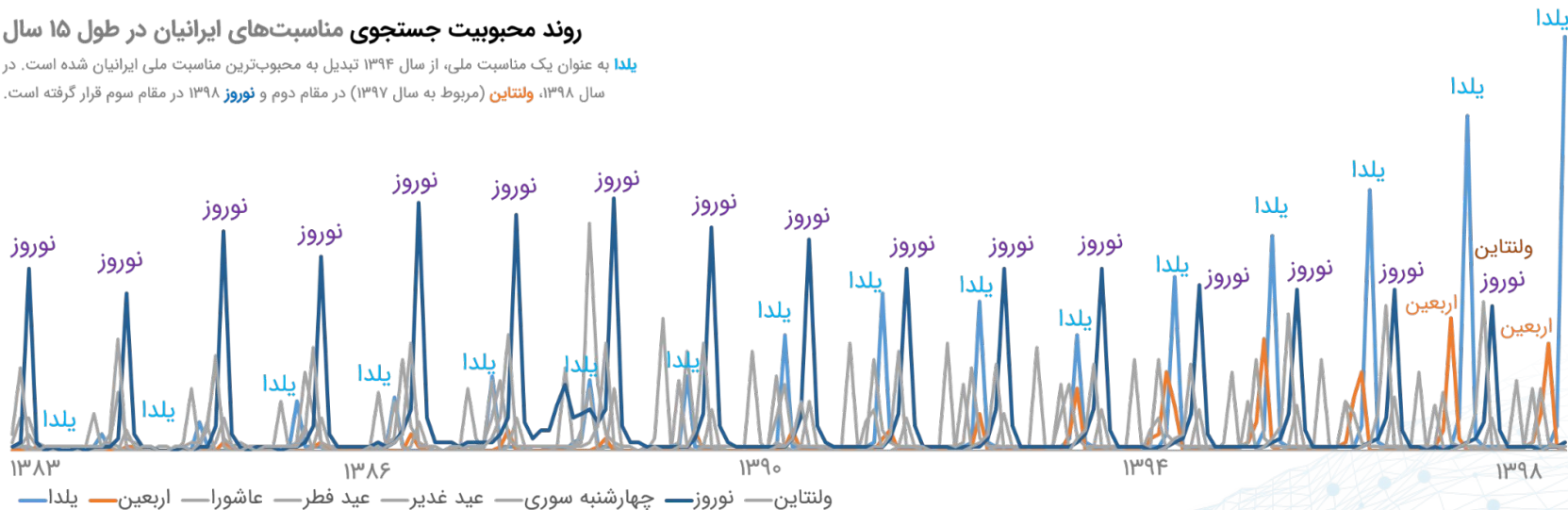
رقابت و سودآوری صنعت برنامه‌های موبایل در ایران

Source: <https://vrgl.ir/FC0pB>

Preattentive attributes in graphs

رند محبوبیت جستجوی مناسبت‌های ایرانیان در طول ۱۵ سال

یلدا به عنوان یک مناسبت ملی، از سال ۱۳۹۴ تبدیل به محبوب‌ترین مناسبت ملی ایرانیان شده است. در سال ۱۳۹۸، ولنتاین (مربوط به سال ۱۳۹۷) در مقام دوم و نوروز ۱۳۹۸ در مقام سوم قرار گرفته است.



منبع: گوگل‌ترندز - بازه زمانی: از ۱۳۸۳ تا ۱۳۹۸

Virgool.io/irantrends - By: Mahdi Nasseri

Source: <https://vrgl.ir/4p3vM>

Color

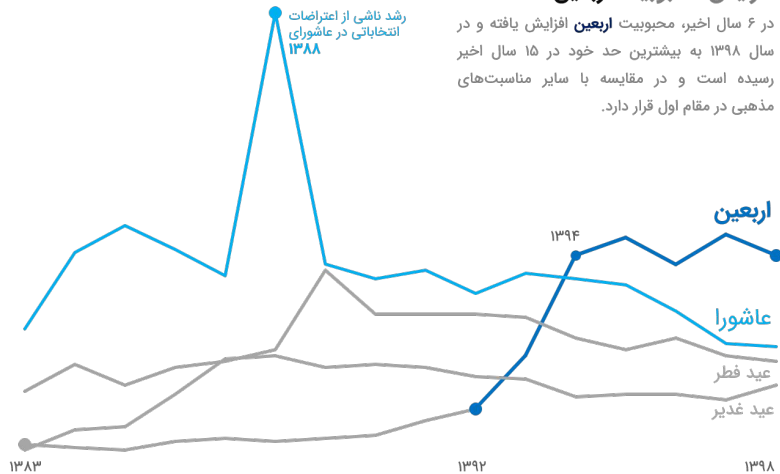
- Use it sparingly
- Use it consistently
- Design with the color-blind in mind
- Be thoughtful of the tone color conveys
- Consider whether to leverage brand colors



Color

افزایش محبوبیت اربعین

در ۶ سال اخیر، محبوبیت **اربعین** افزایش یافته و در سال ۱۳۹۸ به بیشترین حد خود در ۱۵ سال اخیر رسیده است و در مقایسه با سایر مناسبت‌های مذهبی در مقام اول قرار دارد.

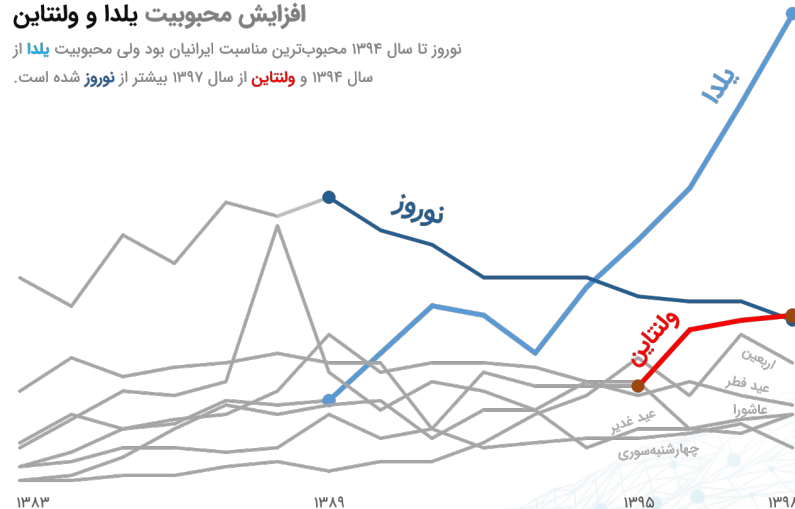


منبع: گوگل ترندز - Virgool.io/iranrends - Mahdi Nasserri

Source: <https://vrql.ir/4p3vM>

افزایش محبوبیت یلدا و ولنتاین

نوروز تا سال ۱۳۹۴ محبوب‌ترین مناسبت ایرانیان بود ولی محبوبیت **یلدا** از سال ۱۳۹۴ و **ولنتاین** از سال ۱۳۹۷ بیشتر از **نوروز** شده است.



منبع: گوگل ترندز - Virgool.io/iranrends - Mahdi Nasserri

Color

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

Top 5 drugs: country-level sales rank

RANK		1	2	3	4	5+
COUNTRY DRUG		A	B	C	D	E
Australia		1	2	3	6	7
Brazil		1	3	4	5	6
Canada		2	3	6	12	8
China		1	2	8	4	7
France		3	2	4	8	10
Germany		3	1	6	5	4
India		4	1	8	10	5
Italy		2	4	10	9	8
Mexico		1	5	4	6	3
Russia		4	3	7	9	12
Spain		2	3	4	5	11
Turkey		7	2	3	4	8
United Kingdom		1	2	3	6	7
United States		1	2	4	3	5

FIGURE 4.15 Use color sparingly



Suffering with data



Storytelling with data

Color

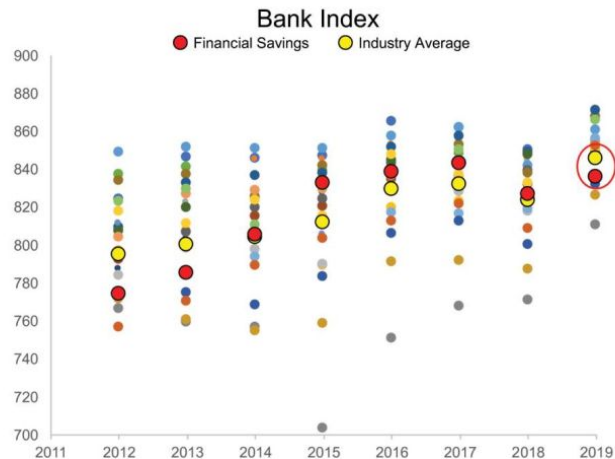


FIGURE 4.4a Bank index

BRANCH SATISFACTION

Financial Savings below industry for first time in 5 years

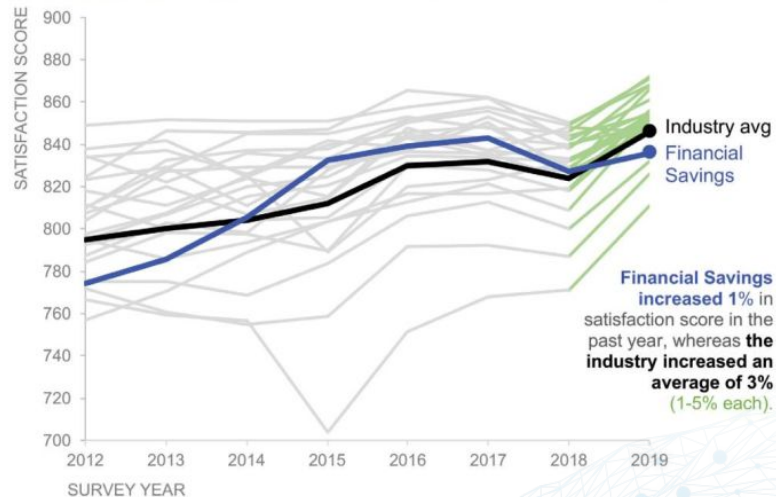


FIGURE 4.4c Could focus on latest year-to-year period of time

✗ Suffering with data

✓ Storytelling with data

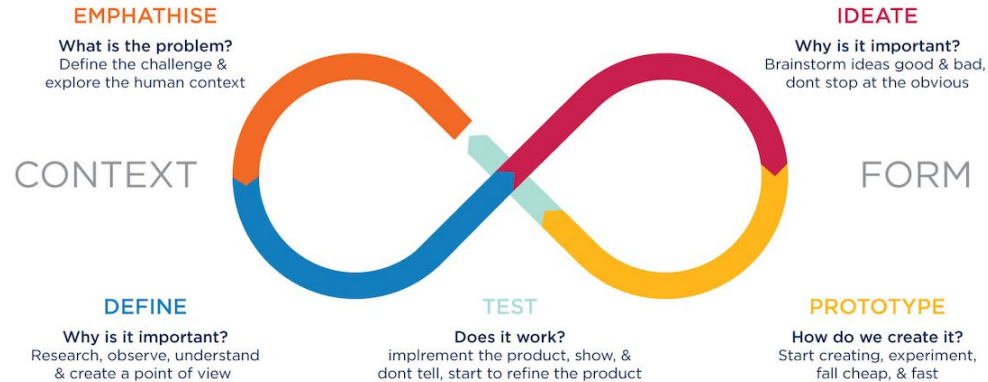
How you'll learn to tell stories with data



Think like a designer

DESIGN THINKING

A FRAMEWORK FOR INNOVATION



Think like a designer

- Affordances
- Accessibility
- Aesthetics
- Acceptance



Think like a designer

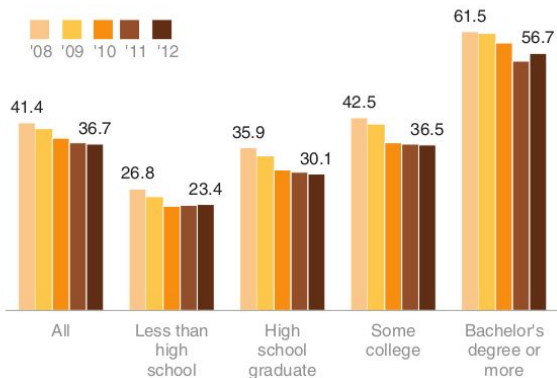
- **Affordances** —————
 - Accessibility
 - Aesthetics
 - Acceptance
- Eliminate distractions
 - Not all data are equally important.
 - When detail isn't needed, summarize.
 - Ask yourself: would eliminating this change anything?
 - Push necessary, but non-message-impacting items to the background.
- **Bold, *italics*, and underlining**
 - CASE and *typeface*
 - Color
 - Inversing elements



Highlight the important stuff

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

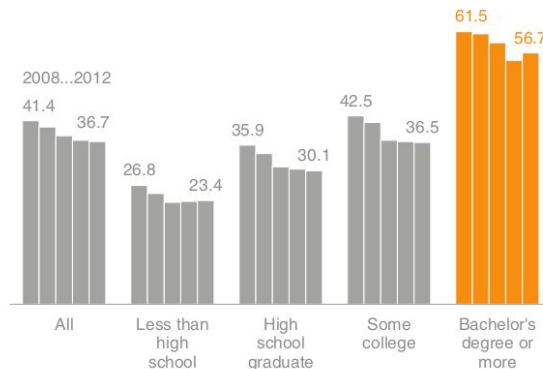
Source: U.S. Census

Adapted from PEW RESEARCH CENTER

FIGURE 5.2 Pew Research Center original graph

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

Adapted from PEW RESEARCH CENTER

FIGURE 5.3 Highlight the important stuff

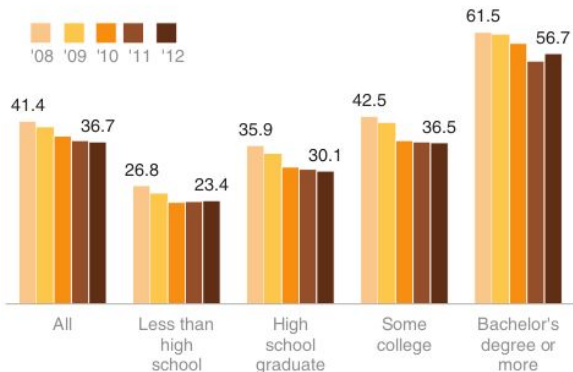
✗ Suffering with data

✓ Storytelling with data

Eliminate distractions

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



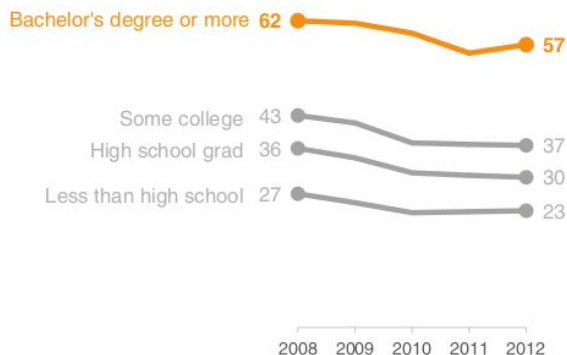
Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

Adapted from PEW RESEARCH CENTER

New marriage rate by education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

Adapted from PEW RESEARCH CENTER

FIGURE 5.5 Before-and-after

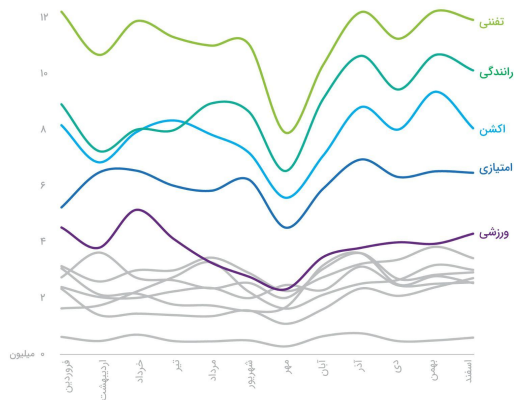
✗ Suffering with data

✓ Storytelling with data

Eliminate distractions



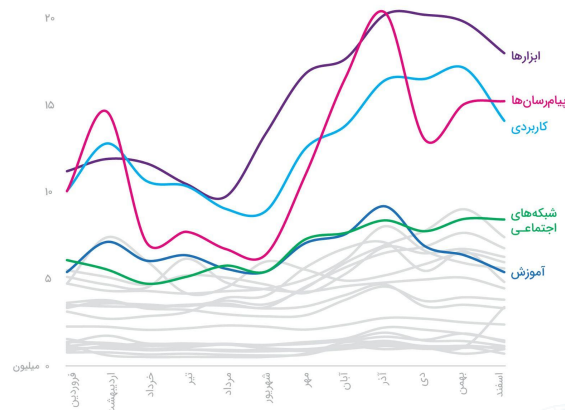
دانلود و به روز رسانی
از دسته‌های مختلف بازی در سال ۱۳۹۸



بازی‌های تفنی، راندگی، اکشن، امتیازی و ورزشی به ترتیب بیشترین تعداد دانلود و به روز رسانی را در سال ۹۸ داشتند.



دانلود و به روز رسانی
از دسته‌های مختلف برنامه در سال ۱۳۹۸



ابزارها، پیام‌رسان‌ها، کاربردی، شبکه‌های اجتماعی و آموزش به ترتیب بیشترین تعداد دانلود و به روز رسانی را در سال ۹۸ داشتند.

Think like a designer

- Affordances
 - **Accessibility** —————
 - Aesthetics
 - Acceptance
- Don't overcomplicate
 - Make it legible.
 - Keep it clean.
 - Use straightforward language.
 - Remove unnecessary complexity.



Text is your friend

Peak Break-up Times
According to Facebook status updates

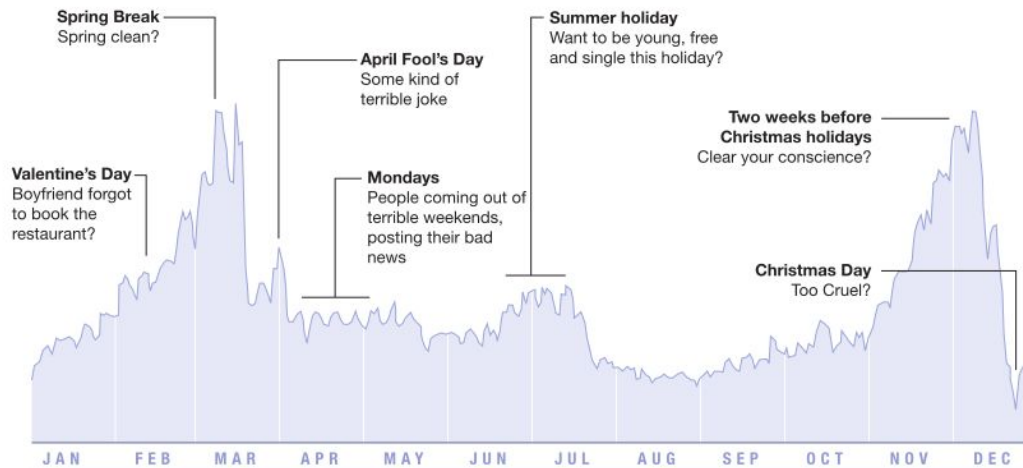
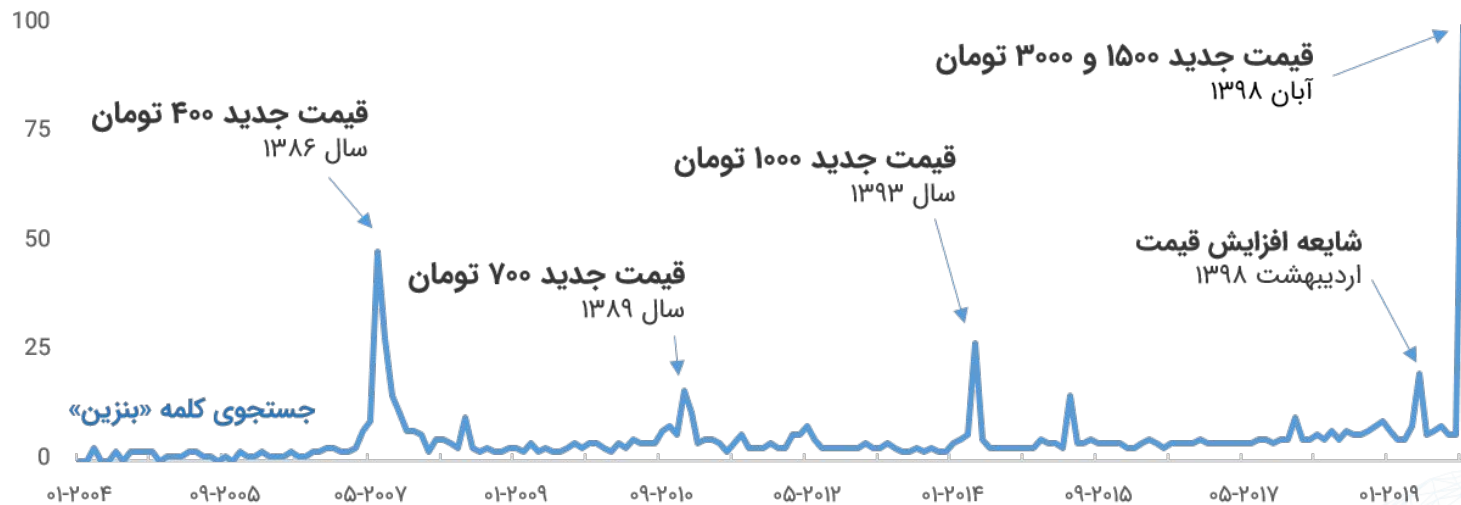


FIGURE 5.7 Words used wisely

Text is your friend



منبع: گوگل ترندز
بازه‌ی زمانی: ۱۵ سال
تاریخ: از ۲۰۰۴ تا ۲۰۱۹

Source: <https://vrql.ir/X8AkG>

Text is your friend

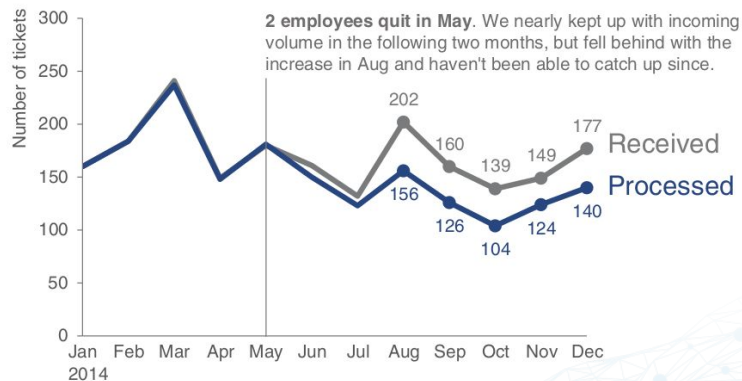


FIGURE 5.8 Let's revisit the ticket example

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

Ticket volume over time



Data source: XYZ Dashboard, as of 12/31/2014. | A detailed analysis on tickets processed per person and time to resolve issues was undertaken to inform this request and can be provided if needed.

FIGURE 5.10 Add action title and annotation

Think like a designer

- Affordances
 - Accessibility
 - **Aesthetics** —
 - Acceptance
- Be smart with color.
 - Pay attention to alignment.
 - Leverage white space.



Aesthetics

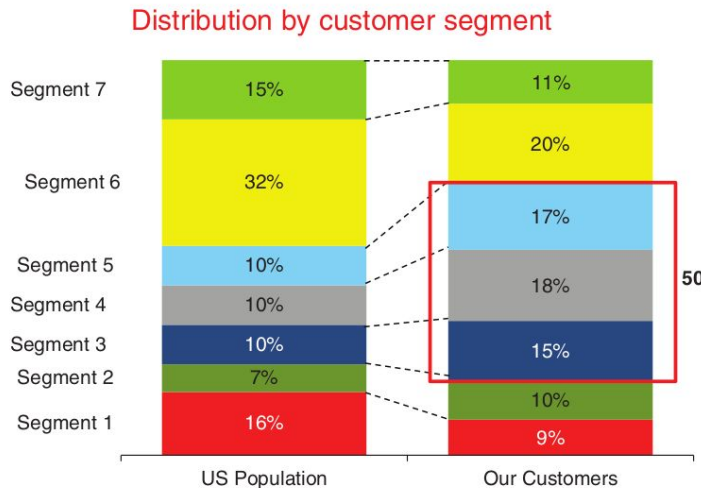


FIGURE 5.12 Unaesthetic design

Distribution by customer segment

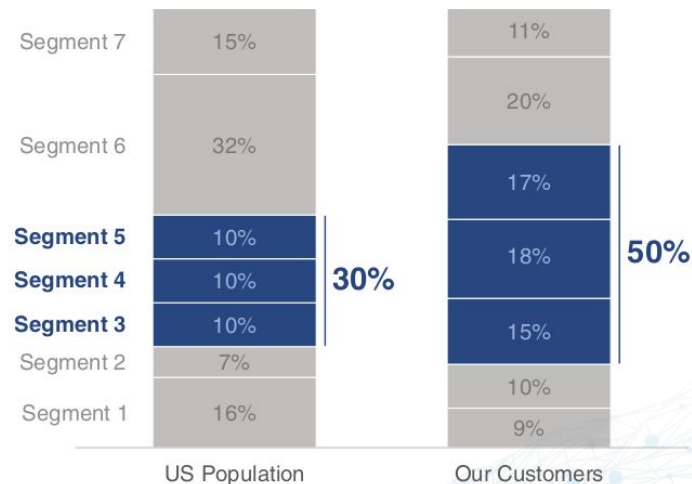
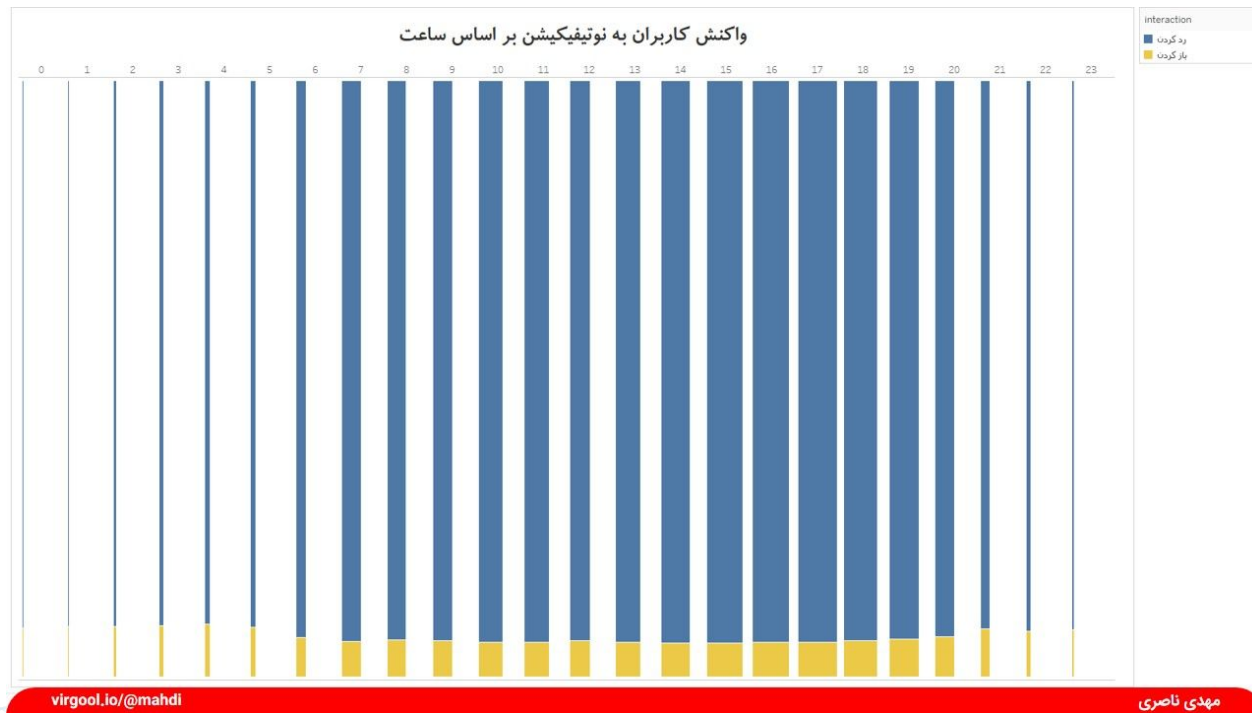


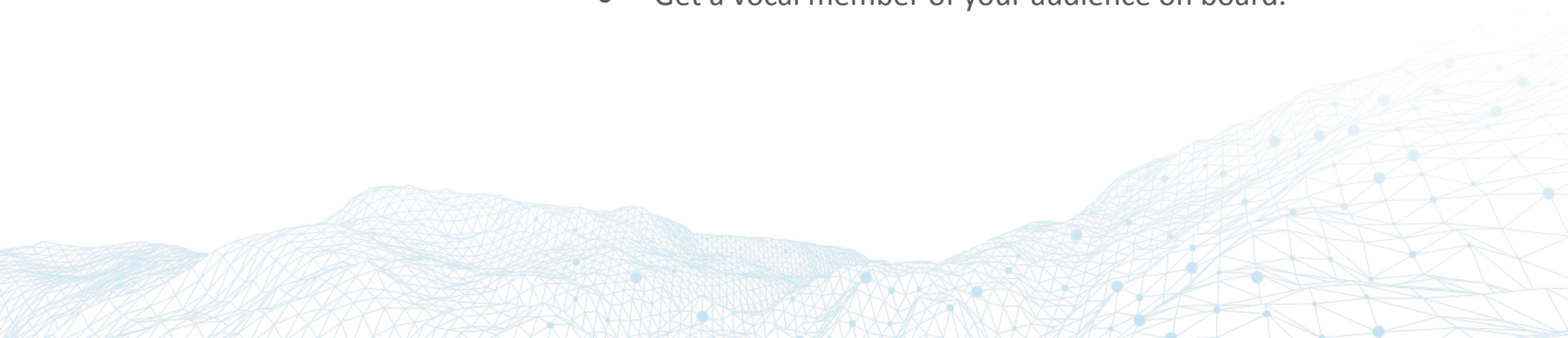
FIGURE 5.13 Aesthetic design

Aesthetics



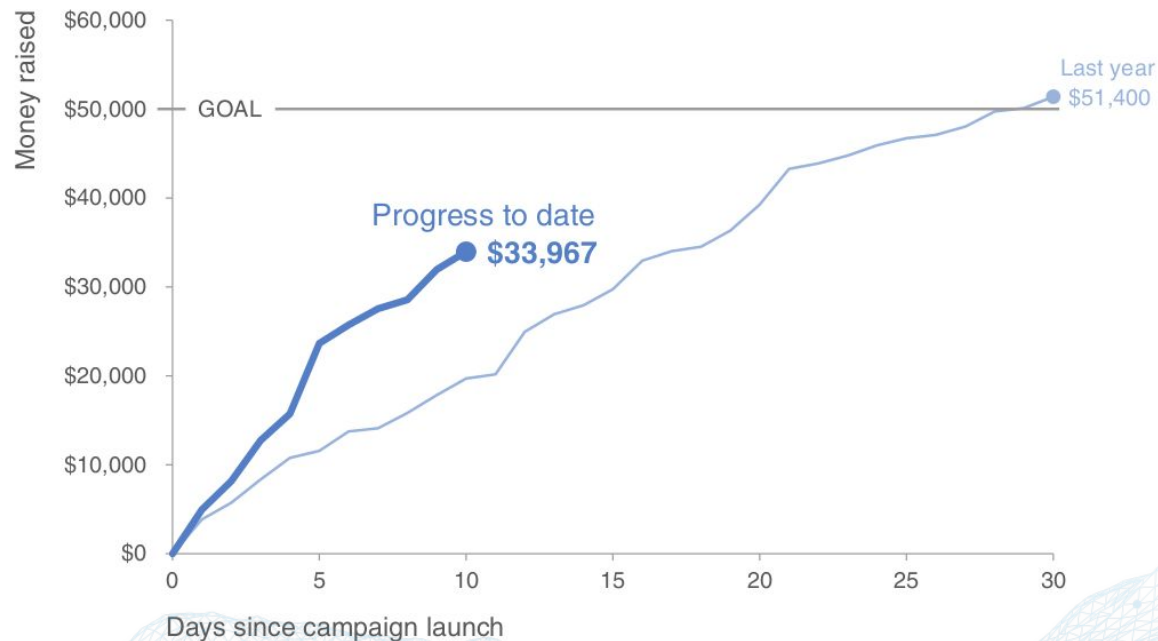
Think like a designer

- Affordances
- Accessibility
- Aesthetics
- **Acceptance** —————
 - Articulate the benefits of the new or different approach.
 - Show the side-by-side.
 - Provide multiple options and seek input.
 - Get a vocal member of your audience on board.



#1: line graph

Annual giving campaign progress



#2: annotated line graph with forecast

Sales over time



Data source: Sales Dashboard; annual figures are as of 12/31 of the given year.

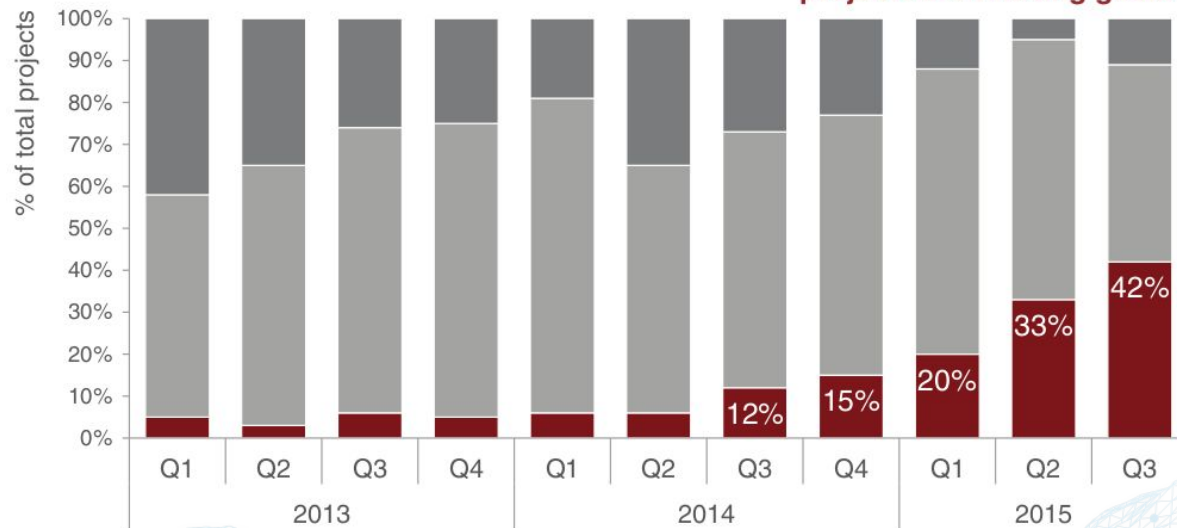
*Use this footnote to explain what is driving the 10% annual growth forecast assumption.

#3: 100% stacked bars

Goal attainment over time

■ Miss ■ Meet ■ Exceed

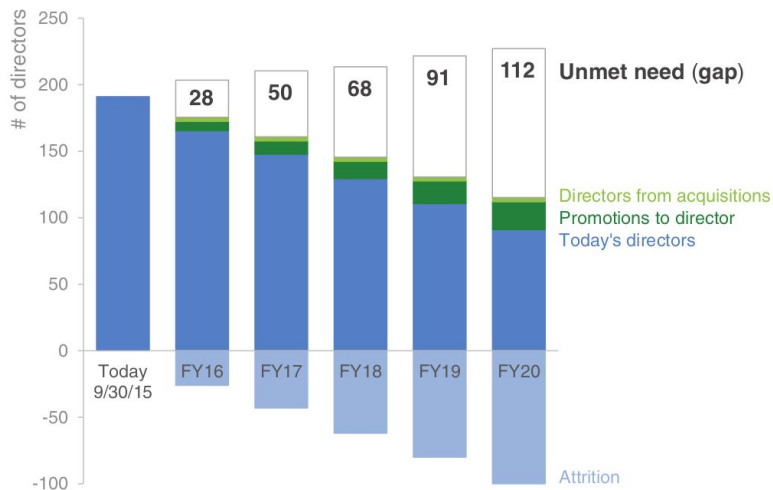
As of Q3 2015, **more than 1/3 of projects are missing goals**



Data source: XYZ Dashboard; the total number of projects has increased over time from 230 in early 2013 to nearly 270 in Q3 2015.

#4: leveraging positive and negative stacked bars

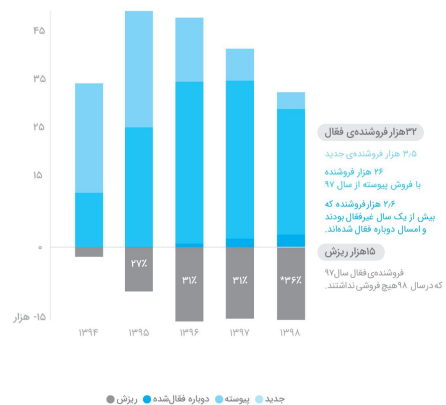
Expected director population over time



A footnote explaining relevant forecast assumptions and methodology would go here.



۳۲ هزار برنامه‌ی فعال فروشنده
در سال ۱۳۹۸

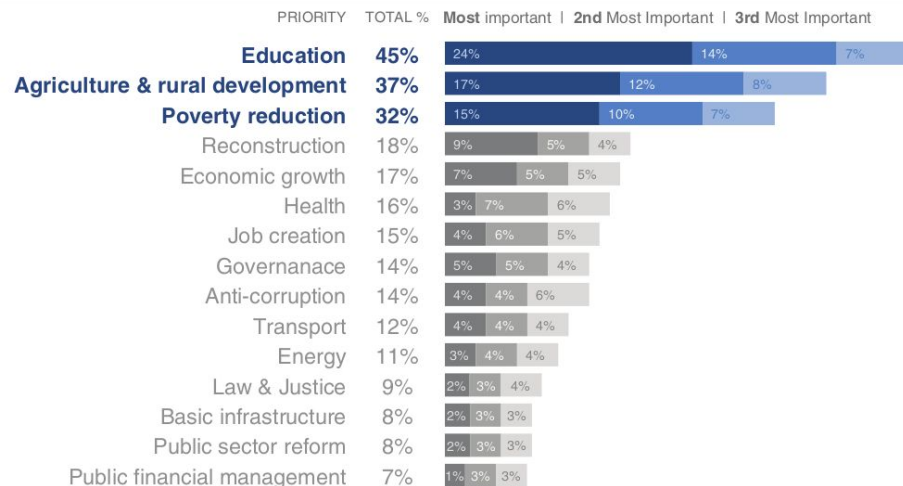


منبع: گزارش سال ۹۸ بازار

در طول عمر بازار تا انتهای سال ۹۸، بیش از ۸۴ هزار برنامه در بازار فروش داشتند که ۳۲ هزار عدد از آن‌ها در سال ۹۸ هم فعال بودم‌اند.

#5: horizontal stacked bars

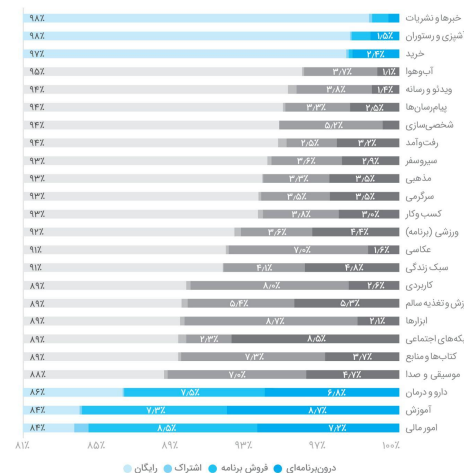
Top 15 development priorities, according to survey



N = 4,392. Based on responses to item, When considering development priorities, which one development priority is the most important? Which one is the second most important priority? Which one is the third most important priority? Respondents chose from a list. Top 15 shown.



سهم برنامه‌های فروشنده از کل
برنامه‌های هر دسته



منبع: گزارش سال ۹۸ بازار

در دسته‌های امور مالی، آموزش و دارو و درمان، توسعه‌دهندگان تمایل کمتری به انتشار برنامه‌های رایگان دارند.

Part 5.

Let's practice!



Matplotlib



Seaborn



Altair



Vega



Bokeh



Plotly



Data Visualization with Python

```
import plotly.graph_objects as go

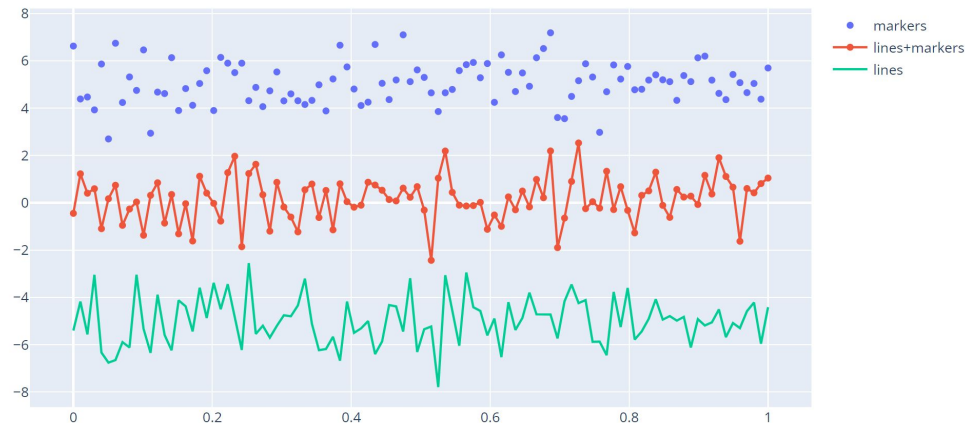
# Create random data with numpy
import numpy as np
np.random.seed(1)

N = 100
random_x = np.linspace(0, 1, N)
random_y0 = np.random.randn(N) + 5
random_y1 = np.random.randn(N)
random_y2 = np.random.randn(N) - 5

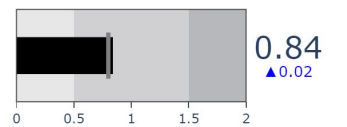
fig = go.Figure()

# Add traces
fig.add_trace(go.Scatter(x=random_x, y=random_y0,
                        mode='markers',
                        name='markers'))
fig.add_trace(go.Scatter(x=random_x, y=random_y1,
                        mode='lines+markers',
                        name='lines+markers'))
fig.add_trace(go.Scatter(x=random_x, y=random_y2,
                        mode='lines',
                        name='lines'))

fig.show()
```



Data Visualization with Python



businesses

0.96

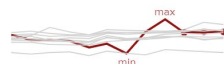
▼ -0.03



community

0.86

-



electronic devices

0.77

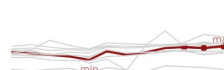
▲ 0.01



for the home

0.85

▲ 0.02



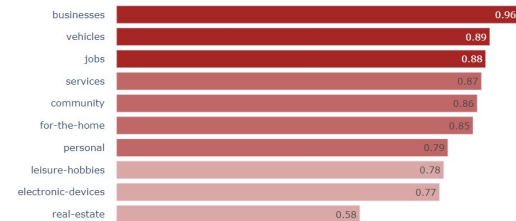
jobs

0.88

▼ -0.01



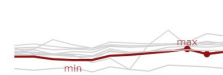
Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep
2020 2021



leisure hobbies

0.78

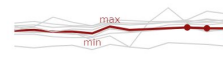
▲ 0.03



personal

0.79

-



real estate

0.58

▼ -0.02



services

0.87

▼ -0.03



vehicles

0.89

▲ 0.02



Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep
2020 2021

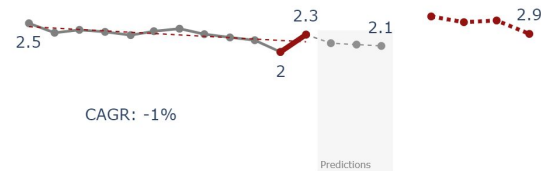
— Other Categories — Category Monthly — Total Monthly

Data Visualization with Python



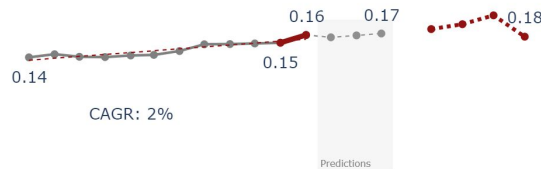
Metric 1

2.33
▲14%



Metric 2

0.16
▲6%



Metric 3

12.0
▼-18%



Metric 4

2.41
▲4%



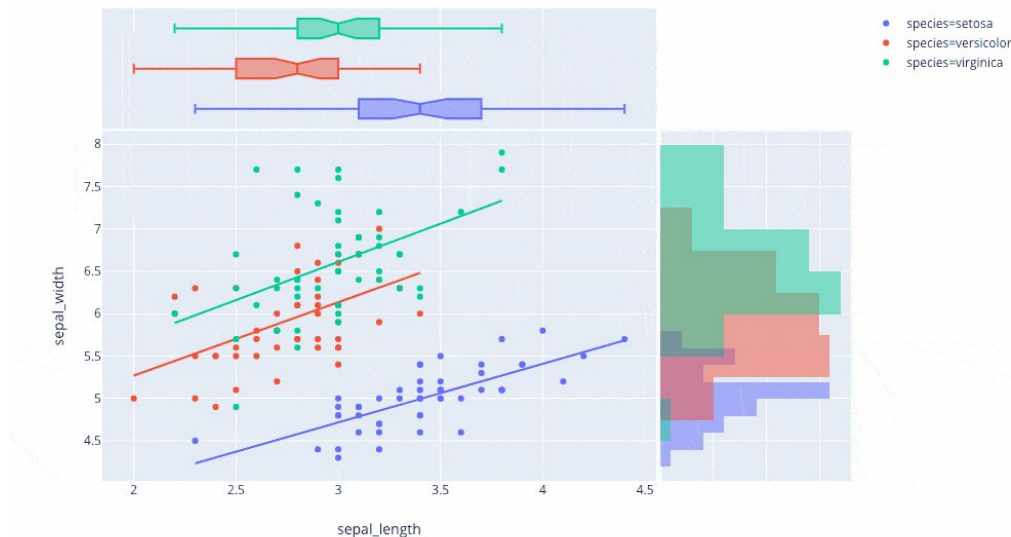
—●— Total Montly - - - - Total Trend - - ● - - Total predication ● - - - - Total Weekly

Data Visualization with Python



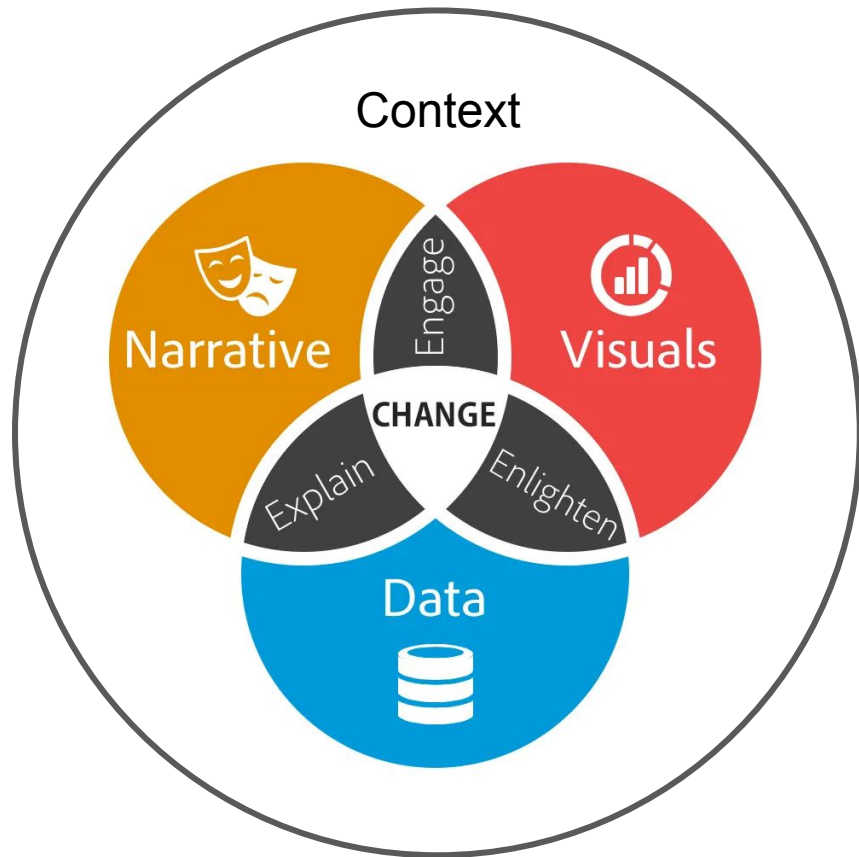
plotly

```
px.scatter(iris, x="sepal_width", y="sepal_length", color="species", marginal_y="histogram",  
           marginal_x="box", trendline="ols")
```



Part 6. Create a Story

Data Story Narrative



The magic of story



Intuitive Decision Making Rational

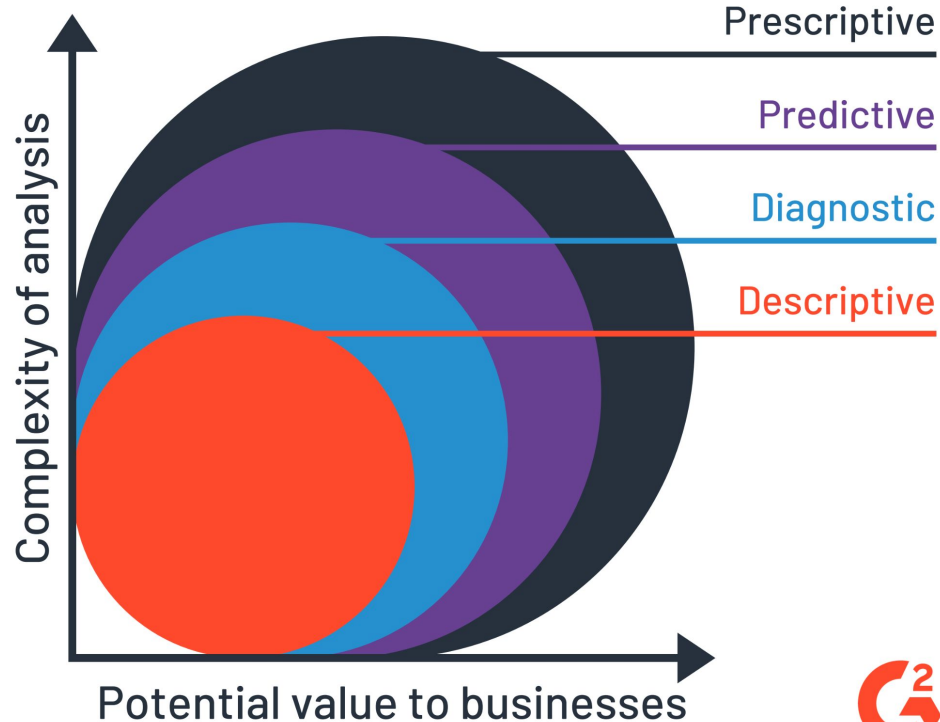


"When you two have finished arguing your ~~opinions~~, I actually have ~~data~~!"

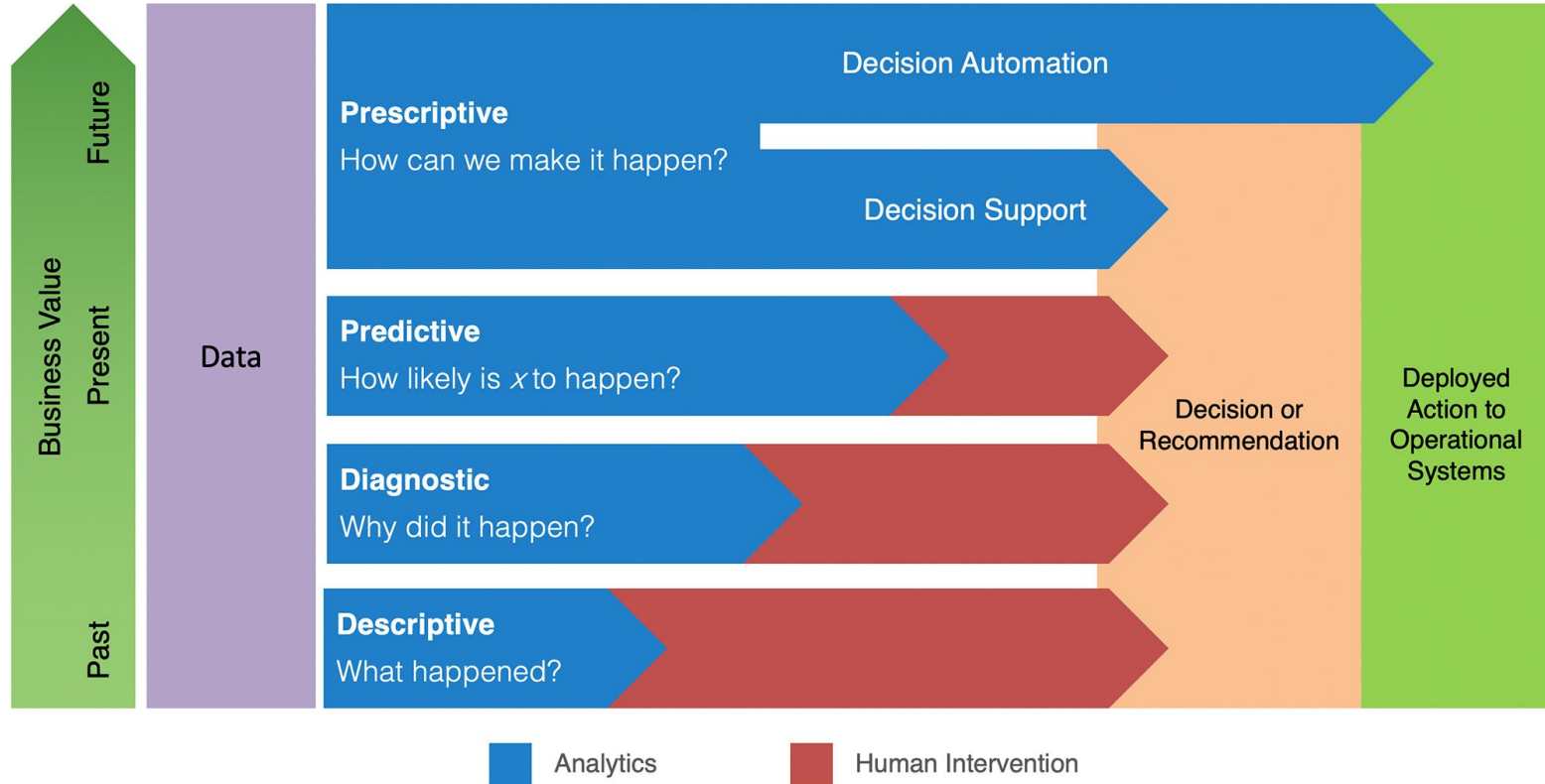
Data

Story

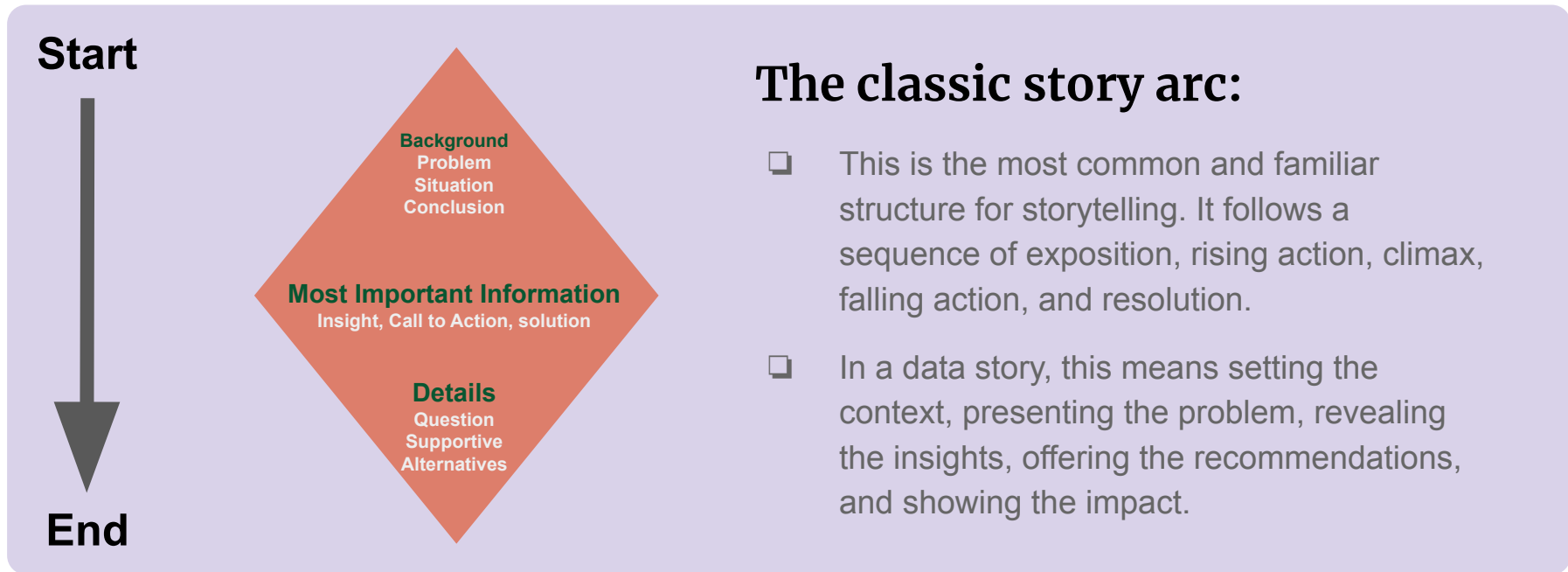
4 types of Data Analytics



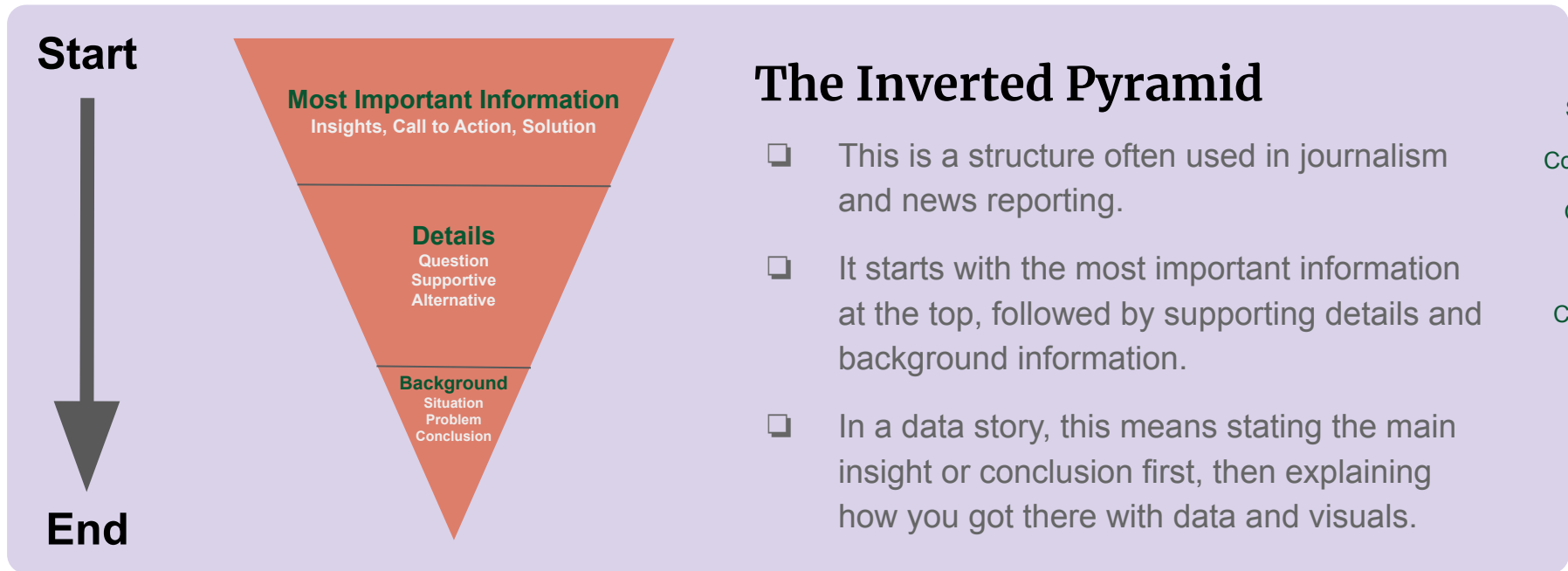
A Data-Story: A mix of Analytics & Human



A Data-Story: Top 3 Structures



A Data-Story: Top 3 Structures

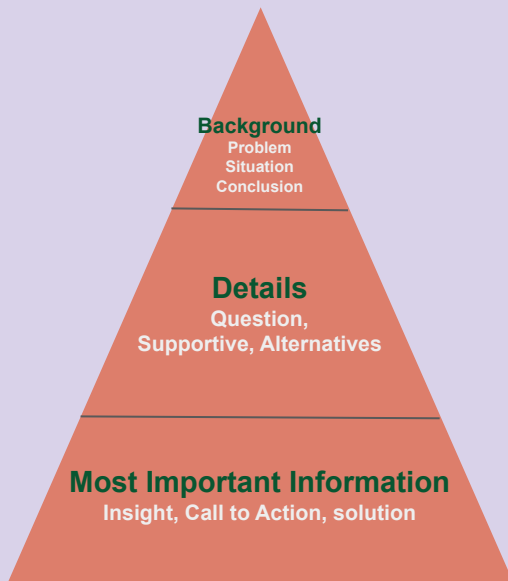


A Data-Story: Top 3 Structures

Start



End



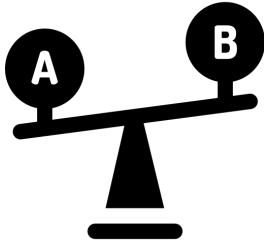
The Story Pyramid

- ❑ Often used for summarizing or writing business data stories.
- ❑ In this structure, you set the scene, present the problem, ask a relevant question, reveal the insight or conclusion, offer a recommendation or solution, suggest an action or next step, and show the impact or benefit.

A Data Story: Top 4 type of Story Nodes



Achievements



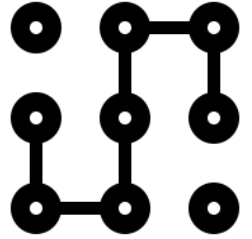
Comparison



Question and Answer



Trends

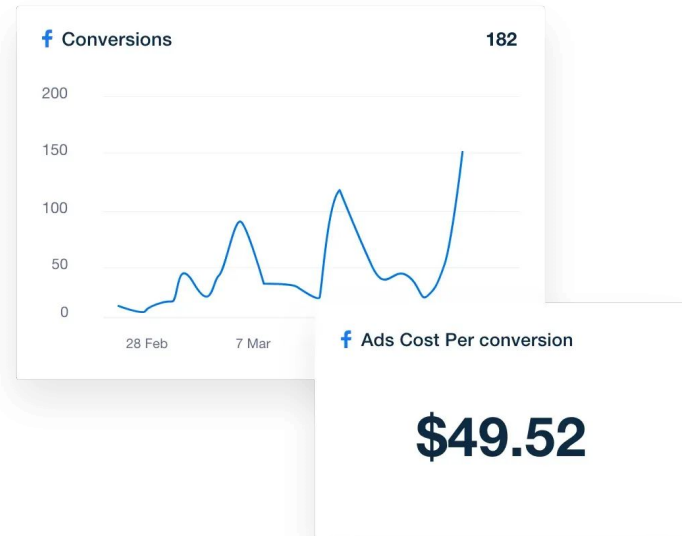


Patterns

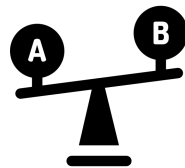
Story Nodes: Achievements



- Highlights the accomplished milestones and progress over time or across dimensions.
- It can showcase positive impacts and growth
- In a data story, it means providing a clear picture of significant achievements in a specific field or project.

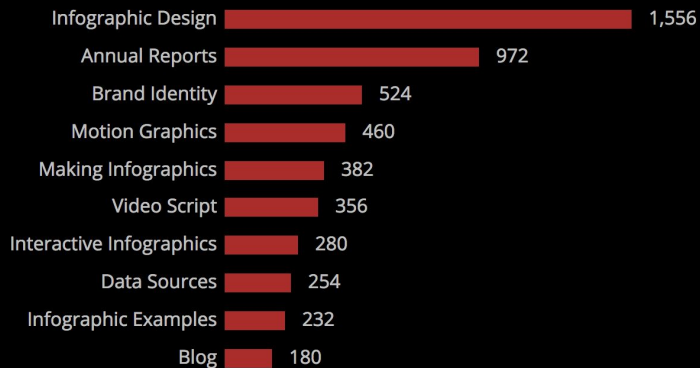


Story Nodes: Comparison



- Highlights the differences or similarities between two or more groups, scenarios, or outcomes.
- It can be used to show contrast, correlation, or causation.
- In a data story, this means using data and visuals to compare and contrast different aspects of your topic.

Top 10 Most Visited Pages of April



Story Nodes: Question & Answer



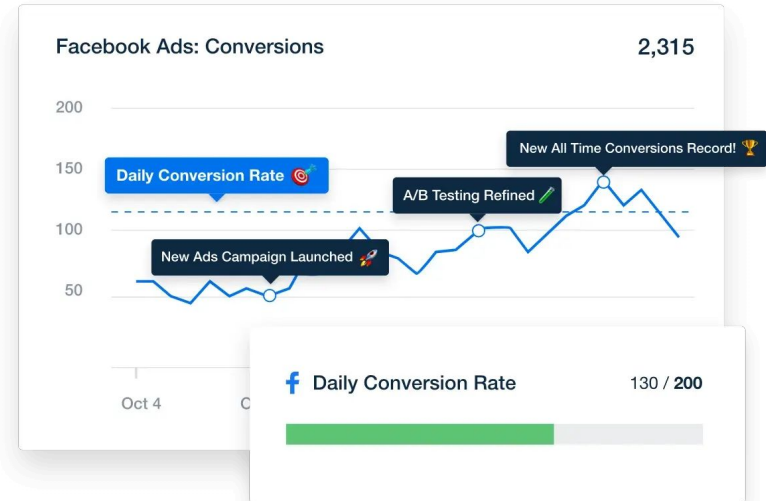
- Engages the audience by posing questions and answering them with data.
- It can be used to address common or anticipated questions from your audience, or to guide them through a discovery process.
- In a data story, this means framing your insights as answers to relevant questions that your audience may have.



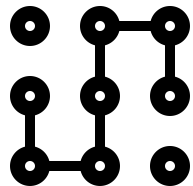
Story Nodes: Trends



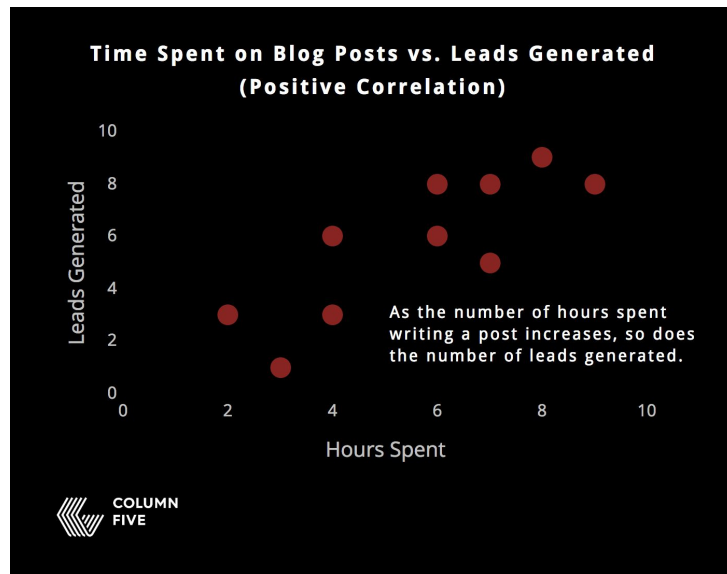
- Shows how something changes over time or space.
- It can be used to show trends, patterns, or progress.
- In a data story, this means using data and visuals to show how your topic evolves or moves along a timeline or a map.



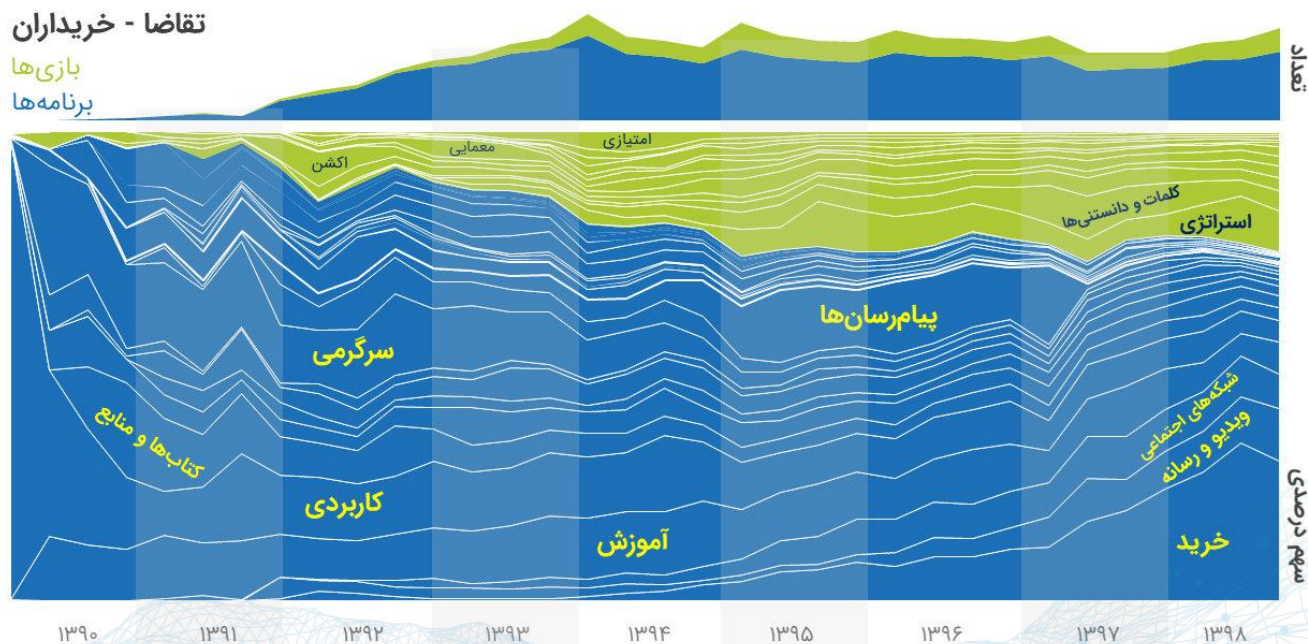
Story Nodes: Patterns



- Reveals recurring trends or correlations across instances or timeframes
- Using data and visuals to highlight significant patterns and deepen understanding.



Pulling It All Together

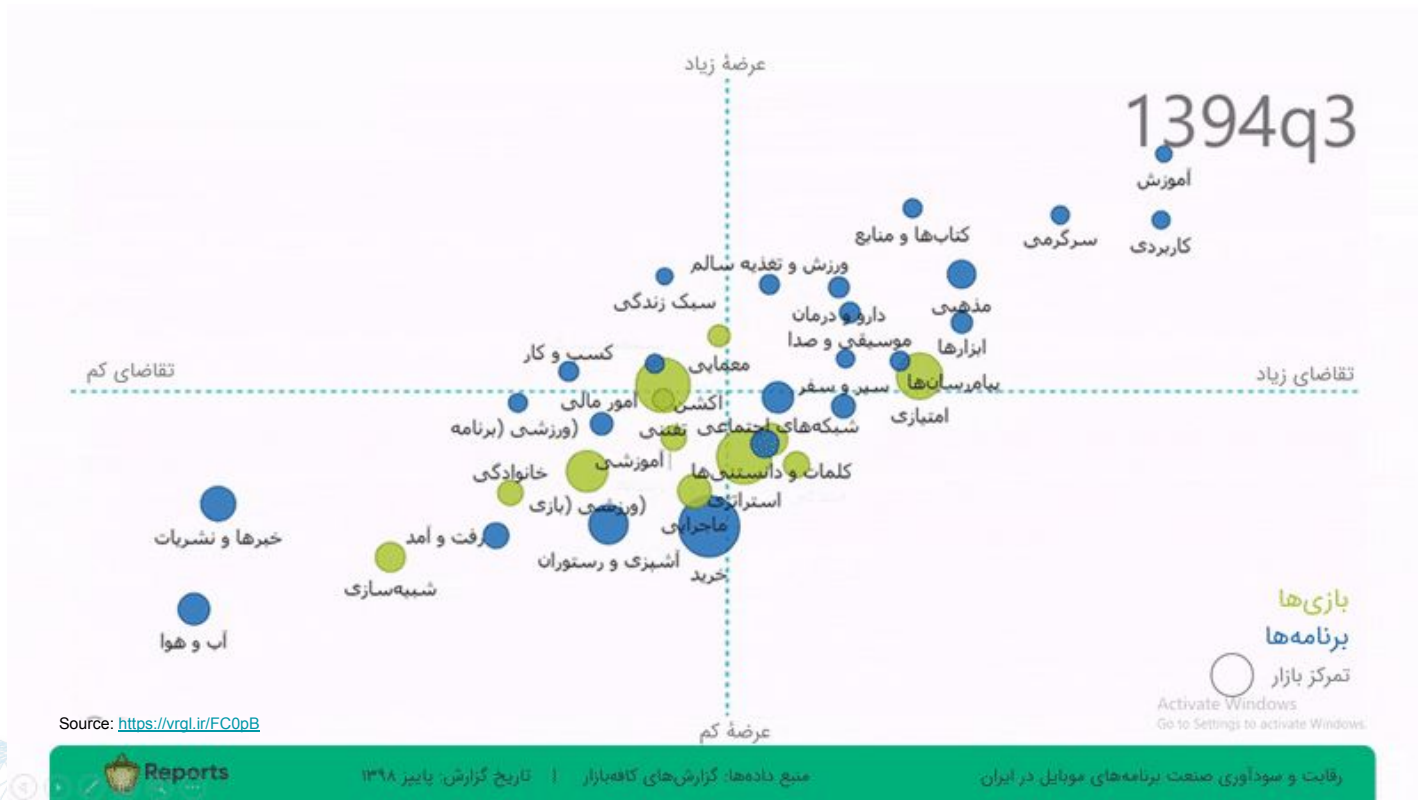


منبع دادهها: گزارشهای کافهبازار | تاریخ گزارش: پاییز ۱۳۹۸

رقابت و سودآوری صنعت برنامههای موبایل در ایران

Source: <https://vrol.ir/FC0pB>

Pulling It All Together



Golden Magic: Logical Coherence

- **Consistency** and **Connectivity** between ideas.
- Essential for clear, convincing, credible data story.

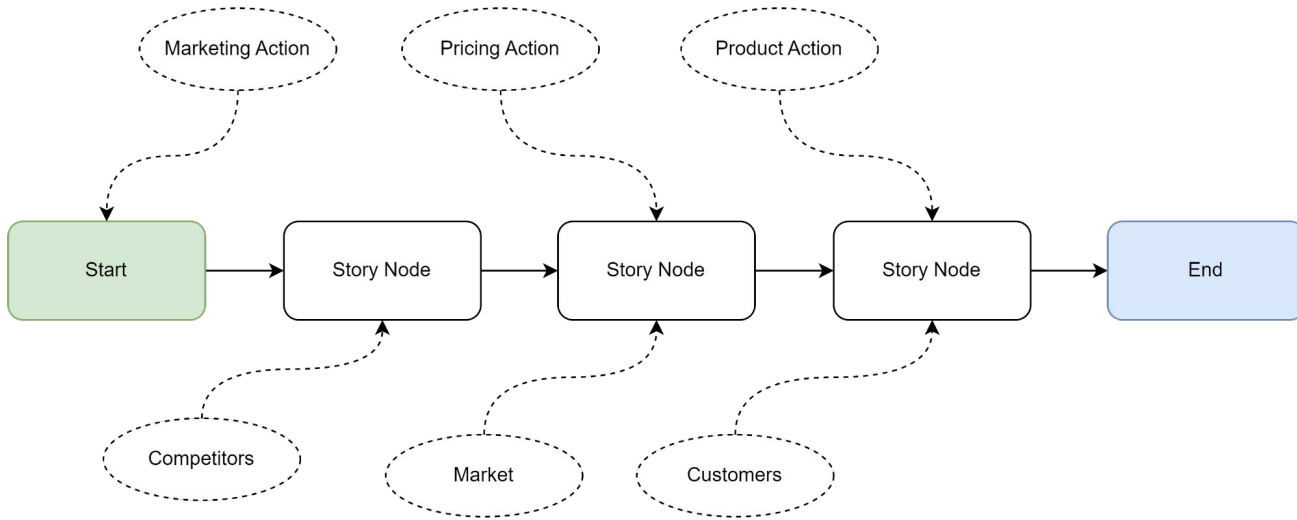
How to create?

- Casual Chain
- Snowflake
- Collage



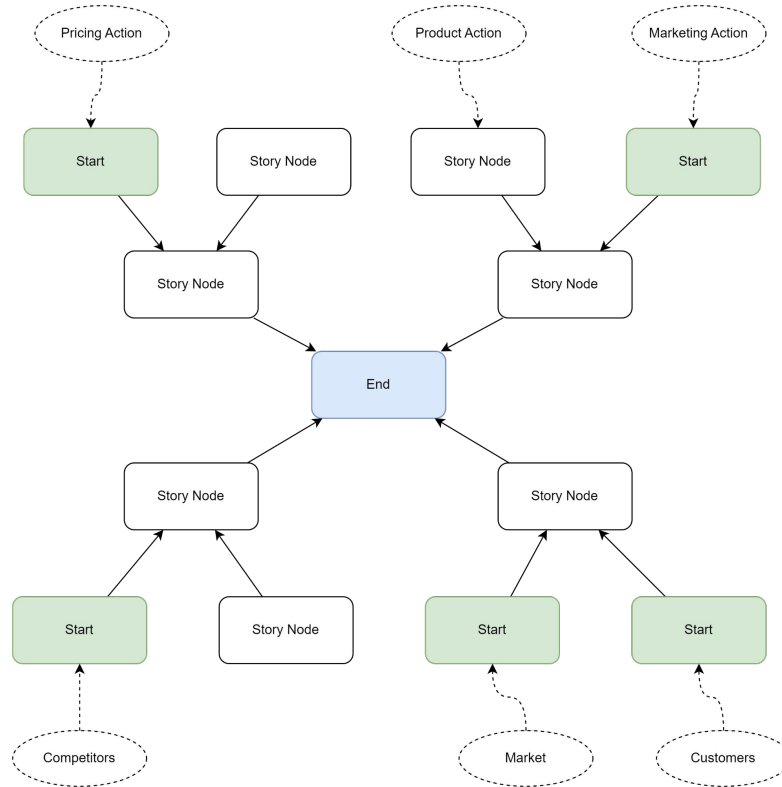
How to create Logical Coherence?

Casual Chain



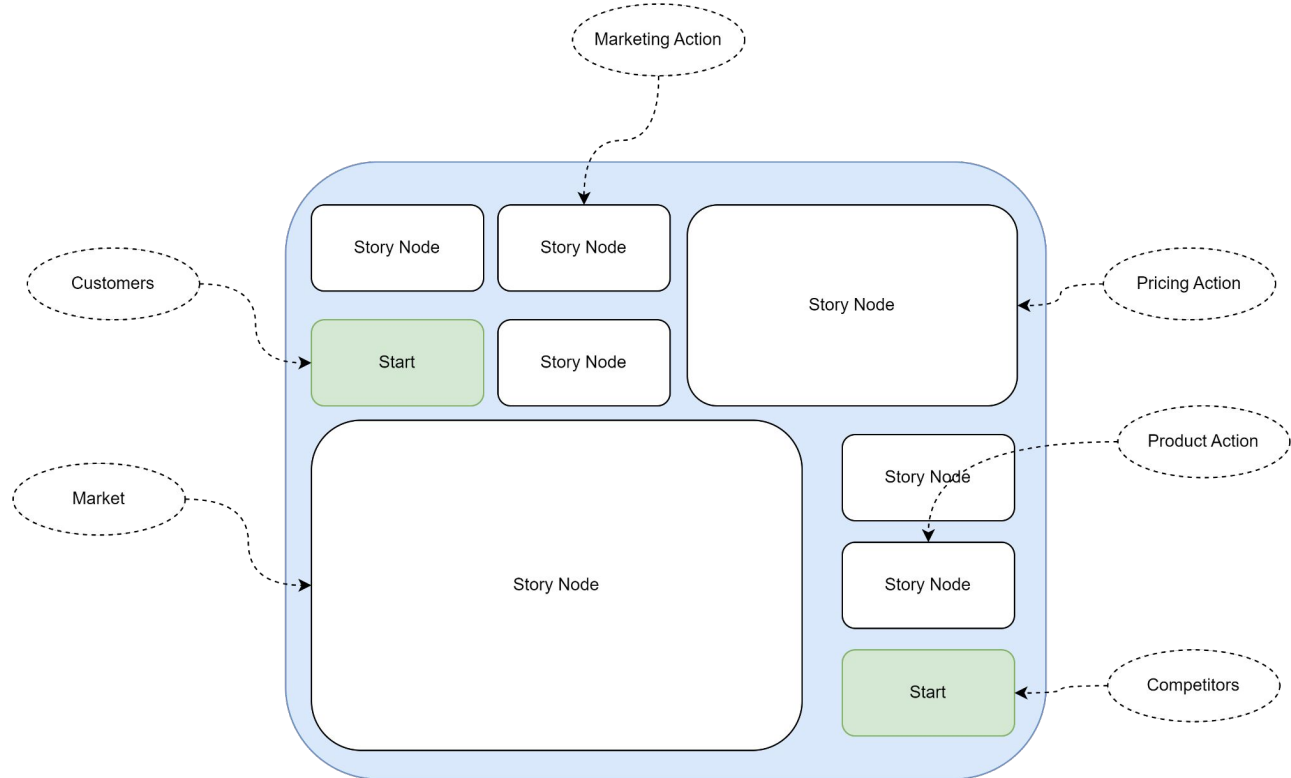
How to create Logical Coherence?

Snowflake



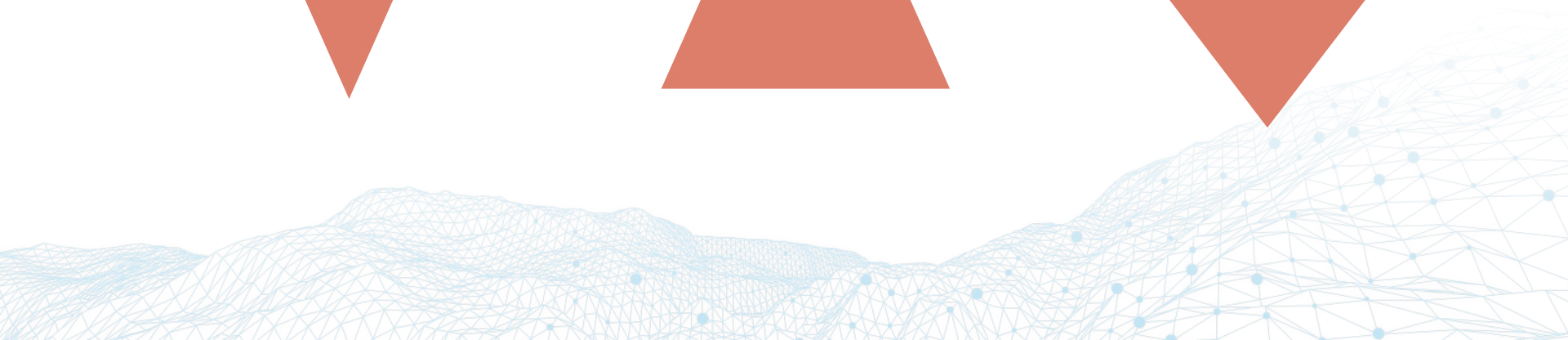
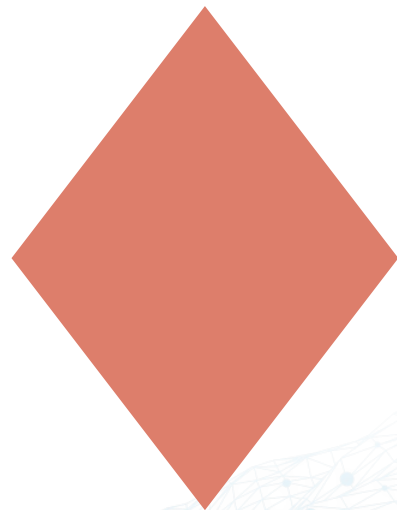
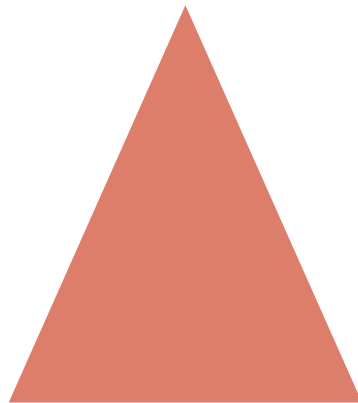
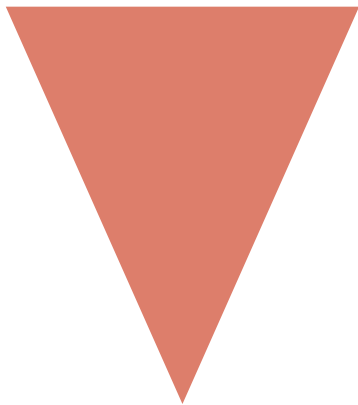
How to create Logical Coherence?

Collage



Part 7.

Let's practice!



A Story



Let's **invest** in employee training to
improve the in-store customer experience

FIGURE 7.4k My Big Idea summarized in a pithy, repeatable phrase

Back-to-school shopping accounts for

30%

of our annual revenue. Because of this, it is a huge driver of our overall annual success.

Data source: monthly Sales report. Based on prior three years (2017, 2018, 2019) of annual back-to-school sales compared to total annual sales.

FIGURE 7.4I Back up and set the plot

What we'll cover today

- 1 Discuss what we've learned**
from our survey analysis¹ and
- 2 Suggest specific recommendations**
on changes to make for the upcoming back-to-school shopping season to improve customer satisfaction and increase sales.

¹Comprehensive details on survey methodology and related info can be found in the Appendix on pages 15-20.

FIGURE 7.4m What we'll cover today

Back-to-school shopping: **consumer sentiment**

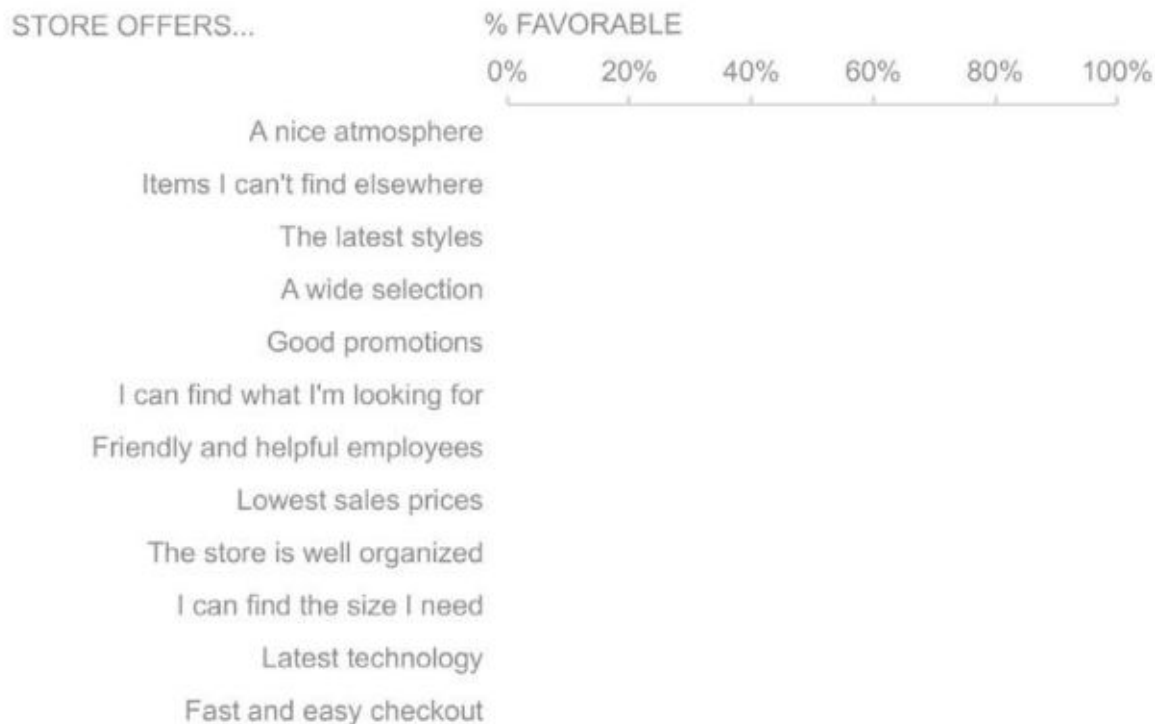


FIGURE 7.4n Set up the graph

Back-to-school shopping: **consumer sentiment**

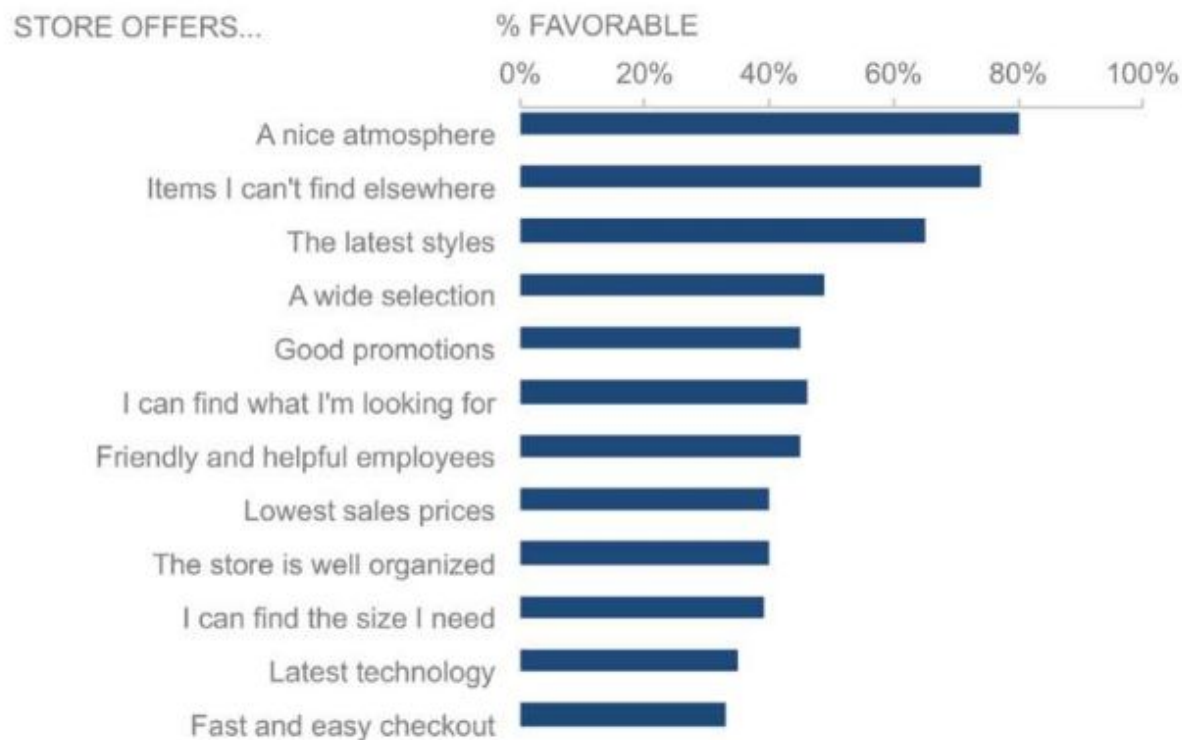


FIGURE 7.4o Focus on our business

Back-to-school shopping: **consumer sentiment**

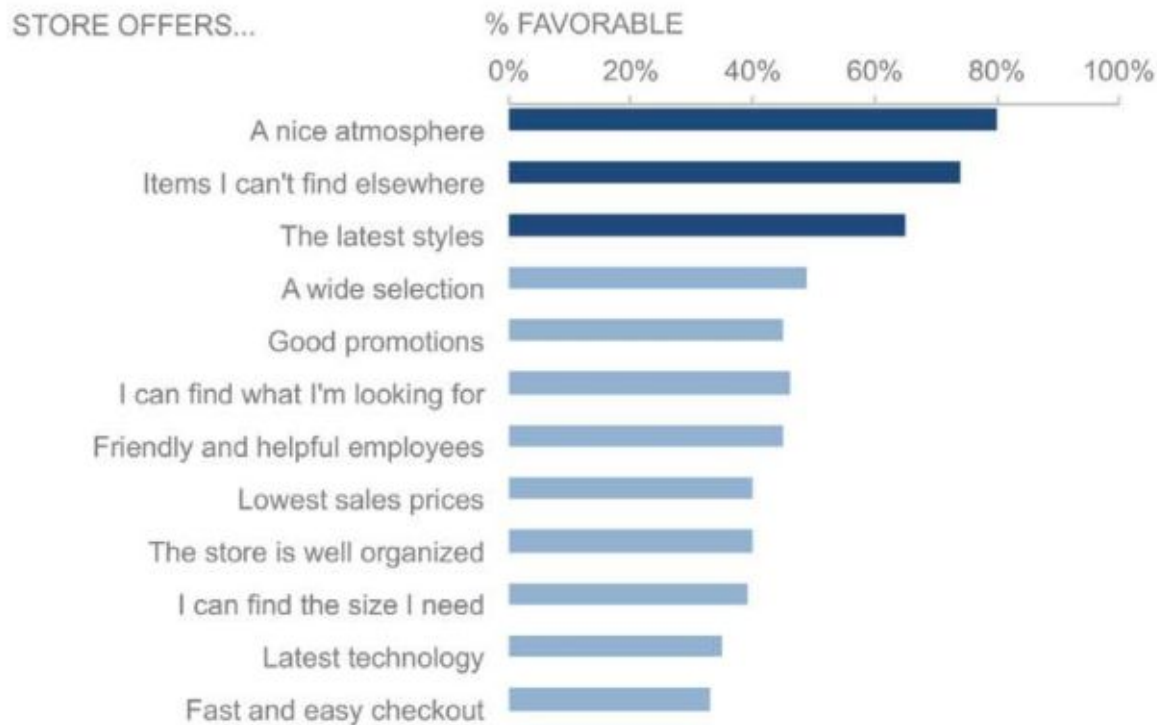


FIGURE 7.4p Focus on highest scoring items

Back-to-school shopping: **consumer sentiment**

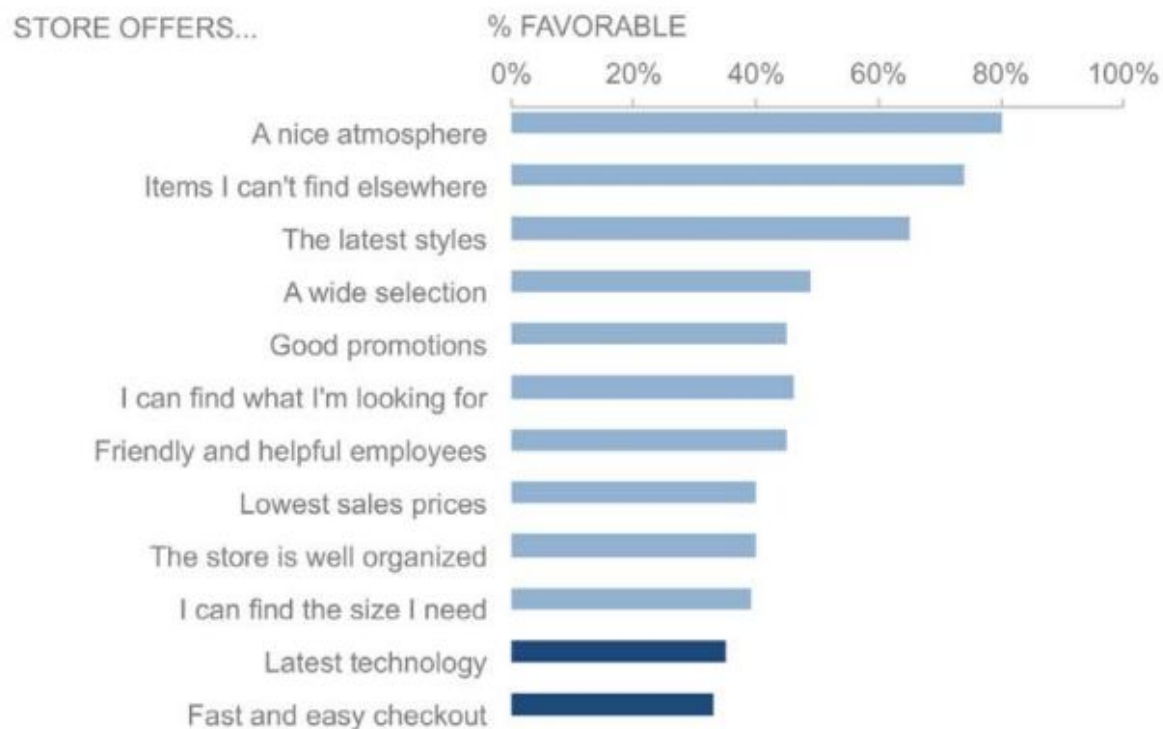


FIGURE 7.4q Focus on lowest scoring items

Back-to-school shopping: **consumer sentiment**



FIGURE 7.4r Add competitor data

Back-to-school shopping: **consumer sentiment**

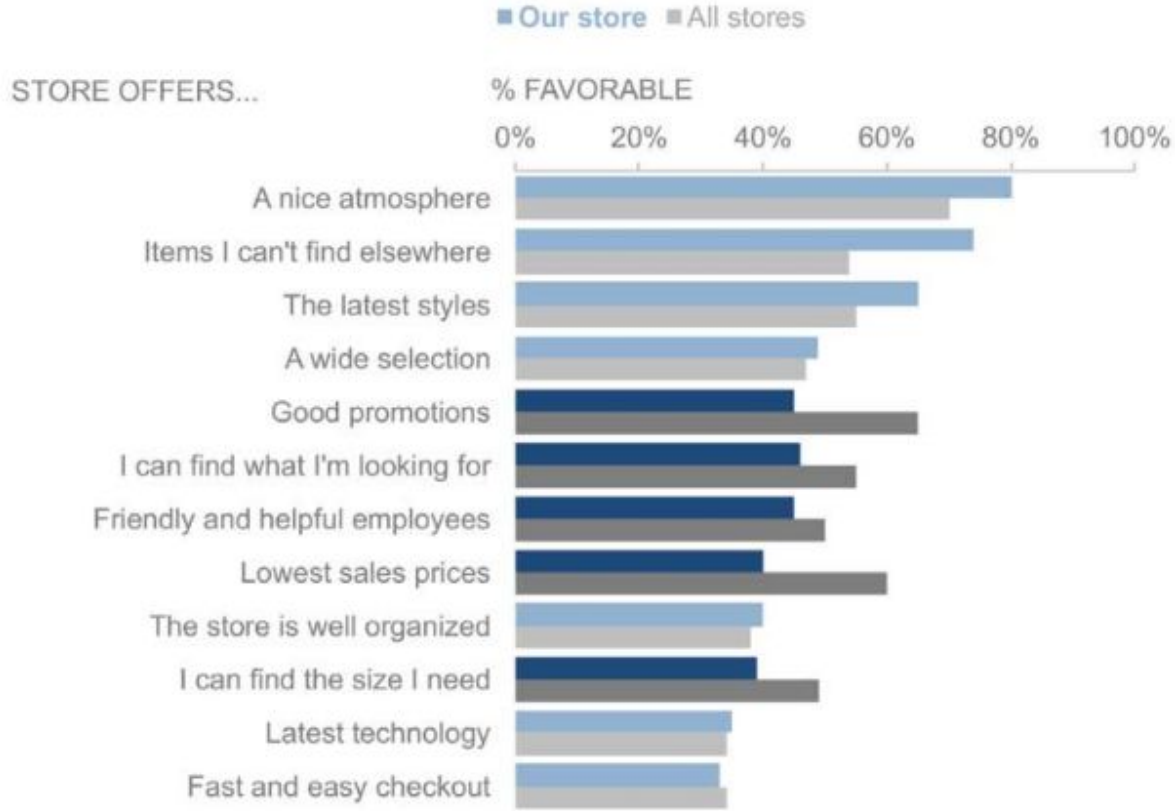


FIGURE 7.4s Highlight where we underscore competition

Back-to-school shopping: **consumer sentiment**

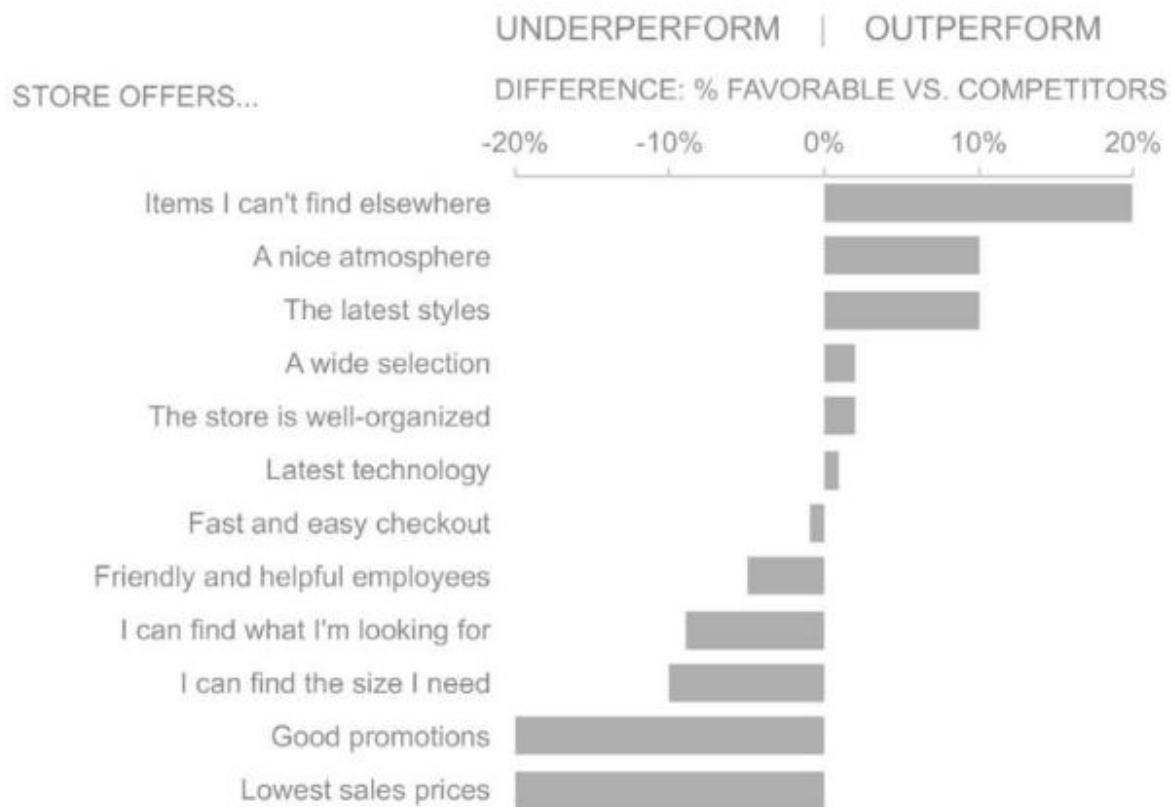


FIGURE 7.4t Shift from focus on absolute to difference

Back-to-school shopping: **consumer sentiment**

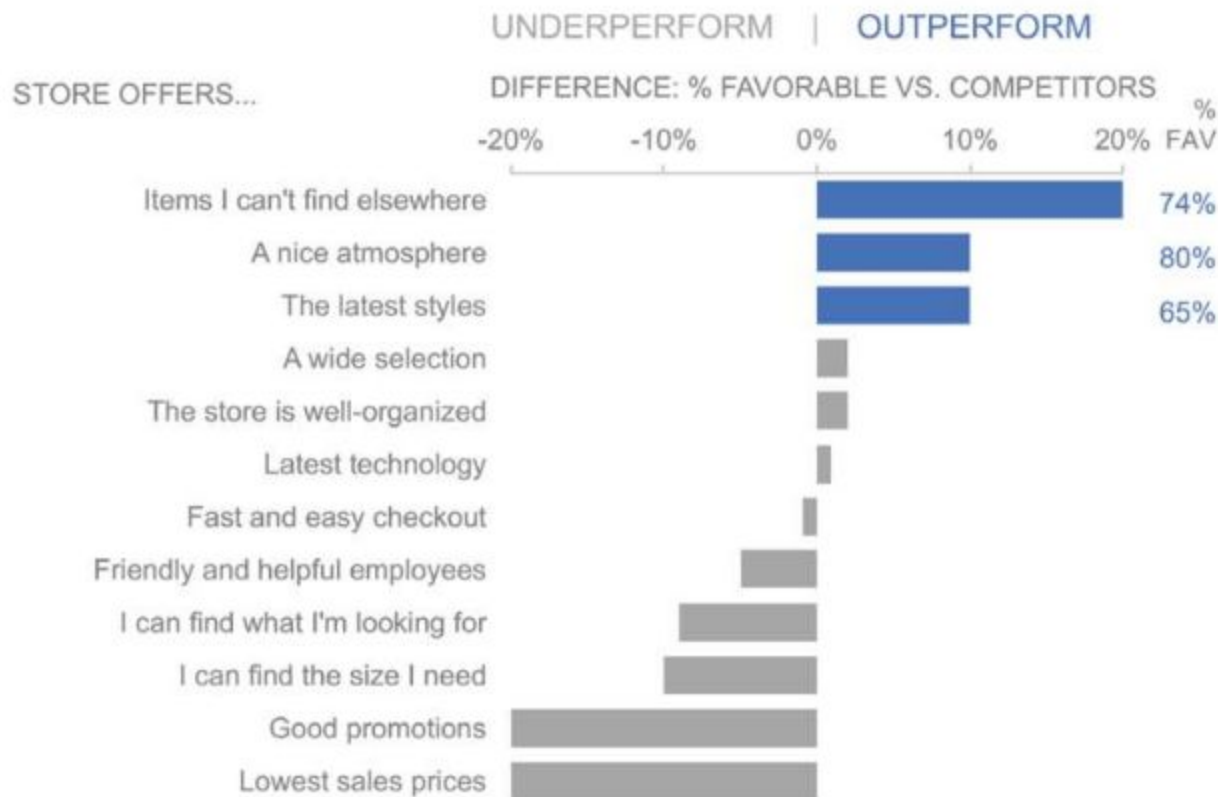


FIGURE 7.4u Focus on items we outperform

Back-to-school shopping: **consumer sentiment**



FIGURE 7.4v Focus on items we underperform

Back-to-school shopping: **consumer sentiment**

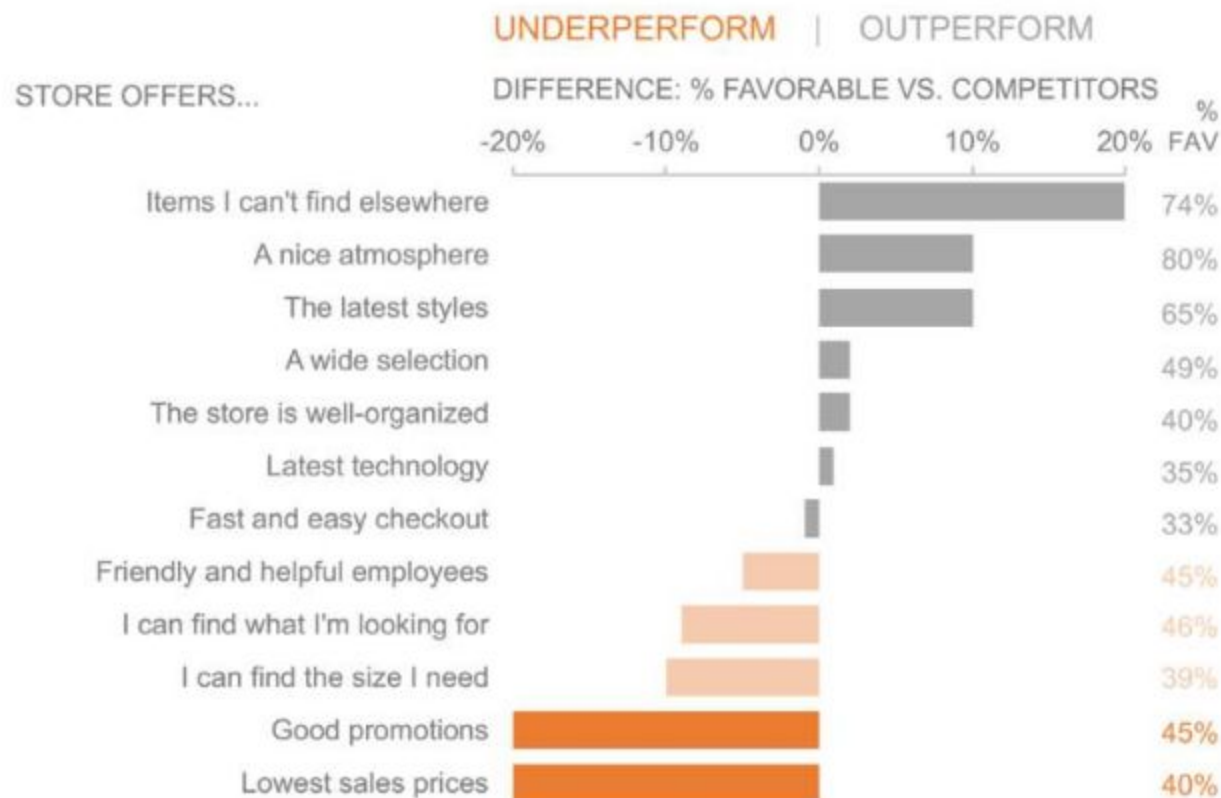


FIGURE 7.4w Underperform the most in promotions

Back-to-school shopping: consumer sentiment



FIGURE 7.4x Recommend focusing on areas we can control

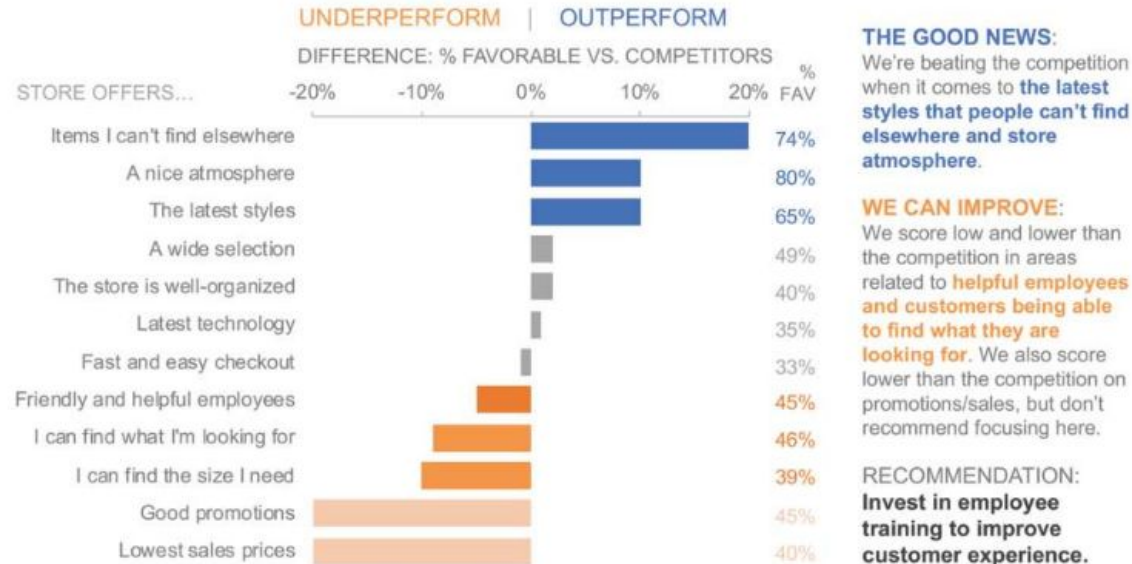


Let's **invest** in employee training to
improve the in-store customer experience

FIGURE 7.4y Repeat my Big Idea summarized in a pithy, repeatable phrase

Action needed: invest in employee training

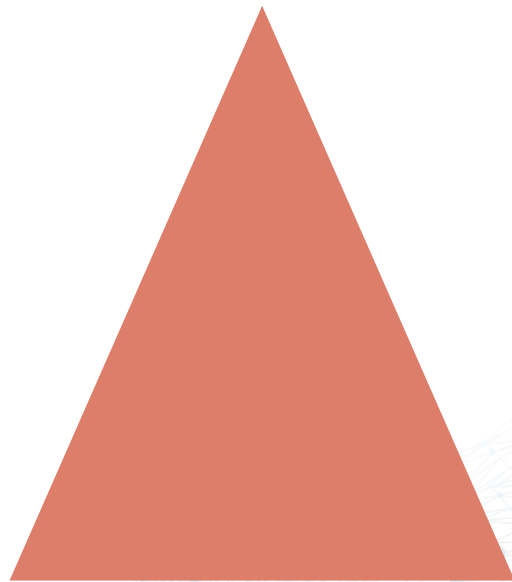
Back-to-school shopping: consumer sentiment



Data Source: 2019 Back-to-School shopping survey (represents 21,862 survey responses).
Additional survey and methodology details available upon request. Reach out to Insights Team.

FIGURE 7.4z Final annotated version

Another Story



Another Story: NPS

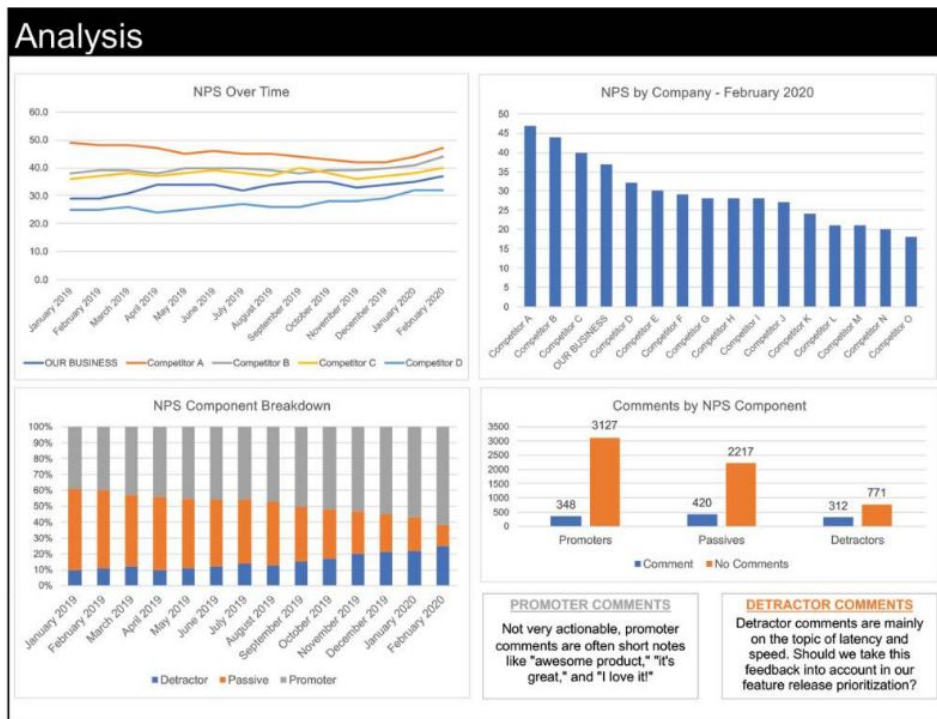


FIGURE 7.6a Typical graphs presented in monthly meeting

Let's learn from our detractors

Monthly NPS Update

Presented by: Customer Insights Team

Date: March 1, 2020

FIGURE 7.6c Title slide



Goal today

1 Build a common understanding of recent feedback.

2 Revisit product strategy given detractor feedback.

FIGURE 7.6d Goal today

NPS: flat to increasing over time

NPS over time

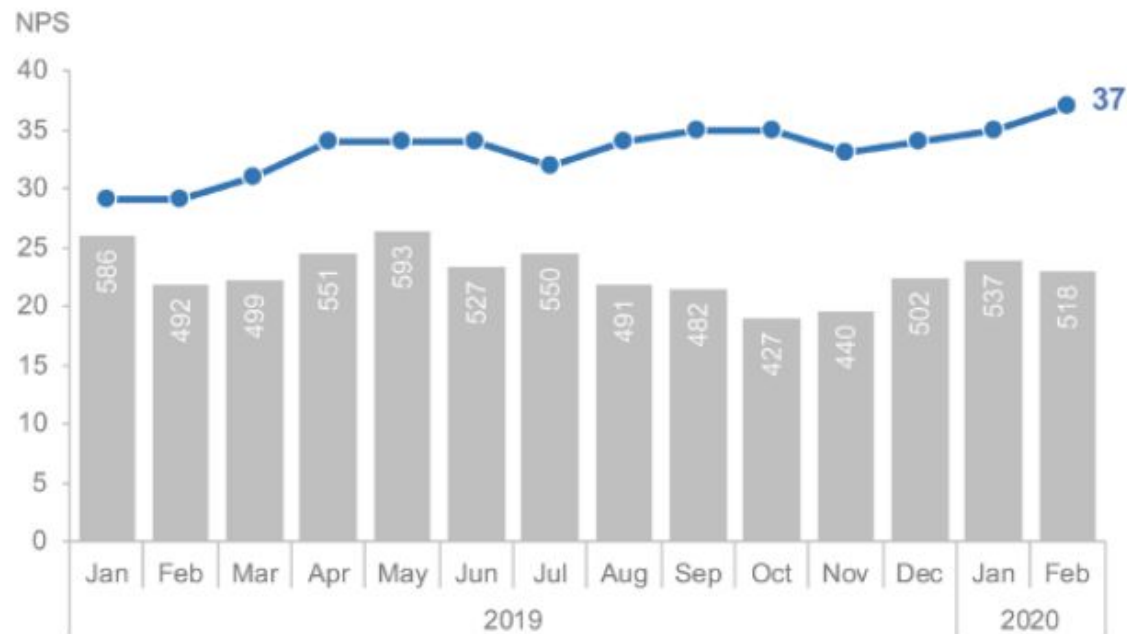


FIGURE 7.6e NPS: flat to increasing over time

NPS: we rank 4th against the competition

NPS by company – February 2020

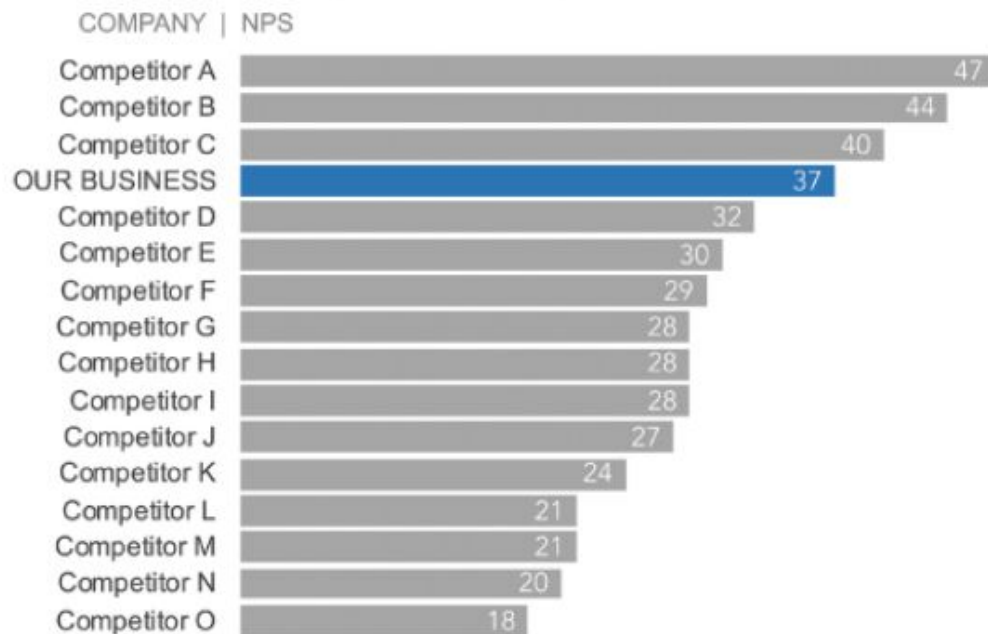


FIGURE 7.6f We rank 4th against the competition

Let's look at NPS components

NPS component breakdown over time



FIGURE 7.6g Let's look at NPS components

Proportion of passives decreasing

NPS component breakdown over time

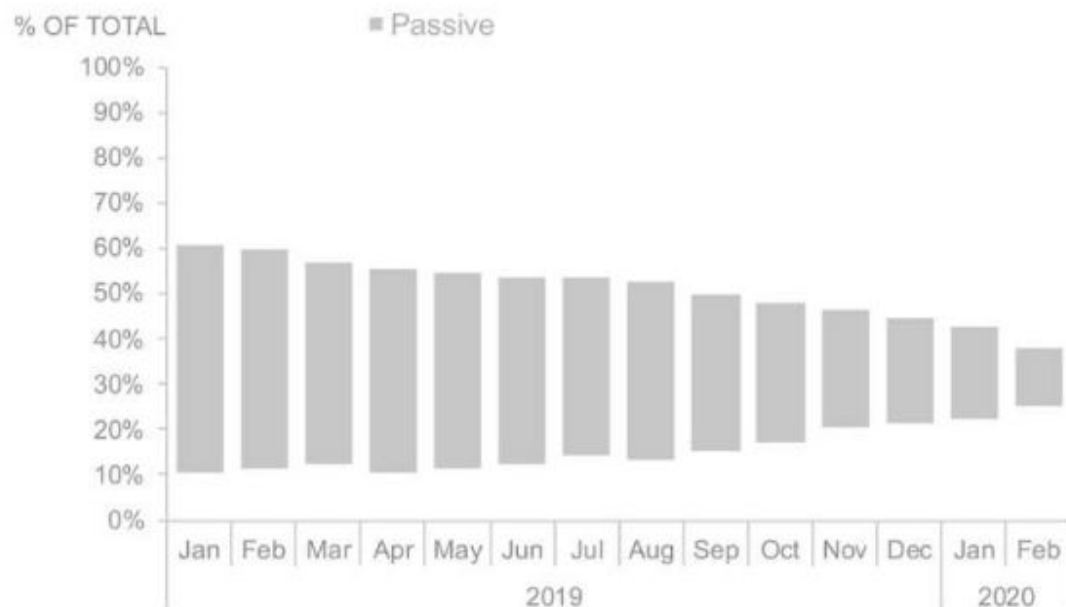


FIGURE 7.6h Proportion of passives decreasing

Proportion of promoters increasing

NPS component breakdown over time

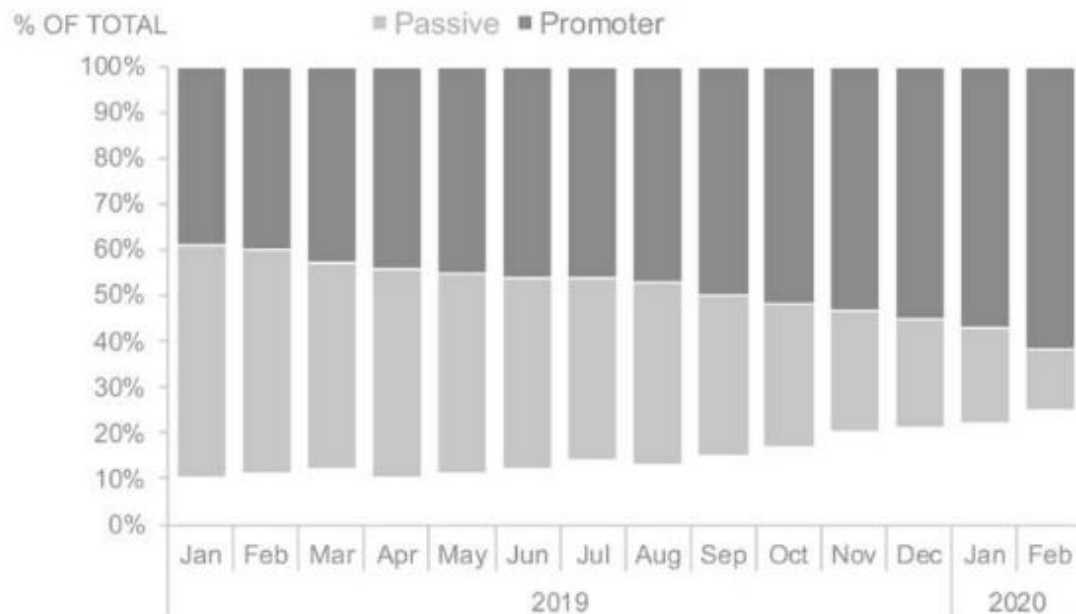


FIGURE 7.6i Proportion of promoters increasing

Proportion of detractors increasing

NPS component breakdown over time

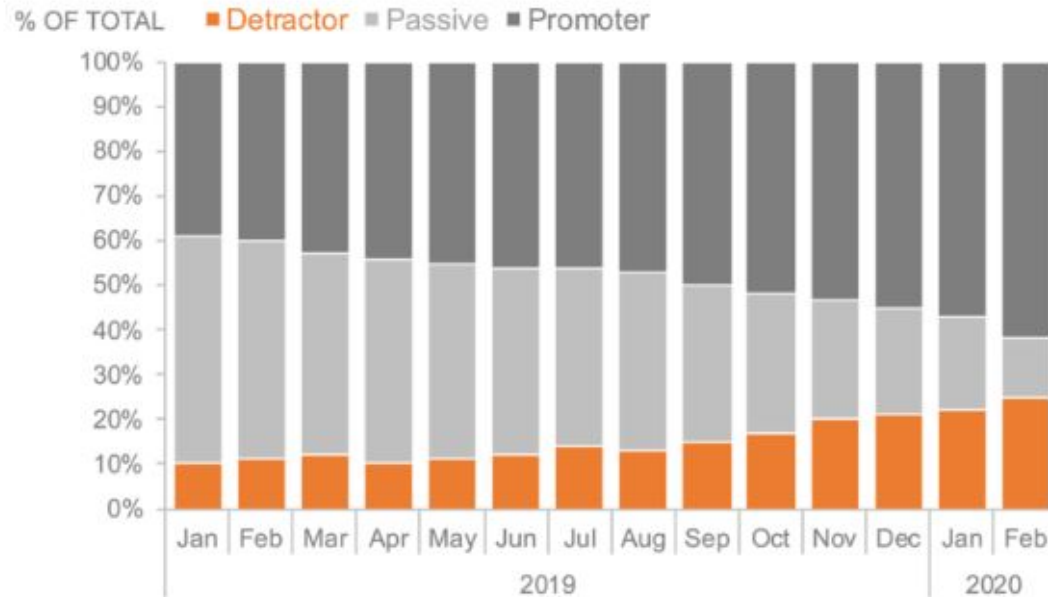


FIGURE 7.6j Proportion of detractors increasing

Detractors: **nearly doubled since Aug**

NPS component breakdown over time

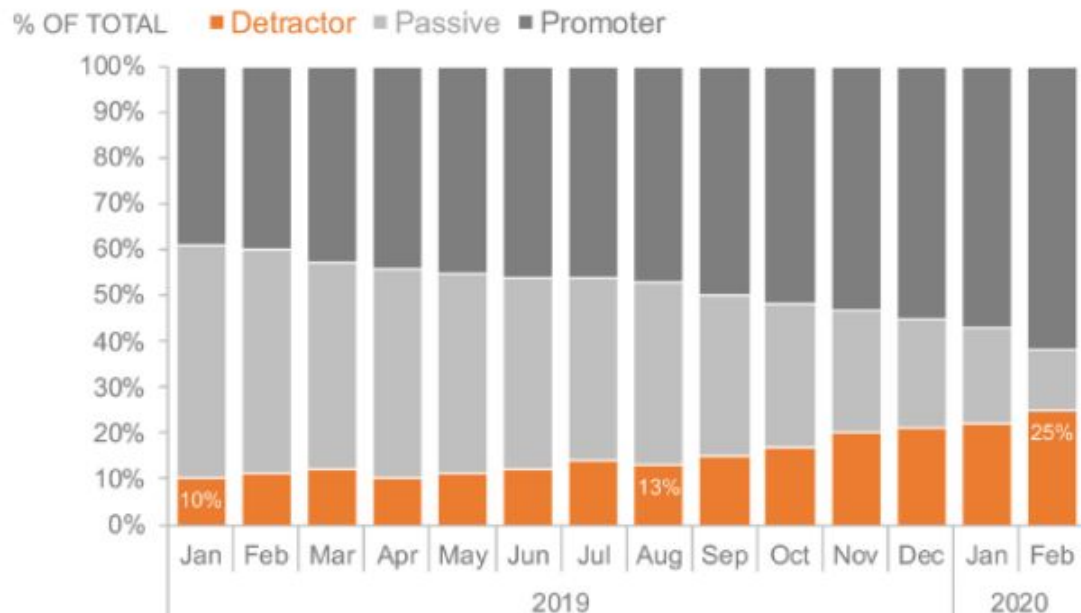


FIGURE 7.6k Detractors: nearly doubled since Aug

Detractors: relatively **more comments**

Comments by NPS component

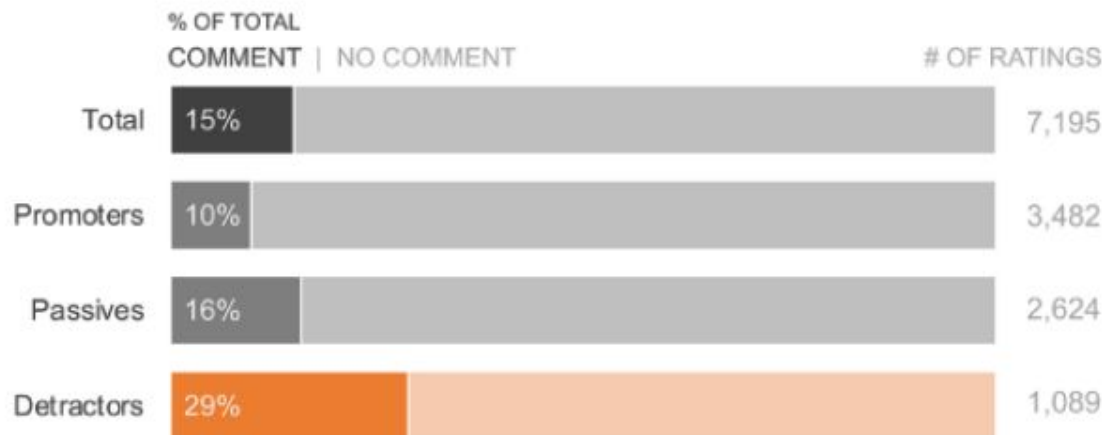


FIGURE 7.6I Detractors: relatively more comments

Comments provide **insights into issues**

1 in 3 detractor comments are about **speed or latency**

This was the single biggest comment theme. The next most common theme, unexpected restarts, accounted for only 6% of detractor comments.

FIGURE 7.6m Comments provide insights into issues

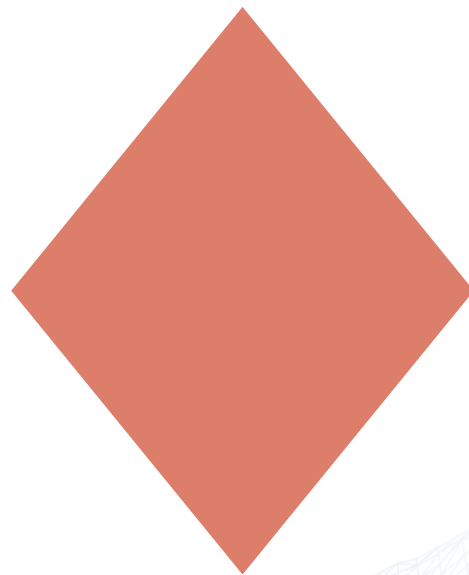
RECOMMENDATION:

Revisit our product and feature release strategy in light of this feedback and **prioritize latency improvements.**

Let's discuss.

FIGURE 7.6n Recommendation

Another Story



3: Logic in order

How satisfied have you been with each of these features?

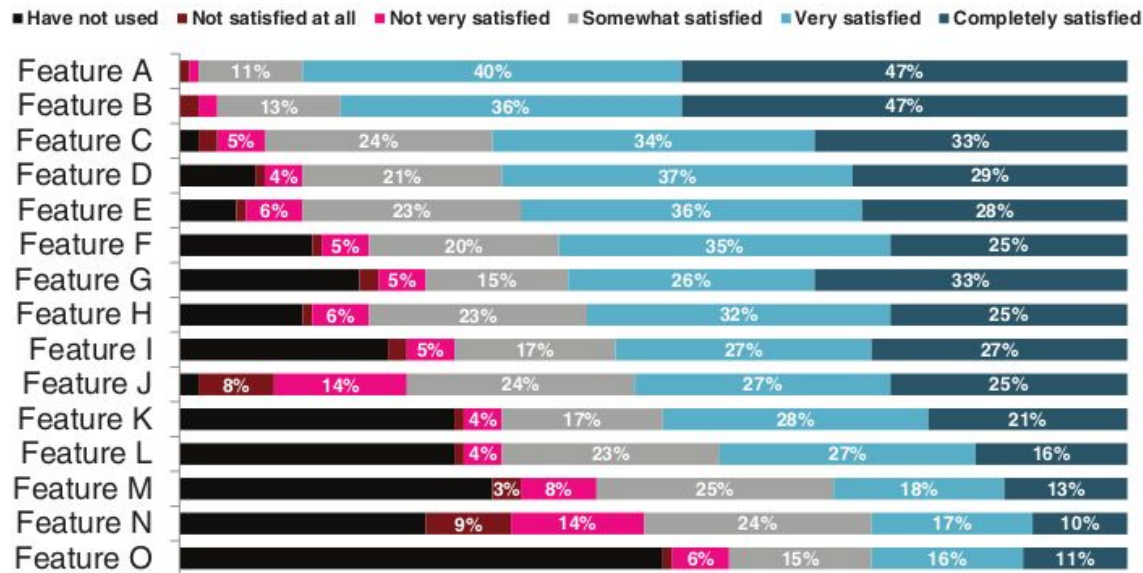
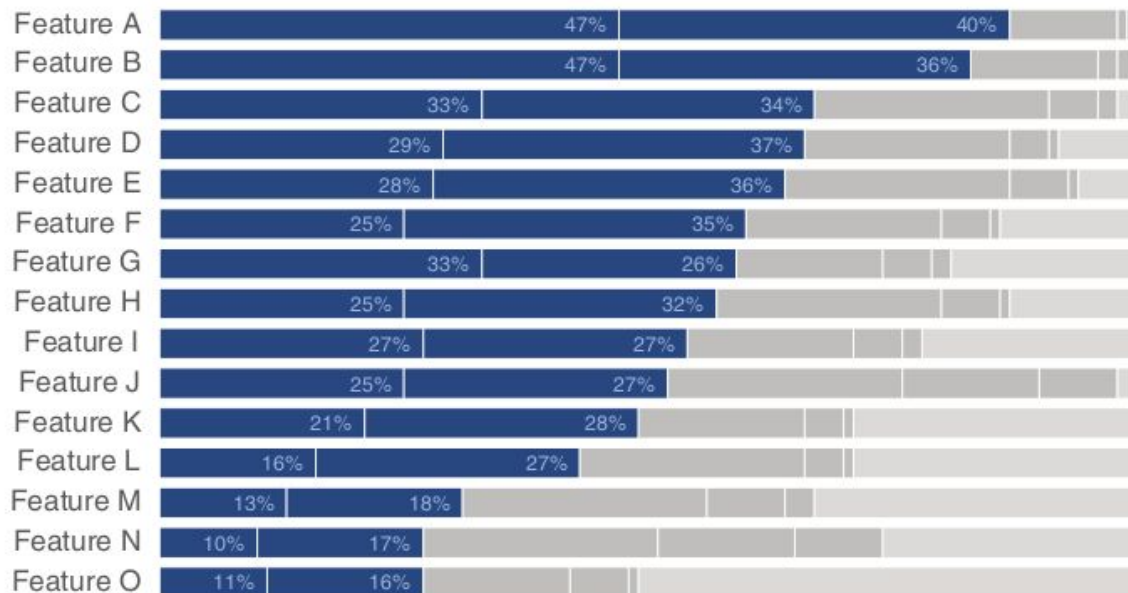


FIGURE 9.12 User satisfaction, original graph

Features A & B top user satisfaction

Product X User Satisfaction: **Features**

■ **Completely satisfied** ■ **Very satisfied** ■ Somewhat satisfied ■ Not very satisfied ■ Not satisfied at all ■ Have not used



Responses based on survey question "How satisfied have you been with each of these features?".

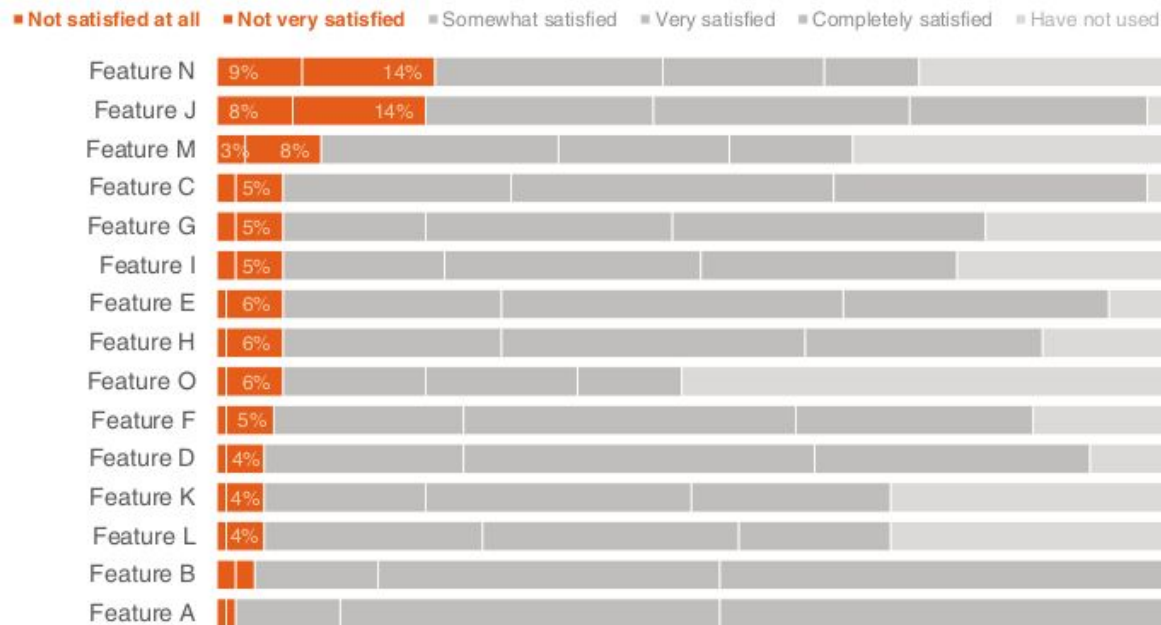
Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.13 Highlight the positive story

Users least satisfied with Features N & J

Product X User Satisfaction: **Features**



Responses based on survey question "How satisfied have you been with each of these features?".

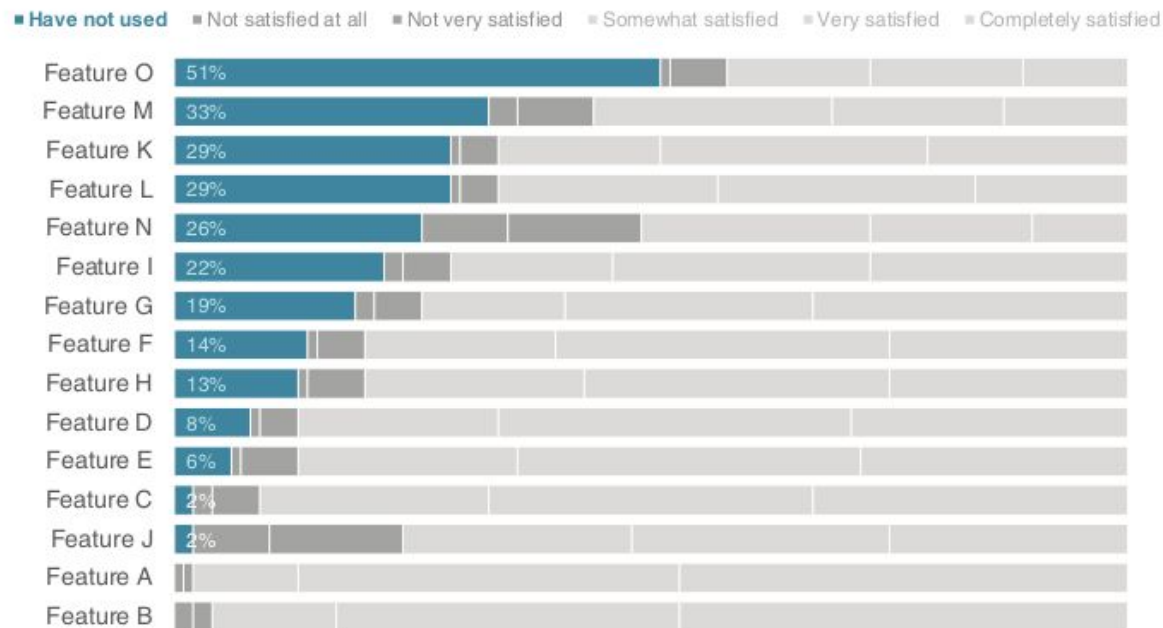
Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.14 Highlight dissatisfaction

Feature O is least used

Product X User Satisfaction: **Features**



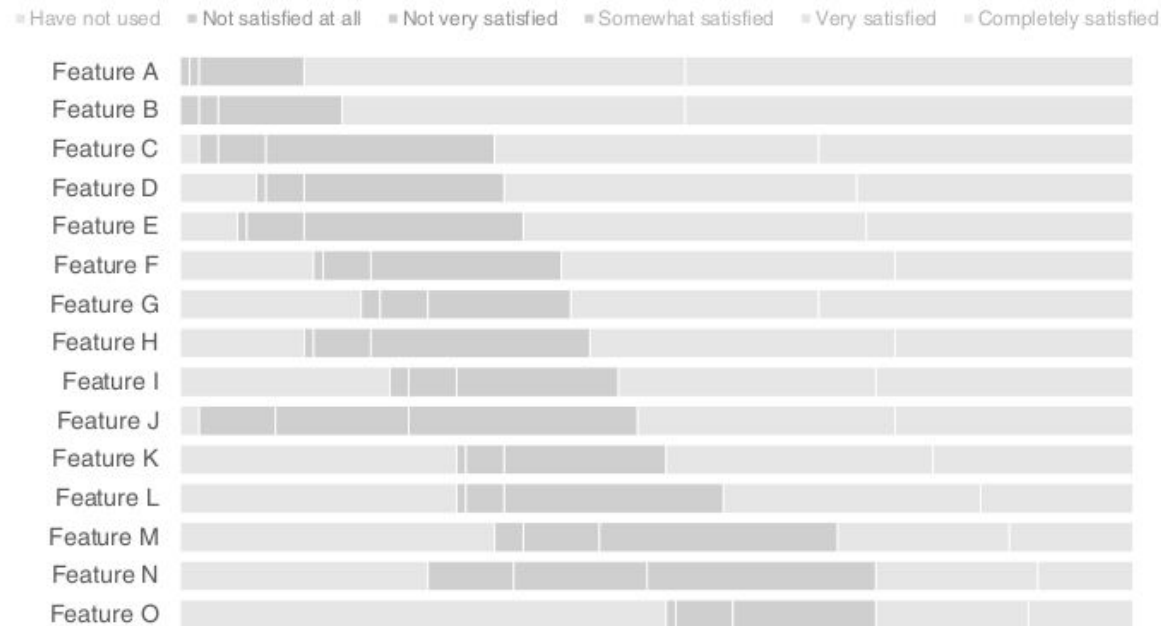
Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent? Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.15 Focus on unused features

User satisfaction varies greatly by feature

Product X User Satisfaction: **Features**



Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

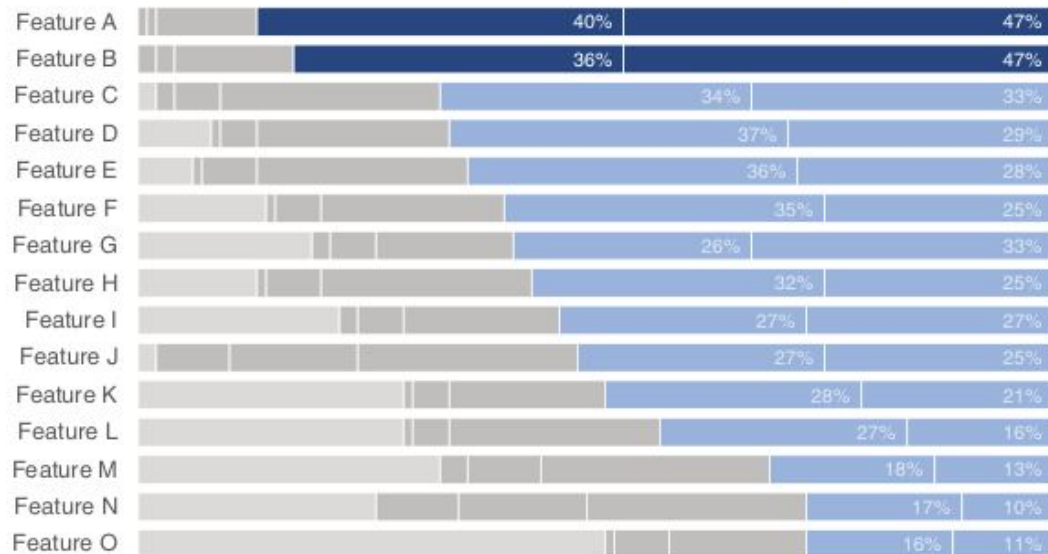
Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.16 Set up the graph

User satisfaction varies greatly by feature

Product X User Satisfaction: **Features**

■ Have not used ■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



Features A and B continue to top user satisfaction

Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

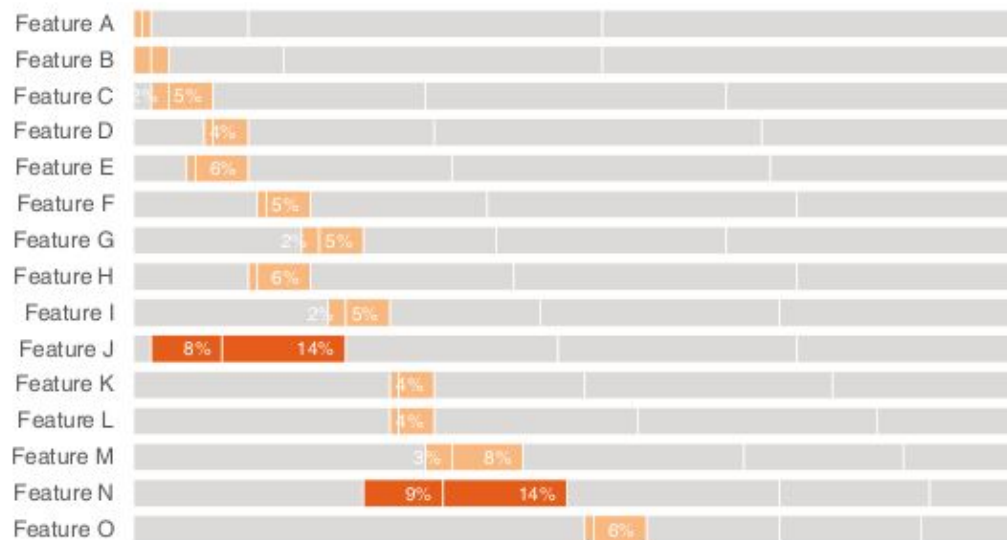
Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.17 Satisfaction

User satisfaction varies greatly by feature

Product X User Satisfaction: **Features**

■ Have not used ■ **Not satisfied at all** ■ **Not very satisfied** ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



Users are least satisfied with Features J and N; what improvements can we make here for a better user experience?

Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

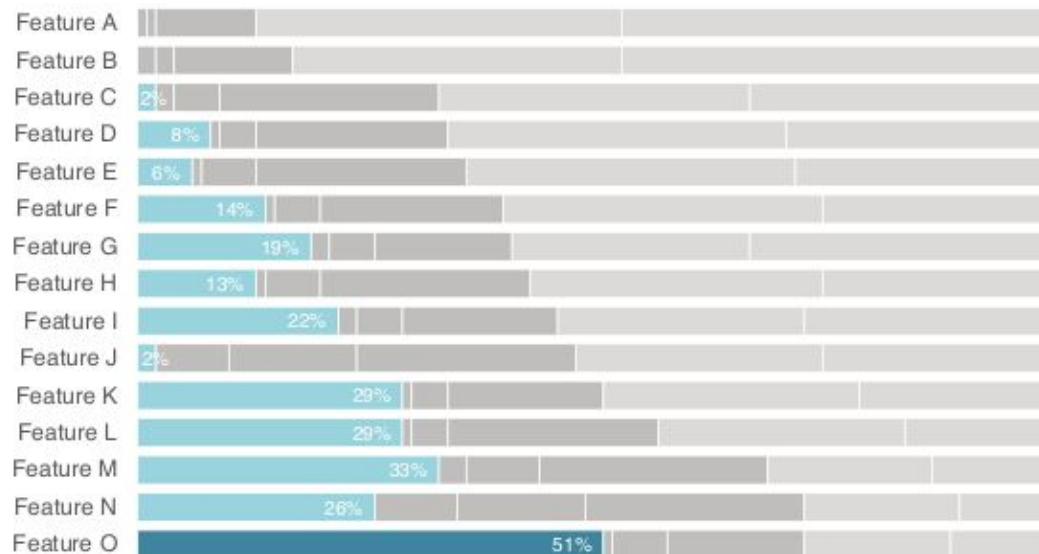
Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.18 Dissatisfaction

User satisfaction varies greatly by feature

Product X User Satisfaction: **Features**

■ Have not used ■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



Feature O is least used. What steps can we proactively take with existing users to increase utilization?

Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

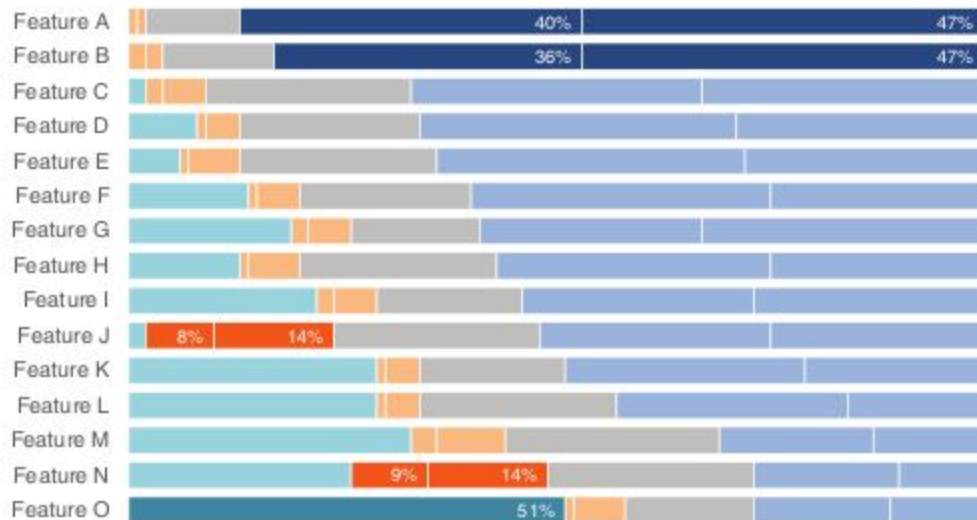
Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.19 Unused features

User satisfaction varies greatly by feature

Product X User Satisfaction: **Features**

■ Have not used ■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



Features A and B continue to top user satisfaction

Users are least satisfied with Features J and N; what improvements can we make here for a better user experience?

Feature O is least used. What steps can we proactively take with existing users to increase utilization?

Responses based on survey question "How satisfied have you been with each of these features?".

Need more details here to help put this data into context: How many people completed survey? What proportion of users does this represent?

Do those who completed survey look like the overall population, demographic-wise? When was the survey conducted?

FIGURE 9.20 Comprehensive visual

Another Story



Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.4 Original graph

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.5

Moonville: active users over time



Data source: ABC Report. For purpose of analysis *active user* is defined as the number of unique users in the past 30 days.

FIGURE 9.6

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.7

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.8

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.9

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.10

Moonville: active users over time



Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

FIGURE 9.11

Another Story



Exercise: sales channel update

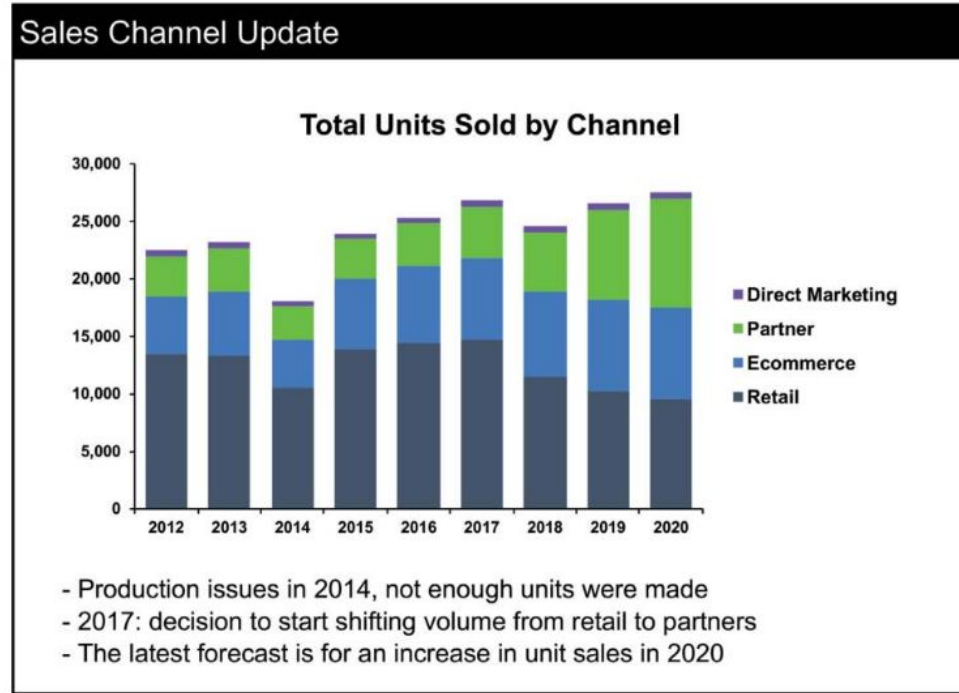


FIGURE 7.2a Sales channel update

Total units sold

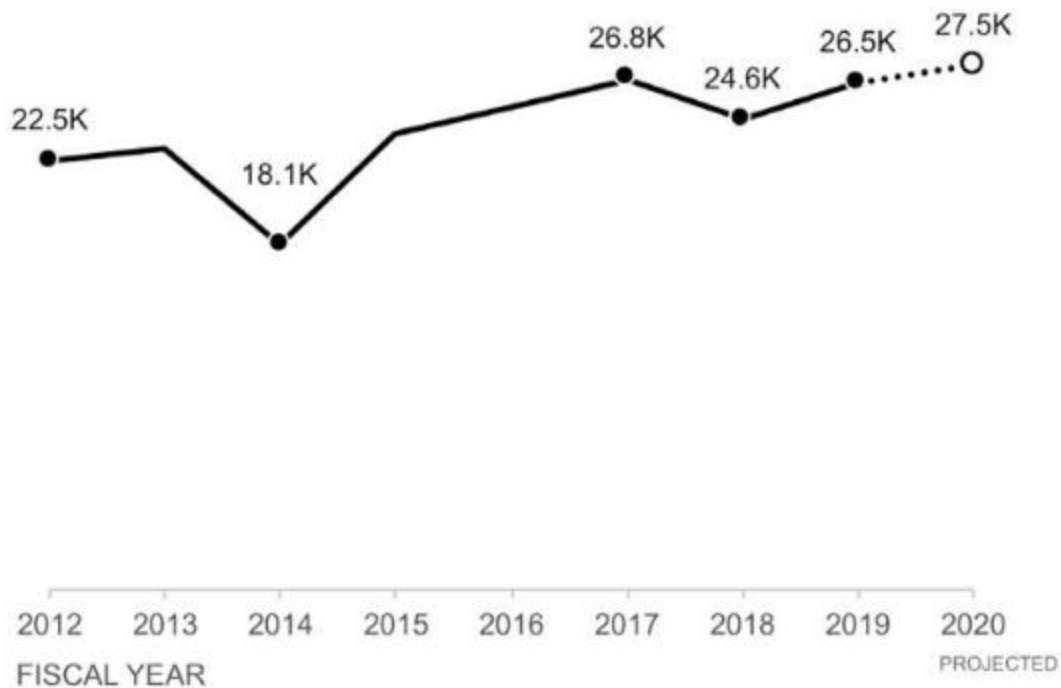


FIGURE 7.2b Visualize units sold over time with a line

Story

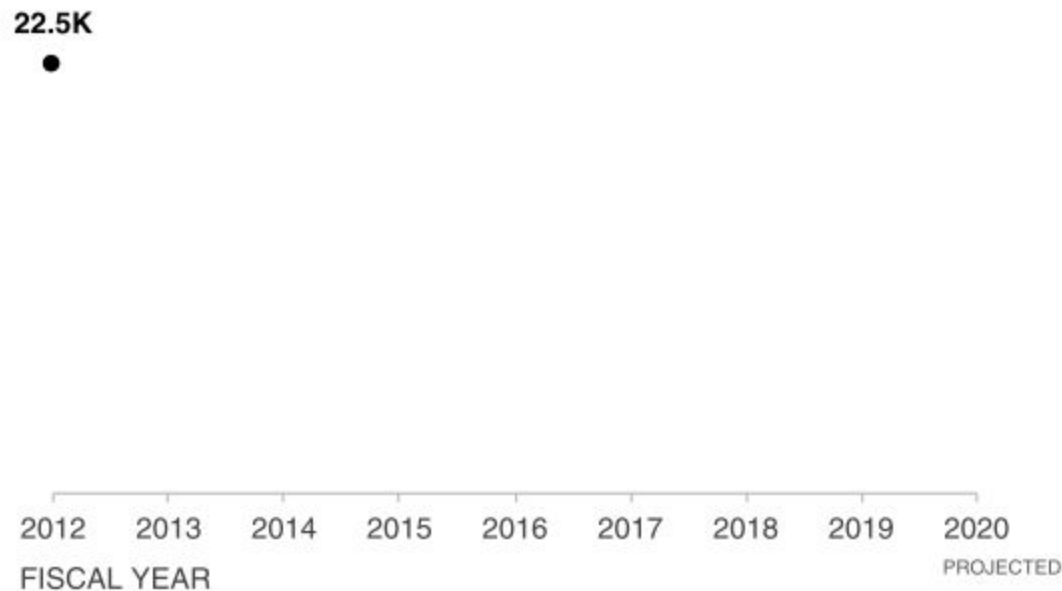
Total units sold



FIGURE 7.2c In a live progression, first set up graph

Total units sold

22.5K



Fiscal Year	Total Units Sold
2012	22.5K
2013	
2014	
2015	
2016	
2017	
2018	
2019	
2020 (Projected)	

2012 2013 2014 2015 2016 2017 2018 2019 2020
FISCAL YEAR PROJECTED

FIGURE 7.2d Live progression

Total units sold

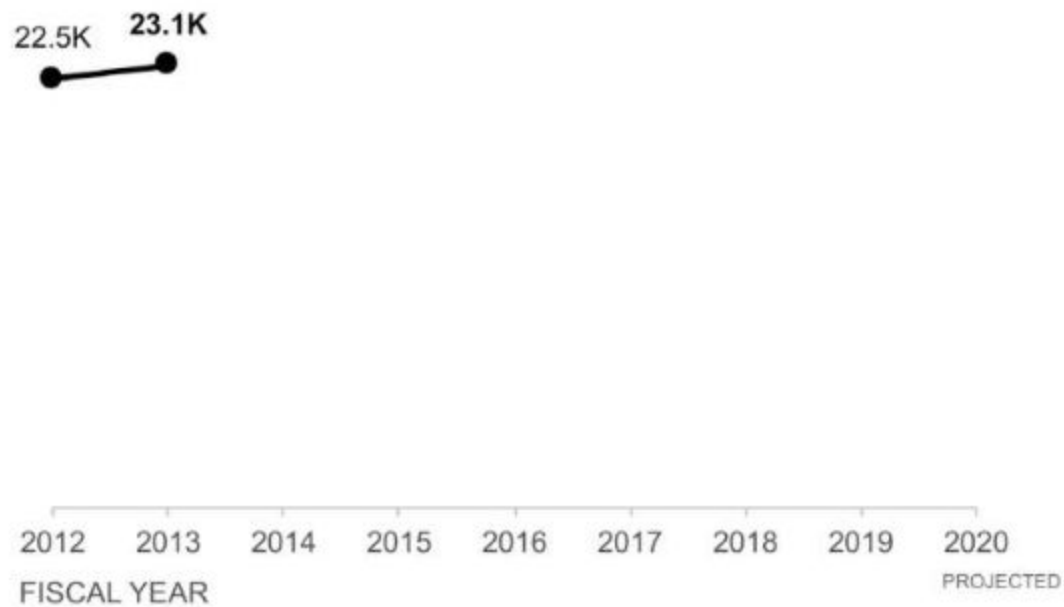


FIGURE 7.2e Live progression

Total units sold

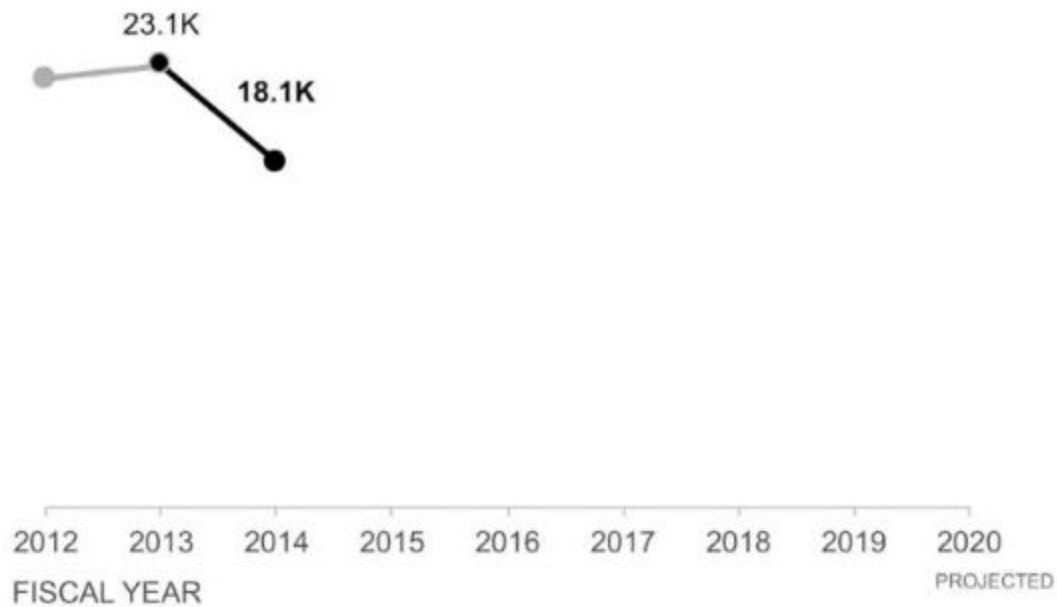


FIGURE 7.2f Live progression

Total units sold

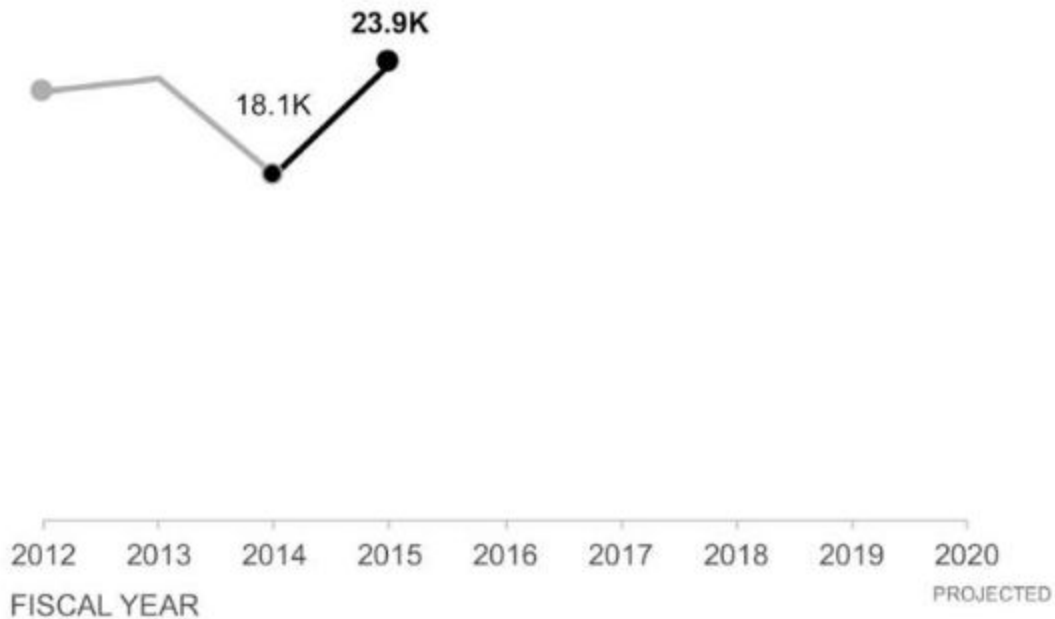


FIGURE 7.2g Live progression

Total units sold

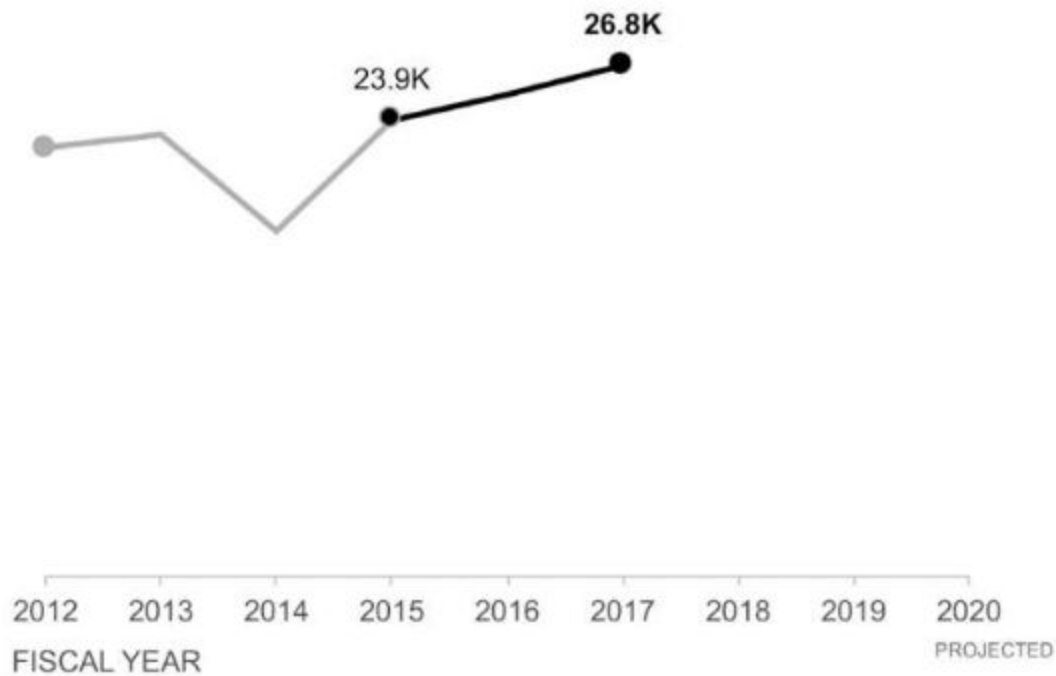


FIGURE 7.2h Live progression

Total units sold

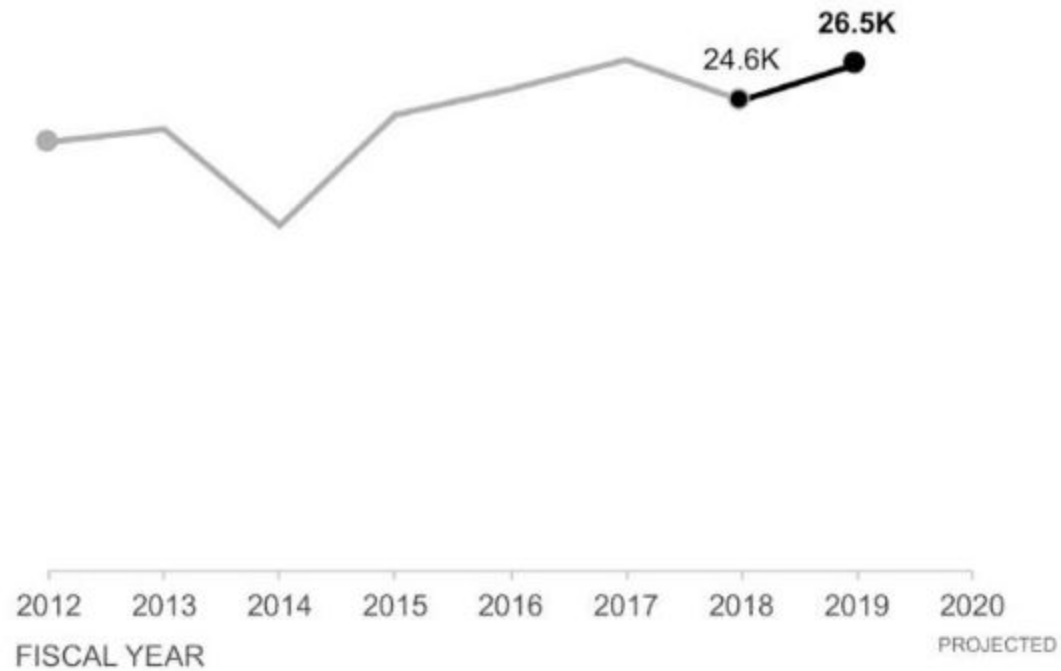


FIGURE 7.2j Live progression

Total units sold

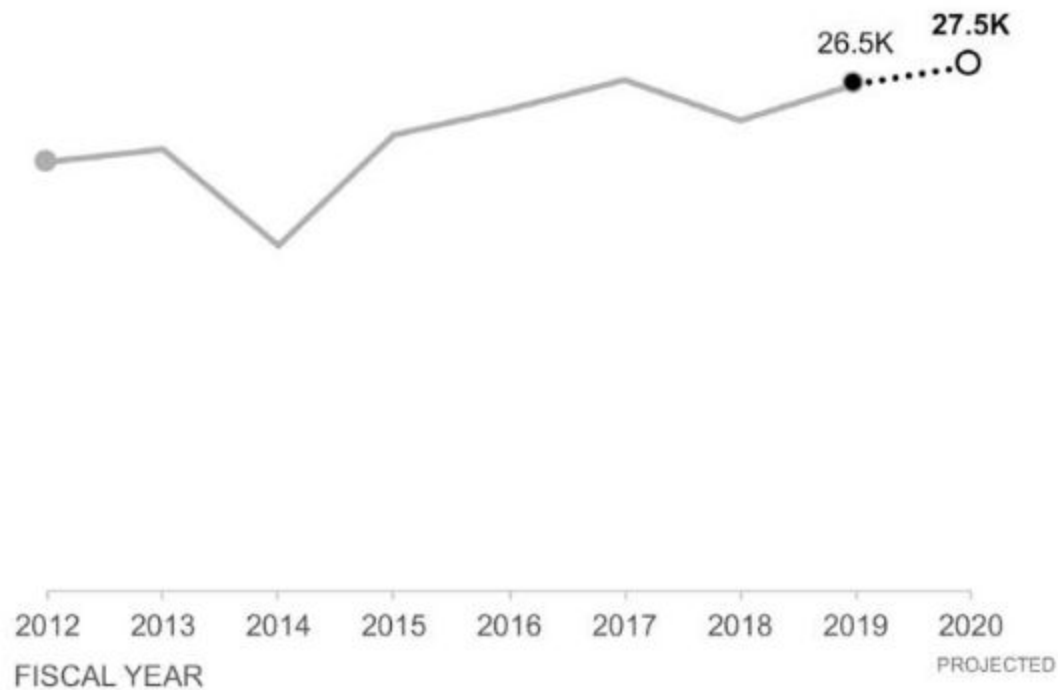


FIGURE 7.2k Live progression

Total units sold

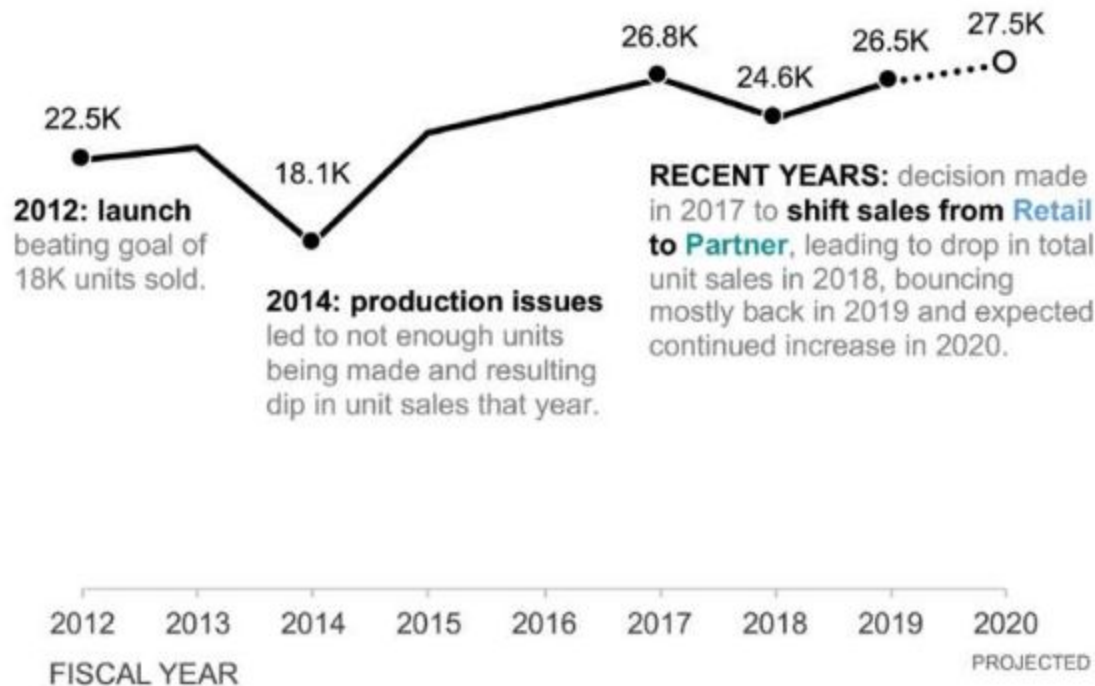


FIGURE 7.2I Annotated line graph

Units sold by channel

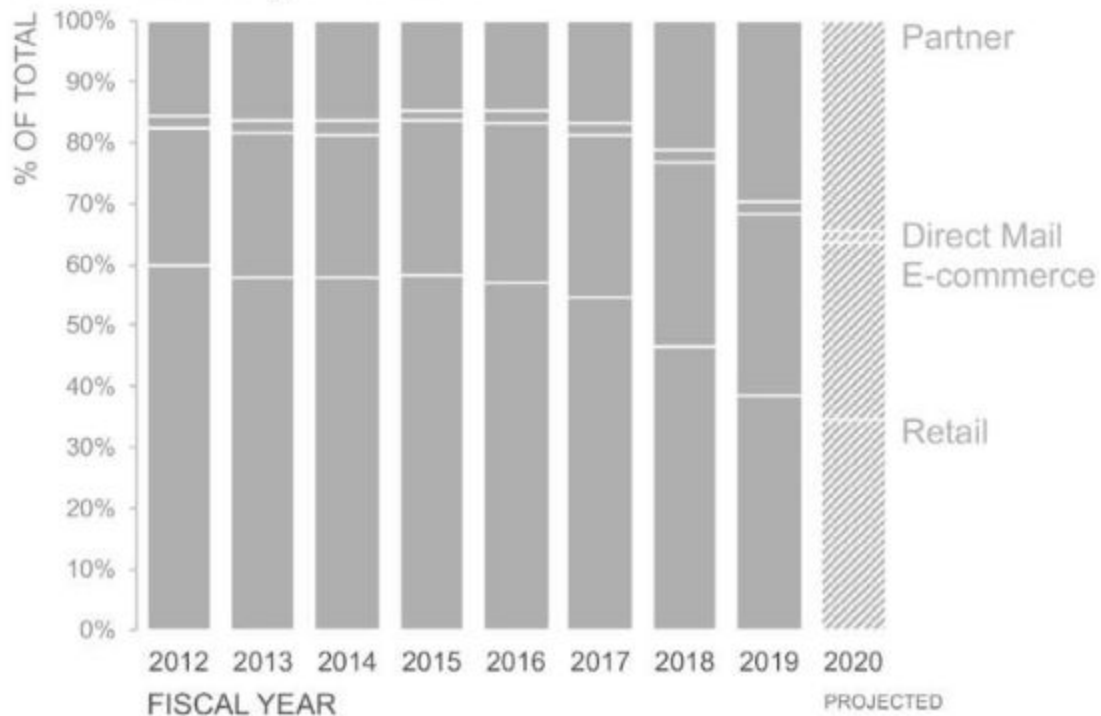


FIGURE 7.2m Another view: channel breakdown

Units sold by channel

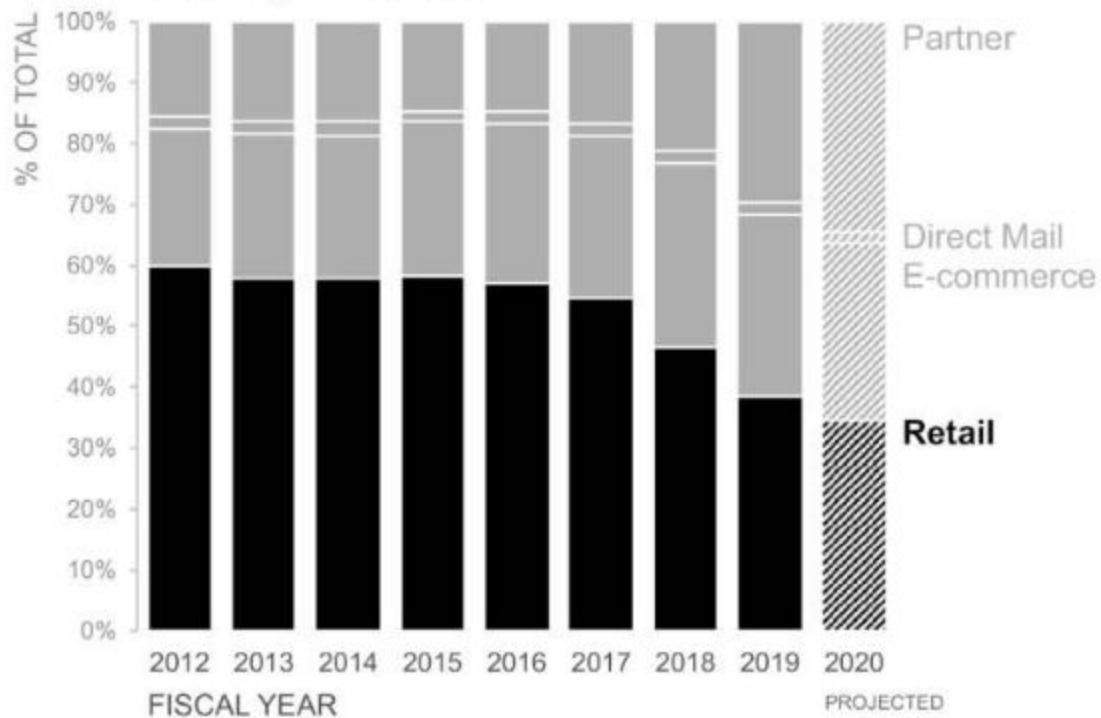


FIGURE 7.2n Focus on retail

Units sold by channel

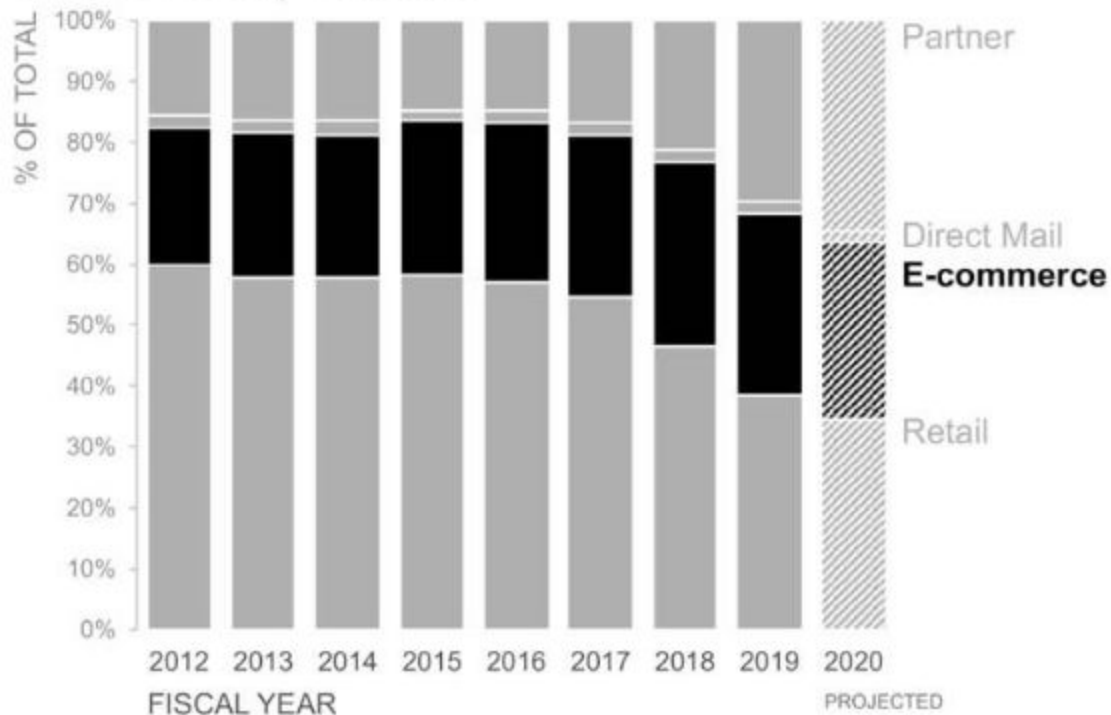


FIGURE 7.2o Focus on e-commerce

Units sold by channel

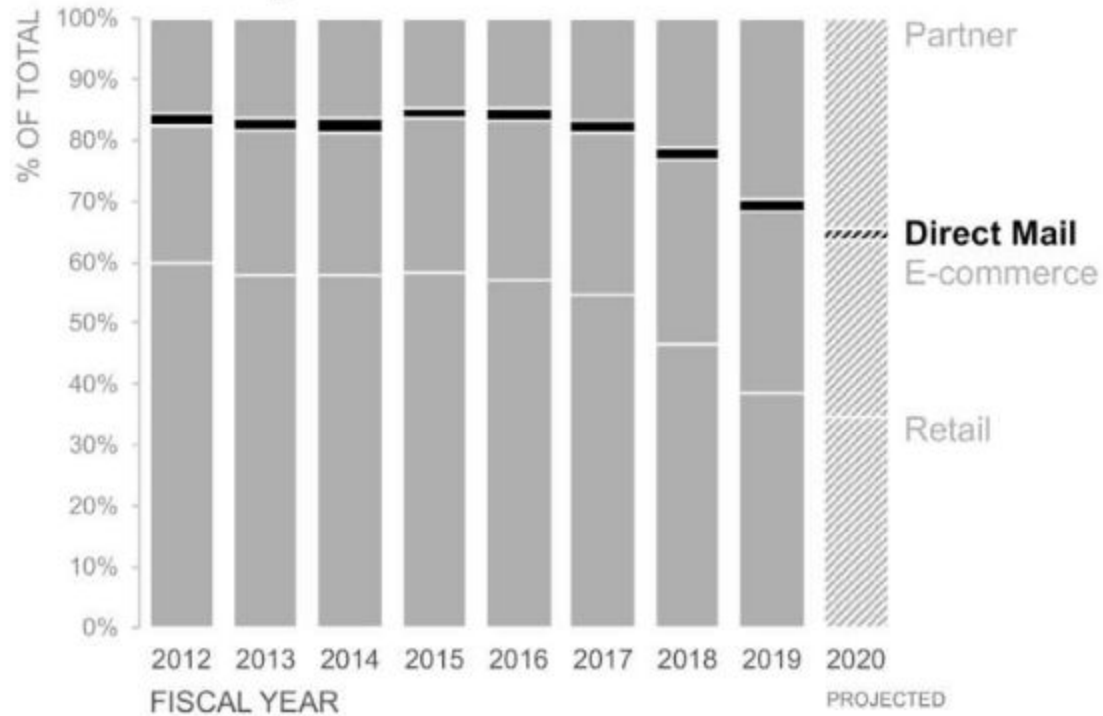


FIGURE 7.2p Focus on direct mail

Units sold by channel

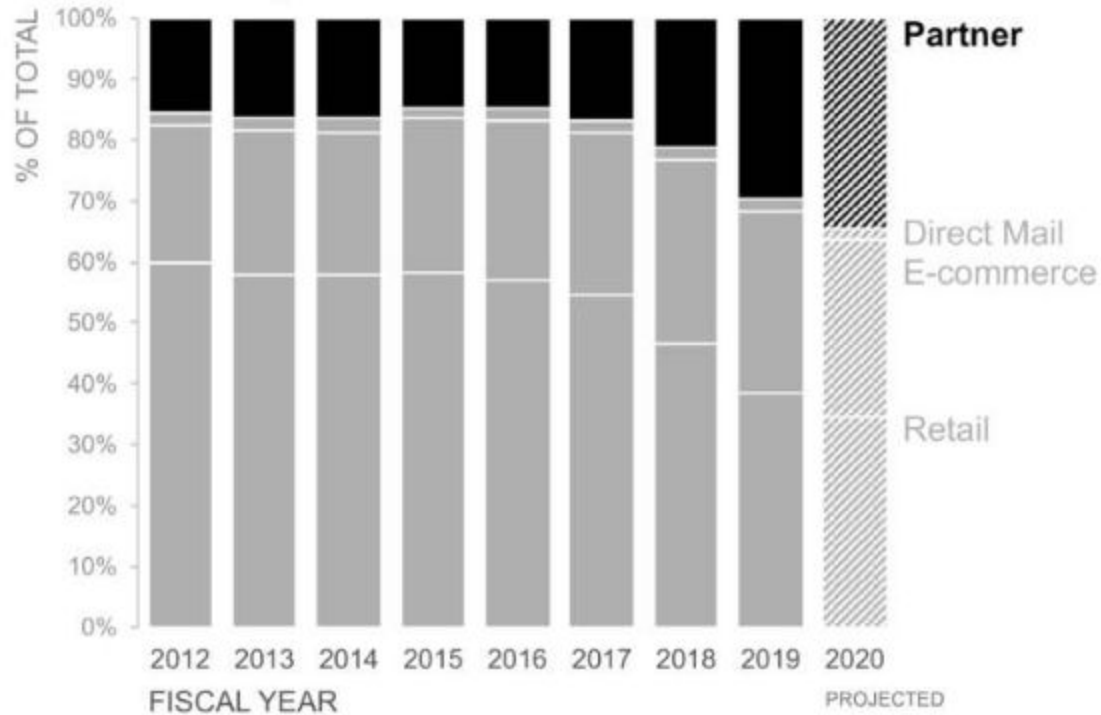


FIGURE 7.2q Focus on partner

Units sold by channel

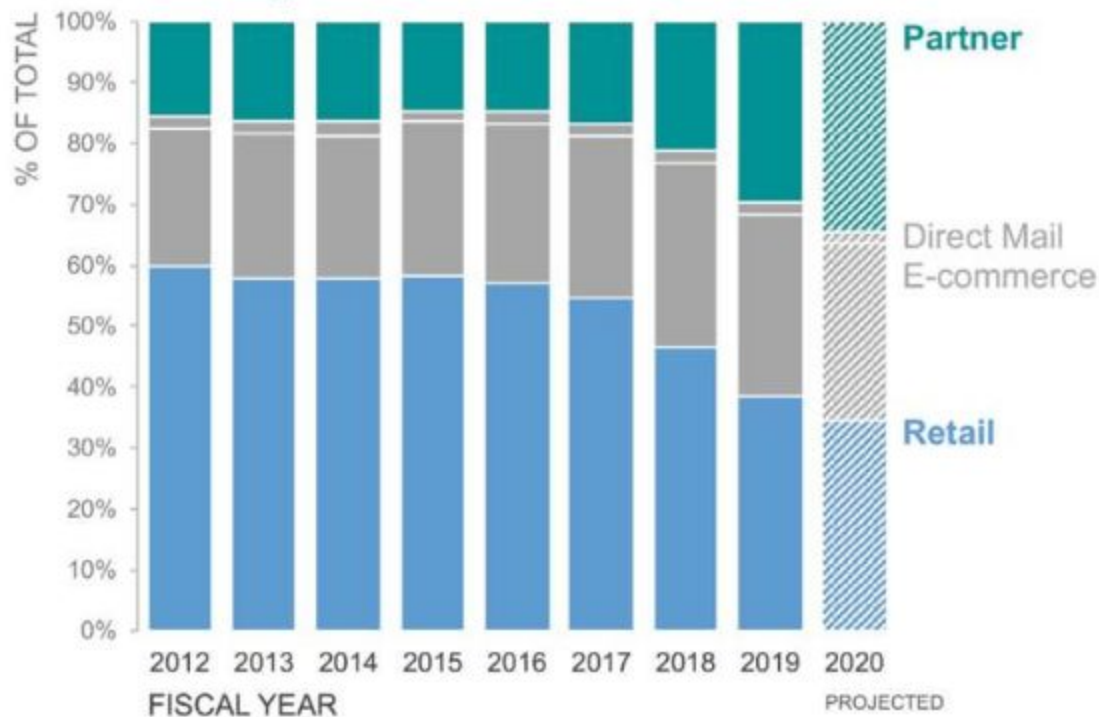


FIGURE 7.2r The desired shift in channels has happened: success!

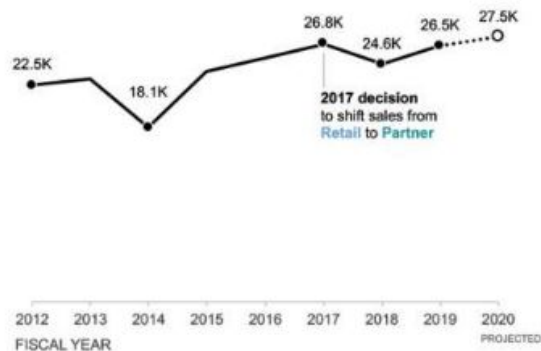
Send-out

Total sales increasing as channels shift

2020 EXPECT INCREASE IN SALES

The 2012 launch yielded 22.5K units sold, beating initial 18K goal by 25%. In 2014, production issues led to not enough units being made and resulting in a 22% drop in annual unit sales. Unit sales increased through 2017, when the **decision was made to shift channel sales** from retail to partners. This led to a drop in 2018, bouncing mostly back in 2019, and expected to continue to increase in 2020.

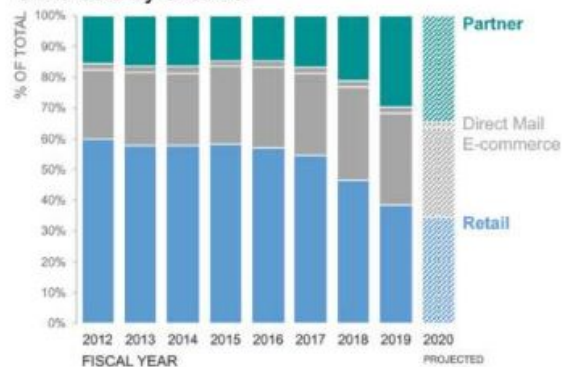
Total units sold



CHANNEL SHIFT OVER TIME

The relative composition of sales channels has shifted markedly over time. Unit sales from the **Retail** channel, which historically accounted for 60% of total sales, has decreased to 38% in 2019 due to intentional shift towards the **Partner** channel, where contribution to total unit sales has increased from less than 20% in our early years to 30% in 2019. We expect these trends to continue.

Units sold by channel



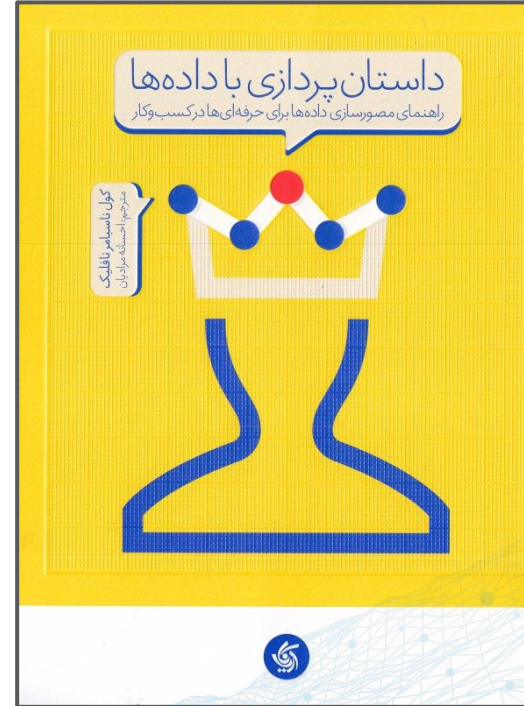
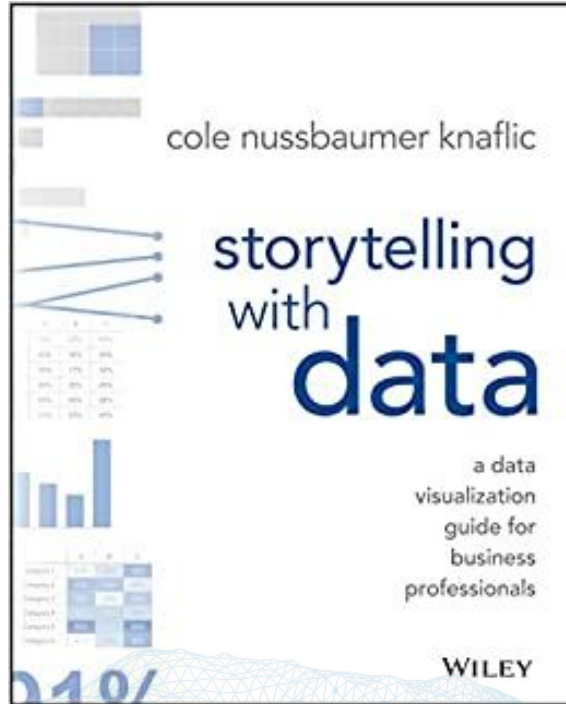
Data is from Sales Database through 12/31/19. 2020 projections assume no major market changes. For more details, reach out to Sales Analytics team.

FIGURE 7.2s Single slide for distribution

Final Thoughts



Resources + Experiences



Storytelling With Data

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